How Long Can a Collatz Orbit Follow the Critical Parity Word?

William C. Houze, Ph.D.

Independent Scholar // Researcher

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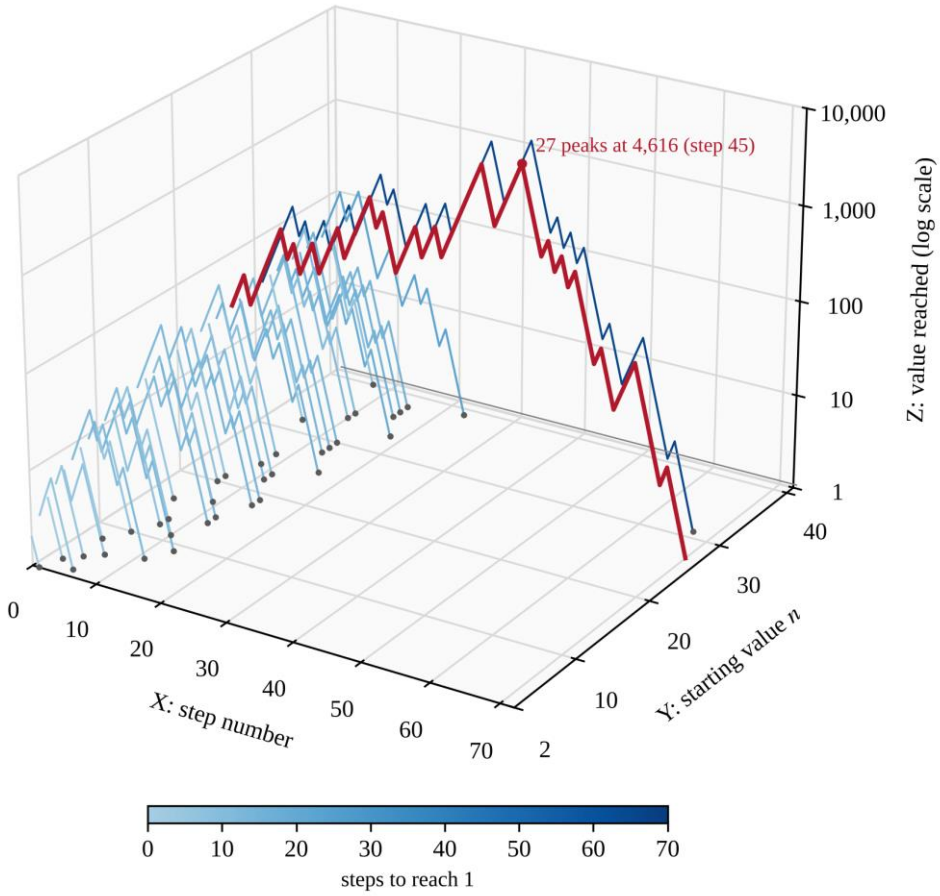
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Precis

An exit theorem for the critical coefficient corridor, and the record of how a scholar in the humanities directed AI systems to the several proofs, presented for mathematicians and for the general reader alike.



The Collatz problem as an XYZ landscape.

Abstract

This paper. An independent scholar trained in the humanities, not in mathematics, directed frontier AI systems to a theorem about the Collatz problem, proved in the Lean proof assistant, that was not located in the literature examined. This paper reports both the theorem and how it came about. The work grew out of a project charter the author approved and a theorem-first ("Ramanujan") charge he gave, and it produced two kinds of result. The directed result is Theorem 1. GPT's sub-agents formulated and froze its exact statement in response to the charge; GPT then proved it, Claude cross-checked the proof, and the theorem was proved in Lean 4 and the proof independently rebuilt. The unrequested results emerged from the work toward Theorem 1 although no one had proposed them. They are chiefly a sharper bound, Theorem 4, and a second proof of Theorem 1 by a largely different route, both originated by Claude. Confidence in the results rests on a verification regime the author designed, not on his mathematical judgment or on the systems' assurances. The episode also examines the author's role as Maestro: a non-specialist who directs the inquiry and organizes checks capable of detecting errors he cannot independently recognize. It provides evidence from one case that a person without specialist mathematical training can direct AI-assisted work toward a defensible research contribution, while the originality and wider significance of the results remain subject to specialist assessment. The mathematical contribution is a specialized quantitative result, an exact exit bound for one boundary parity pattern, and it has not yet been refereed by a human specialist.

For mathematicians. For the shortcut Collatz map $T(n) = n/2$ (n even), $T(n) = (3n+1)/2$ (n odd), let $C_j(n) = 3^{a_j(n)}/2^j$, where $a_j(n)$ counts the odd terms among $n, T(n), \dots, T^{j-1}(n)$. Exactly one parity sequence keeps C_j inside $[1,3]$ for every j: the upper mechanical (Sturmian) word of slope $\log_3 2$. We prove that every integer $n \geq 2$ leaves this corridor within $\lfloor \log_2(n^2) \rfloor + 1$ steps, and that the bound is attained at $n = 3$. Equivalently, the 2-adic integer ξ whose parity vector is this word satisfies $|n - \xi|_2 \geq n^{-2}$ for every integer $n \geq 2$, and the same argument gives a short proof that ξ is irrational (a case of a recent result of De Jesús). For seeds whose bit length lies away from the convergent denominators of $\log_3 2$ we obtain the sharper bound $k + q_m + 3$. The proof combines a factor-complexity pigeonhole argument with an elementary exclusion of balanced Collatz cycles. The main theorem and the balanced-cycle classification are proved in Lean 4 with Mathlib, using only the standard axioms. We also show that $[1,3]$ is critical for the method: for every $K > 3$ the number of admissible parity words in $[1, K]$ grows exponentially. None of these results bears directly on convergence of Collatz orbits.

For the general reader. The Collatz problem asks whether a simple halve-or-triple rule, applied over and over, always brings a whole number down to 1. It is unsolved. This paper does not solve it. It proves one small, exact fact about it. Keep a running score of how much the rule's "up" and "down" steps have multiplied a number, and require that score to stay between 1 and 3, like a walker in a narrow hallway. Only one rhythm of steps stays in that hallway forever. We prove that every starting number from 2 upward falls out of that rhythm by a deadline that can be computed with schoolbook arithmetic. The number 3 meets its deadline exactly, so the deadline cannot be reduced by even one step as a rule for every starting number. This does not show that

it is the best possible deadline for larger numbers. The main theorem has also been proved in a computer proof checker.

New in v10. Version 10 completes the paper's open items and investigates its natural next questions.

- *Remark 15.* It proves the growth rate conjectured there: for $3 < K \leq 9/2$ the number of admissible parity words is exactly a power of two, $N_K(L) = 2^{B_K(L)}$, and grows at rate $\log_3(K/3)$.

- *Second proof.* It gives a complete second proof of Theorem 1 by a different route, which replaces an archived working record found to contain an error.

- *The maximum exit time.* It studies $M(k)$, the longest exit time among all seeds below 2^k. It proves $k + 1 \leq M(k) \leq 2k$ and reduces $M(k)$ exactly to the runs of zeros in the binary expansion of ξ. It proves $M(k) \leq k + q_m + 3$ on wide intervals and $\liminf M(k)/k \leq \sqrt{2}$.

- *What remains open.* Whether the constant 2 can be lowered for every k, and whether $M(k) = k + o(k)$, remain open. Exact computation indicates where the present method stops; a proof of that limit is outstanding.

For the general reader: among all starting numbers from 2 with at most k binary digits, the latest first exit from the hallway occurs at a step between $k + 1$ and $2k$. For every k below 20,000, computer checks find that latest exit no later than step $k + 16$. Whether it stays close to k for every k remains open.

The new results have written proofs reviewed by AI systems, including GPT, but not yet by human specialists. Several are also proved in Lean: the exact count of Proposition N1 for every real width, Corollaries N2 and N3, and the finite base of the second proof, together with the earlier Corollary 2 and Proposition 14. Theorem 4, Corollary N4 and Proposition N5 are only partly proved in Lean, and the others are not yet proved in Lean.

How the paper is organized. The paper is written for both audiences. Part I is the mathematical paper, with full statements and proofs. Part II explains the same result for the general reader, using only arithmetic and analogies. Part III, written for both audiences, places the result in the history of the Collatz problem and gives the methodology and the findings: how the work was directed, where the statement of Theorem 1 came from, and which results were asked for and which were not. A shared conclusion, a three-column lexicon of key terms, a statement on the use of AI systems, acknowledgements, references and seven appendices of primary records follow.

Keywords: Collatz problem; $3x + 1$ problem; Sturmian words; mechanical words; factor complexity; balanced cycles; 2-adic integers; formal verification; Lean 4; human–AI collaboration; AI-assisted mathematics; research provenance.

MSC 2020: 11B83; 68R15; 68V15.

1 Introduction

1.1 The thesis of this paper

This paper documents how an independent scholar trained in the humanities, not in mathematics, directed frontier AI systems to a theorem on the Collatz problem, proved in Lean, that was not located in the literature examined. He did so under a project charter he approved and a theorem-first charge he gave. The paper argues that the episode produced two distinct kinds of contribution.

A directed result. The first is Theorem 1 (Hypothesis M). GPT's sub-agents formulated and froze its exact statement after the charge, with earlier work by both systems available to them. GPT then proved it, Claude cross-checked the proof, and it was proved in Lean, with an independent public rebuild.

Unrequested results. The second consists of results that no one proposed but that emerged from the work toward Theorem 1. They are chiefly the sharper Theorem 4 and a second proof of Theorem 1 by a largely different route, both originated by Claude.

A verification regime. Both kinds rest on a verification regime the author designed: frozen statements, adversarial review between systems, independent replication and formal proof. Confidence in the results therefore depends neither on the director's mathematical expertise, which he does not claim, nor on the machines' assurances.

The contribution to mathematics is specialized: an exact bound on how long an orbit can follow one boundary parity pattern, with a sharper companion bound and a proved limit of the method. The contribution to the study of human–AI collaboration is the record itself. It shows what the human directed, what the machines added unasked, and what still awaits human mathematicians. Part III (Sections 16–18) places the result in the history of the problem and develops these claims; the mathematics in Parts I and II can be read without it.

This paper calls the human director the Maestro. The term identifies responsibility for selecting the research objective, coordinating the systems, requiring evidence, and deciding what can responsibly be reported. It describes a division of labor in which the director's expertise lies principally in organizing the inquiry, while the mathematical arguments are generated and examined through the participating systems and verification procedures. The question raised by this case is whether that division permits a person without specialist mathematical training to direct work of contributory value within a specialized field.

The episode belongs to a broader development in AI-assisted mathematics. During 2026, the Aletheia research report described work with Gemini Deep Think on mathematical research tasks, including collaboration with mathematicians [41]. Alon and colleagues presented a human-verified exposition of an OpenAI-generated counterexample to the Erdős unit-distance conjecture [25, 37]. In October, OpenAI released further mathematical results together with Lean formalizations for many of the proofs [42]. The community record of AI contributions to Erdős problems distinguishes complete solutions, partial results, rediscoveries, and unsuccessful attempts [30]. These examples establish a substantial research context without implying that

every reported result has the same standing or that all the research systems used are publicly available.

Closest to the present case in subject, Chang reported a structured collaboration in which a human moderator directed Claude and GPT through an extended exploration of the Collatz problem itself [48]. That work reports many structural, computational and conditional results and states that it does not prove the conjecture. The present paper differs in reporting a single explicit theorem with a formal proof and in directing the work through frozen statements and preregistered tests. The two cases together suggest that such collaborations can be documented and compared.

Participation also extends beyond traditional academic settings. In October 2026, Thomas Bloom explicitly addressed contributors outside academia who use AI tools, while emphasizing correct formalizations, understandable exposition, attribution, and disclosure [43]. Being outside academia does not itself establish a lack of mathematical expertise. The present case addresses that narrower circumstance directly: its director is a scholar in the humanities who does not claim the ability to referee the mathematics. The contribution examined here is a bounded theorem concerning Collatz parity words, together with an account of how a non-specialist directed the work and arranged its checks. It offers evidence about the feasibility of the Maestro role, without resolving the Collatz conjecture or establishing how reliably other people could reproduce this outcome.

1.2 Two audiences, one result

This paper, posted as a preprint on SSRN, reports a single theorem about the Collatz problem and its supporting results. It is written for two groups of readers. The first is mathematicians, who need exact statements, complete proofs, precise credit to earlier work and a clear account of how the formal verification was done. The second is the intelligent, interested general reader. Such a reader has heard of the Collatz problem, perhaps through its reputation as easy to state and hard to solve, and wants to know honestly what has been proved and what has not.

Results on famous open problems are easily misreported, and that is why we address both audiences. A precise statement about one narrow pattern can be retold as progress on the whole problem. Giving the plain-language account alongside the technical one, and marking the limits in both, makes it harder to misread the paper in either direction.

1.3 The result in brief

Apply the shortcut Collatz map T to a starting number n and record, after j steps, the *coefficient* $C_j(n) = 3^{a_j(n)}/2^j$. Here $a_j(n)$ is the number of odd steps so far. The coefficient is the multiplicative part of the orbit, with the additive "$+1$" terms set aside. We call the condition $1 \leq C_j \leq 3$ the *critical corridor*. Exactly one infinite pattern of odd and even steps keeps the coefficient in this corridor forever. Our main result, Theorem 1, says that every $n \geq 2$ leaves the corridor within $\lfloor \log_2(n^2) \rfloor + 1$ steps, and that $n = 3$ attains this bound.

In plain terms, think of the coefficient as a score that must stay between a floor of 1 and a ceiling of 3. Only one rhythm of steps keeps the score in that hallway forever. The theorem gives every

starting number a deadline of roughly twice its length in binary digits. By that deadline, the number has broken the rhythm and left the hallway.

The problem in one picture. In the Precis, Figure 1 shows the problem to which the result belongs. That figure is shown again below.

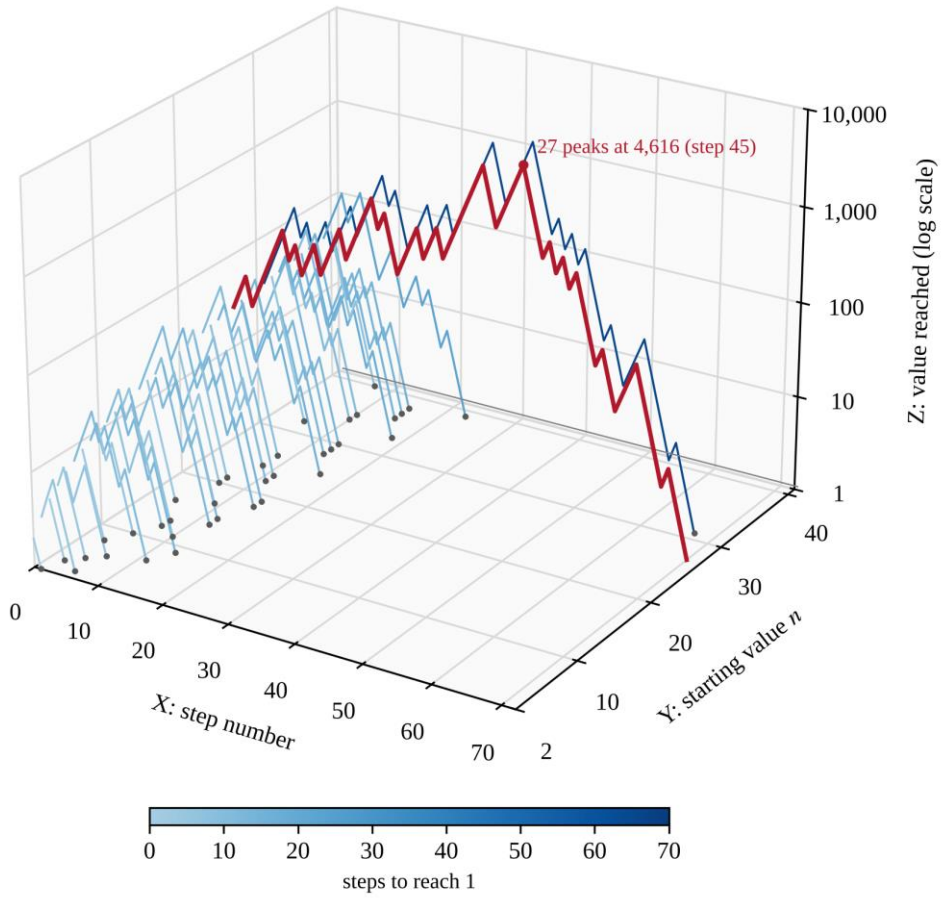


Figure 1. The Collatz problem as an XYZ landscape. X is the step number, Y the starting value n from 2 to 40, and Z the value reached, on a logarithmic scale, under the shortcut map $T(n) = n/2$ (n even), $T(n) = (3n + 1)/2$ (n odd). Each line is one complete orbit, drawn until it reaches 1 (grey dot); darker lines take more steps. The orbit of 27 (red) climbs to 4,616 at step 45 and reaches 1 at step 70. Under the standard map, which does not halve after $3n + 1$, the same orbit peaks at 9,232 and takes 111 steps. For the general reader: each line follows one starting number through the game, and all of them fall to 1, but by very different routes. The layout follows a display that Google Search produced at the author's request; it is redrawn here from complete trajectories.

Each line is the complete orbit of one starting value from 2 to 40 under the shortcut map, followed step by step until it reaches 1. The orbits climb and fall irregularly, many merge into shared paths, and every one shown ends at 1. The orbit of 27 is the classic example: it climbs to 4,616 before falling to 1 after 70 steps. The conjecture asserts that every orbit, without exception, ends at 1, and no picture can show that. The three axes are those of Descartes' coordinate geometry; despite the shared letters, they are unrelated to the author's X/Y/Z principle (Section 17.2).

During the project the statement of Theorem 1 was called *Hypothesis M*, the name it carried while it was still unproved. The review record also calls it Theorem M. This paper uses "Theorem 1" throughout. Table 1 lists every result in the paper with a plain-language reading, its support and its relation to earlier work.

New in v10. Section 7 now proves the exact count suggested in Remark 15 (Proposition N1). Section 6A gives a complete second proof of Theorem 1. In 2-adic form (Corollary 2), Theorem 1 says that $|n-\xi|_2 \geq n^{-2}$ for every integer $n \geq 2$, where ξ is the 2-adic integer whose parity vector is the critical word. Since $|\xi-1|_2 = 1/2$, this gives $|\xi-y|_2 \geq 1/2\, y^{-2}$ for every integer $y \geq 1$. For the intercept-zero point studied by De Jesús [50], this supplies the effective-measure conclusion sought in his Open Problem 2 at that intercept, by a different method. It does not construct the 2-adic Mahler theory he proposes, and it does not address other intercepts. Section 7A studies the largest exit time among all seeds below 2^k and states precisely what is and is not proved about it.

Table 1. Results at a glance. "Not located" means not found in the sources examined; see the conclusion for the literature comparisons still to be completed.

Result	**Statement**	**In plain words**	**Support**	**Earlier work**
Theorem 1 (Theorem M)	$\sigma_3(n) \leq \lfloor \log_2 n^2 \rfloor + 1$ for all $n \geq 2$; equality at $n = 3$.	Every number from 2 upward leaves the hallway by its deadline; 3 uses its whole allowance.	Written proof; Lean proof of the seed form; independent rebuild.	Explicit bound not located in the sources examined; an effective sharpening of Dubickas [5]; at intercept 0 it supplies the measure sought in De Jesús [50, Open Problem 2], by a different method.
Corollary 2	$r_L^2 \geq 2^L$ for $L \geq 2$; $\lvert n-\xi \rvert_2 \geq n^{-2}$ for $n \geq 2$.	For $L \geq 2$, the smallest number that follows the drumbeat for L steps is at least $2^{L/2}$.	Written proof; certificate for $2 \leq L \leq 10^6$. Proved in Lean (integer form), Lean round 2.	Equivalent forms of Theorem 1.
Corollary 3	ξ is not rational.	No fraction-like 2-adic number follows the drumbeat forever.	Written proof.	Known: De Jesús [4], Cor. 14.

Result	Statement	In plain words	Support	Earlier work
Theorem 4 (Theorem P)	$\sigma_3(n) \le k + q_m + 3$ when $q_m + 4 \le k \le q_{m+1} - 4$.	A much tighter deadline for numbers whose binary length lies in certain wide ranges.	Written proof; exact checks at all 9,954 eligible $k \le 10^4$. Partly proved in Lean: Lemma 11 is not yet proved in Lean (Lean round 2).	Not located in the sources examined.
Theorem 5, Lemma 10	Balanced positive cycles are $1 \leftrightarrow 2$; balanced primitive cyclic words have rotations $1u0$, $0u1$.	The only loop with evenly spread odd steps is 1, 2.	Written proofs; Lean proofs.	Known: Knight [11]; Halbeisen–Hungerbühler; Steiner; Borel–Laubie; Lothaire.
Props. 13–14	$N_K(L)$ grows exponentially for every $K > 3$; exact formula for $N_4(L)$.	In any wider hallway many rhythms survive, so the counting step used in this proof no longer works.	Written proofs; N_4 formula checked for $L \le 2000$. Proposition 14 proved in Lean (formula and rate), Lean round 2.	Not located in the sources examined.
Proposition N1 (new, §7)	N_K(L) = 2^(B_K(L)) for 3 < K ≤ 9/2; growth rate exactly δ_K = log₃(K/3).	In hallways with ceilings up to 4½, every extra choice happens alone and doubles the count.	Written proof (§7; proposed by GPT); AI adversarial review; exact checks L ≤ 3000 at 22 widths; Lean proof of the exact count for every real K (Lean round 2); growth rate not proved in Lean; checked by the author on GitHub, 24 Oct. 2026.	Settles Remark 15; not located (limited search).
Corollaries N2–N4, Prop. N5 (new, §7A)	M(k) = k + 1 + z_k and k + 1 ≤ M(k) ≤ 2k; M(k) ≤ k + q_m + 3 for q_m + 4 ≤ k ≤ q_(m+1) − 4; liminf M(k)/k ≤ √2.	Among all numbers from 2 with at most k binary digits, the latest exit comes between steps k + 1 and 2k; well before 2k is proved for many k.	Written proofs (§7A); AI adversarial review; exact checks for k < 20,000; N2 and N3 proved in Lean; N4 and N5 only partly proved in Lean; checked by the author on GitHub, 24 Oct. 2026.	N4 not located. N2–N3 are reformulations; the measure/zero-run equivalence behind N3 is in De Jesús [50]. N5's bound follows from Ren [52, Thm 4.1] (Appendix G.5, Round 9).
Prop. N7; second proof of Theorem 1 (new, §6A)	σ₃(n) ≤ 2k − 1 for bit length k within 3 of a denominator q ≥ 65; Theorem 1 proved again via Theorem 4, N7 and a finite check.	A second proof of the main theorem, by a different route.	Written proof (§6A); AI adversarial review; exact constants at all windows to q = 24,727. Finite base	Not located in the sources examined (limited search).

Result	Statement	In plain words	Support	Earlier work
			($2 \le n < 2^{22}$) proved in Lean.	

1.4 What is not claimed

The paper does not prove that any orbit reaches 1. It does not exclude orbits that grow without bound, and it does not exclude cycles other than the balanced ones. A number may leave the corridor upward, while still growing, so leaving the corridor is not descent. The same exit inequality also holds for the $5x + 1$ map, which has a nontrivial cycle. So the exit bound cannot, on its own, force descent. Section 7 also shows that the counting argument does not survive any widening of the corridor. Both parts of the paper repeat these limits.

New in v10. This version does not prove either asymptotic statement of Section 7A. Neither $M(k) \le (2 - \varepsilon)k + C$ for all k nor $M(k) = k + o(k)$ is claimed. The intervals and the lim inf bound of Section 7A are not substitutes for them.

1.5 Status of verification and review

Table 2 lists each layer of checking, with a plain-language column. Proved in Lean are Theorem 1, in the seed form stated in Section 8, and Theorem 5 together with Lemma 10. Corollary 2 and Proposition 14 were added during Lean round 2 (Section 8). Theorem 4, Corollary 3 and Proposition 13 have written proofs and lie outside it. On 24 October 2026 the author rebuilt the Lean development on GitHub, together with the new Lean proofs of Lean rounds 1 and 2, and every check passed (Section 8). All review so far has been by AI systems. The reviews identified errors in intermediate work, which were corrected, and reported no unresolved mathematical defect in the final version they examined. The originality review located no earlier statement of Theorems 1 or 4 or of Proposition 13 in the sources it could access, and left the originality of Proposition 14 undetermined. No human specialist has yet refereed the work. The two-audience edition itself was reviewed in a second round by GPT, Grok and Gemini (Table 2). A human reading is the most important remaining step. Part III and the appendices, new in this edition, have been reviewed by GPT, and the changes it called for were made where Claude concurred; Grok and Gemini have since reviewed selected results and their originality (Appendix G.5), but not Part III and the appendices as a whole.

New in v10. This version adds the following.

- Proposition N1, which proves the growth rate conjectured in Remark 15.
- Section 6A, a complete second proof of Theorem 1 including Proposition N7.
- Section 7A on the maximum exit time $M(k)$, with Corollaries N2–N4, Proposition N5 and Remark N6.

Their proofs are written out in full.

Review. Review took place in two Claude sub-agent rounds, a subsequent GPT review and focused check, and a Grok review of selected results, with different coverage by result;

Appendix G.5 records that coverage. GPT reviewed N1's corrections as its original author. GPT supplied the replacement failure argument and the $\sqrt{2}$ strengthening, which Claude checked; Grok then reviewed both, with Proposition N7, the second proof and Corollaries N2–N4. Errors in intermediate assertions were found and corrected. Appendix G records each review, its objections and their resolution.

Lean status. Lean round 1 added proofs in Lean 4.35.0 of two of the new results (Section 8):

- the bounds $k + 1 \leq M(k) \leq 2k$ of Corollary N2, the lower bound unconditionally and the upper bound from the formal Theorem 1;
- the exact count $N_K(L) = 2^{B_K(L)}$ of Proposition N1, for every rational $K = P/Q$ with $3 < K \leq 9/2$.

Both depend only on the standard axioms *propext*, *Classical.choice* and *Quot.sound*. On 24 October 2026 the author ran the public GitHub Actions workflow in his repository *whouxe82/hypothesis-m-lean*. It rebuilt the frozen development from source, then compiled and axiom-gated these proofs against it, using the formal Theorem 1 itself. The run passed (Appendix G.5, Round 11). Corollaries N3–N4, Proposition N5, Remark N6, Proposition N7 and the second proof of Section 6A were not proved in Lean in that round; Lean round 2 proves several of them (below).

Other status. No human specialist has read the new results. The Lean status of the earlier results is unchanged, and the author repeated their fresh rebuild on GitHub on 24 October 2026; every run passed. Table 2 adds the layers of checking used in the completion round.

Lean round 2 (24 October 2026). This round changed no mathematical statement or proof. In it, further results were proved in Lean. GPT wrote thirteen Lean modules against the frozen development, and Claude checked each statement against the paper. The author added a workflow to his public repository that compiles them after the full replay and gates their axioms, and on 24 October 2026 every run passed (Section 8, Appendix G.7). The modules prove in Lean:

- Corollary 2, in integer-congruence form;
- Proposition 14: the exact formula for $N_4(L)$ and its growth rate $2\alpha - 1$;
- the exact count N_K(L) = 2^(B_K(L)) of Proposition N1, for every real K with $3 < K \leq 9/2$;
- Corollary N2 in full, including the identity M(k) = 1 + max S_k;
- Corollary N3, in residue and binary-digit form;
- the finite base of the second proof (Section 6A), for every $2 \leq n < 2^{22}$, without using Theorem 1 or Corollary 2.

Two further modules are partial. One derives the conclusions of Theorem 4 and Corollary N4 from an explicit premise, the convergent-period property of Lemma 11, which is not yet proved in Lean. The other proves the arithmetic of Proposition N5 under stated hypotheses about the convergent denominators. So Theorem 4, N4 and N5 are counted as partly proved in Lean, not as proved in Lean. Corollary 3, Proposition 13, the growth rate of N1, Remark N6, Proposition N7 and the rest of the second proof have no Lean proof yet.

Table 2. The layers of verification and review as of this version.

Layer	**What was done**	**Outcome**	**In plain words**
Written proofs	Complete proofs of every result (Part I).	Complete.	The traditional argument on paper.
Cross-model review	Claude and GPT attacked each other's load-bearing steps over two days.	Several proposals found equivalent to the target, or false, and discarded; the final proof stands.	The two helpers tried to break each other's work.
Formal proof	Lean 4.35.0 with Mathlib: Theorem 1 (seed form), Theorem 5 and Lemma 10; axioms *propext*, *Classical.choice*, *Quot.sound* only.	Proved in Lean.	A computer proof checker accepted every step.
Independent rebuild	GitHub Actions run 39807720548 from commit *449929d*: hashes of 33 frozen sources, build, statement check, axiom gate. Repeated by the author on GitHub on 24 October 2026 (five runs, all passed for the frozen development; run 40109923980 also checked the Lean round 1 proofs). Repeated again on commit 13bd411 (run 40131960242, passed).	Passed (2 min 46 s).	The check re-run from scratch on fresh hosted computers. This shows the recorded build is reproducible, not that the statement means what is intended.
Exact computation	Certificate $r_L^2 \geq 2^L$ for $2 \leq L \leq 10^6$; CHECKS 1–9; Theorem 4 at all 9,954 eligible $k \leq 10^4$.	Reproduced on separate machines with matching hashes; no violations.	Very many cases tested without rounding. Evidence, not proof.
Internal AI review	Claude referee agent; GPT full-package reviews.	No remaining mathematical gap reported.	The helpers' own checking.
External AI review, stage 1	Grok: adversarial review of the mathematics and the Lean statement, with a fresh run of the checker. Gemini: originality and prior art.	Grok: every item supported. Gemini: no earlier statement of Theorems 1, 4 or Proposition 13 located in the accessible sources; Proposition 14 undetermined; source access limited.	Outside AI checkers, one told to break the work and one to search for earlier publication.
Review of the two-audience edition (v6)	GPT: full review of the two-audience edition, with independent recomputation of	GPT and Grok: no mathematical defect; their corrections to the new text	A second round of outside AI checking,

Layer	What was done	Outcome	In plain words
	every table and figure. Grok: adversarial review of Part I, Part II, the tables and figures, with a fresh run of the checker. Gemini: originality and citations.	applied. Gemini: no citation errors found; source access limited.	on this version of the paper.
v10: exact computation	Generator and separately written checker in exact integer arithmetic: residues to L = 20,000; all seeds below 2^22; N_K(L) for L ≤ 3000 at 22 widths; N7 constants to q = 24,727. Author's Colab replications, 24 October (05:36 and 06:15). Exhaustive recomputation of the method-power test after review.	All checks passed. The 05:36 run matched all nine results recorded for the first script; the 06:15 run matched all ten results recorded for the extended script, including the Section 6A checks.	The new results tested without rounding, on two machines. Evidence, not proof.
v10: AI review	Two rounds of Claude sub-agents with fresh context (statement first), a subsequent GPT review and focused check, and a Grok review of selected results, with different coverage by result, followed by a Gemini originality review (Appendix G.5). GPT reviewed N1's corrections as its original author and supplied two adopted arguments, which Claude checked and Grok reviewed.	Errors in intermediate assertions were found and corrected; the review verdict was accept with corrections.	A helper that had not written the proof tried to break it.
v10: formal proof (Lean round 1)	Lean 4.35.0: Corollary N2 (k + 1 ≤ M(k) ≤ 2k) and Proposition N1 (rational K); standard axioms only. Checked by the author on GitHub Actions in his repository, 24 October 2026 (run 40109923980, passed), against the rebuilt frozen development and the formal Theorem 1. N3–N7 and the second proof not proved in Lean in that round.	Passed.	Two of the new results are now also checked by the computer proof checker, on GitHub's machines.
v10: formal proof (Lean round 2)	Lean 4.35.0 modules written by GPT and checked against the paper by Claude: Corollary 2, Proposition 14, N1 (every real K), N2, N3 and the second proof's finite base; Theorem 4, N4 and N5 partly proved in Lean. Run by the author on GitHub Actions on commit 13bd411, 24 October 2026 (round 2 workflow, run	Passed: 27 gated theorems, standard axioms only; dependency check passed.	Most of the new results, and two older ones, are now also checked by the computer proof checker; a few are only partly proved in Lean.

Layer	What was done	Outcome	In plain words
	40131960252; round 1 workflow, run 40131960287).		
Human review	None yet.	Pending.	Not yet checked by human experts.

1.6 How the work was done

The author directed two AI systems, GPT and Claude, each running sub-agents, first under a project charter and then under a theorem-first charge. The statement of Theorem 1 was recorded and frozen before its principal computation and before the round in which it was proved, and it did not change afterwards. The systems proposed arguments, attacked each other's load-bearing steps and corrected errors. Proposals that turned out to be equivalent to the target, or false, were discarded and recorded. Part III gives the full account, including where the statement came from and which results were not asked for.

1.7 How to read this paper

Mathematicians can read Part I (Sections 2–8) as a self-contained research paper. Section 2 states all results and their relation to earlier work. Sections 3–7 give the proofs, and Section 8 describes the formalization. The general reader can go straight to Part II (Sections 9–15), which assumes nothing beyond arithmetic. Section 12 works through the corridor by hand with specific numbers. Part III (Sections 16–18), written for both audiences, places the result in the history of the problem and gives the methodology and the findings about how the work was directed and what it produced; its Figures 6 and 7 show where the result falls among earlier results and the provenance of Theorem 1. The conclusion (Section 19) addresses both audiences. The lexicon (Section 20) gives each key term twice: a precise meaning for mathematicians and a plain meaning for everyone else. Appendices A–G reproduce or cite the primary records. Figure 2 shows how the results of Part I depend on one another.

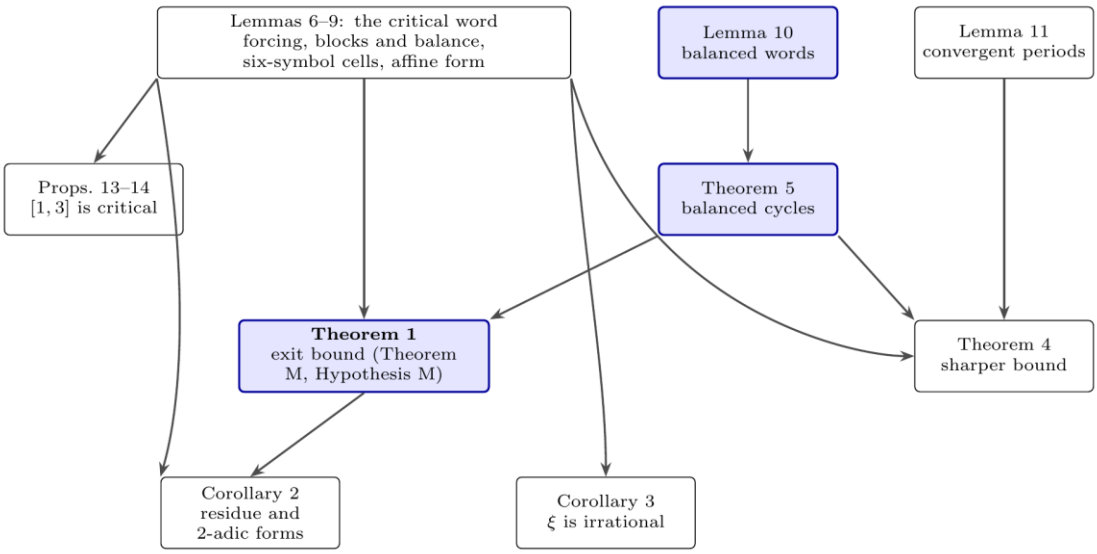


Figure 2. How the results depend on one another. An arrow means "is used to prove". Shaded boxes are proved in the original Lean development; Lean round 2 has since also proved Corollary 2 and Proposition 14 (Section 8), which the figure does not show. For Theorem 1 the Lean proof follows the same plan but replaces Lemmas 6–9 with a rational-coefficient version (Section 8). For the general reader: each box is one step of the argument. Theorem 1, the main result (Hypothesis M), is at the lower left; the facts about the drumbeat sit above it, and the fact that the only balanced loop is $1 \to 2 \to 1$ *feeds in from the right. Section 13 tells the same story in plain words.*

New in v10. Figure 2 shows the results of v6. Figure 4, in Section 7A.6, shows how the results added in v10 depend on them.

Theorem, lemma and proposition numbers follow the earlier versions of the manuscript, so that references to results in the review record remain valid. Section numbers in Part I are one higher than in v6, because of the new Introduction. Part III, Figure 2 and the appendices are new in this edition, and the conclusion and lexicon are now Sections 19 and 20.

Numbering (added in v10). Theorems, corollaries, lemmas, propositions and numbered remarks share one sequence: Theorem 1, Corollaries 2 and 3, Theorems 4 and 5, Lemmas 6–11, Remark 12, Propositions 13 and 14, and Remark 15. Definition 1 is numbered separately. There are therefore no missing Theorems 2 and 3: those numbers belong to the two corollaries, both proved in Section 5. "Lean round 1" and "Lean round 2" name the two rounds of 24 October 2026 in which results were proved in Lean (Section 8, Appendix G); the repository's file and workflow names call them revision 8 and revision 9.

Results added in v10 carry the provisional labels N1–N7 until their numbering is settled. Their new sections are 6A and 7A, and the record of their construction is Appendix G. Figure 4 shows how the results added in v10 depend on the earlier ones, and Figure 3 summarizes what is proved and computed about the maximum exit time.

Part I. The Mathematical Paper

2 Setting and statement of results

Write T for the shortcut Collatz map

$$T(n) = \begin{cases} n/2, & n \text{ even}, \\ (3n+1)/2, & n \text{ odd}, \end{cases}$$

and $x_t = T^t(n)$ for the orbit of a seed $n \geq 1$. Let $a_j(n)$ be the number of odd states among $x_0, \dots, x_{j-1}$ and put

$$C_j(n) = \frac{3^{a_j(n)}}{2^j}, \qquad C_0 = 1.$$

The quantity C_j is the multiplicative part of T^j: an odd step multiplies it by $3/2$ and an even step by $1/2$. It is the classical object behind Terras's coefficient stopping time [24, 6, 13, 20]. Under a two-sided condition $1 \leq C_j \leq K$ (for a fixed constant K) at all small j, an orbit has not yet been forced down by its multiplicative part, and its multiplicative part is not growing fast. We study the *critical corridor*

$$1 \leq C_j \leq 3,$$

in which the odd-step count is the least that keeps $C_j \geq 1$.

Since $\alpha = \log_3 2$ is irrational, the corridor condition at time $j \geq 1$ reads $j\alpha \leq a_j \leq j\alpha + 1$ and determines $a_j = \lceil j\alpha \rceil$ uniquely. Hence there is exactly one infinite parity sequence that stays in the corridor forever,

$$\mathbf{e} = e_1 e_2 e_3 \cdots, \qquad e_j = \lceil j\alpha \rceil - \lceil (j-1)\alpha \rceil,$$

which begins $1101101101011011010\cdots$. It is the upper mechanical word of slope α and intercept 0; equivalently $\mathbf{e} = 1\, c_\alpha$, where c_α is the characteristic Sturmian word of slope α [16, Ch. 2]. Mechanical parity words in the $3x+1$ setting were studied by López and Stoll [15], building on the conjugacy map of Bernstein and Lagarias [2].

Every finite parity word of length L is realized by exactly one residue class modulo 2^L [24, 13], so arbitrarily long finite prefixes of $\mathbf{e}$ are followed by some positive integers. Our main result says that they cannot be followed for long by small ones.

Definition 1. For $n \geq 1$ let $\sigma_3(n) = \min\{j \geq 1 : C_j(n) < 1 \text{ or } C_j(n) > 3\}$, with the convention $\min\varnothing = \infty$; this is the first exit from the critical corridor.

Theorem 1 (Critical corridor exit). For every integer $n \geq 2$,

$$\sigma_3(n) \leq \lfloor \log_2(n^2) \rfloor + 1.$$

The bound is attained at $n = 3$, and fails for $n = 1$.

Relation to earlier work. The qualitative statement that no positive integer follows **e** forever is known: it follows at once from Dubickas's theorem that a divergent $3x + 1$ orbit has parity complexity eventually above $1.7095\,m$ [5], together with the fact that a periodic orbit has a periodic parity word. Our proof is an effective and sharpened form of Dubickas's argument (repeated factors, a congruence modulo 2^m, and a size bound); see also the density results of Garcia and Tal [8]. De Jesús's drift-sensitive refinement of Dubickas's argument [4] uses a related size bound, but it bounds the factor complexity of an entire aperiodic word with a rational 2-adic image; it does not bound how long a positive integer can follow a finite parity prefix. In a separate note De Jesús studies the same 2-adic number (ξ is the value he calls Ξ, at slope $\log_2 3$ and intercept 0) [50]. For a finite prefix he proves a linear lower bound for the least integer following it, unless that integer's orbit meets a nontrivial positive cycle [50, Theorem 5.5]; he excludes an integer following the whole infinite word [50, Corollary 5.6]; and he poses as open an effective 2-adic irrationality measure $|\xi - y|_2 \geq C \cdot y^{(-\mu)}$ with any finite μ, to be obtained by a 2-adic Mahler method [50, Open Problem 2]. Corollary 2 below gives such a measure at intercept 0, with $\mu = 2$ and $C = 1/2$, by a different method. The new ingredients are the height bound $x_t = C_t n + E_t$ with $C_t \leq 3$, the six-symbol cell estimate of Lemma 8, which yields the constant 2, and the balanced-cycle step, which removes the need to assume divergence. For this single word, Theorem 1 is also a proved, much stronger instance of Rozier's conjectural Lower Bound Hypothesis [19], which at the critical slope would give only $j \lesssim 20\log_2 n$. We have not located Theorem 1, Theorem 4 or Proposition 13 in the sources we examined, including the existing Lean formalizations [12, 21].

Both exits are allowed: for $n = 7$ the orbit $7 \to 11 \to 17 \to 26$ leaves through the top, $C_3 = 27/8 > 3$, while its value is still rising. Theorem 1 is therefore a statement about one boundary pattern of the map, not about descent.

Let $r_L \in [0, 2^L)$ be the least residue whose first L parities are $e_1 \ldots e_L$, and let $\xi = \lim r_L \in \mathbb{Z}_2$ be the 2-adic integer whose parity vector is **e**. Lemma 6 gives $\sigma_3(n) = v_2(n - \xi) + 1$, and Theorem 1 takes the following equivalent forms.

Corollary 2.

1. $r_L^2 \geq 2^L$ for every $L \geq 2$.
2. $|n - \xi|_2 \geq n^{-2}$ for every integer $n \geq 2$.

Trivially $|r_L - \xi|_2 \leq 2^{-L} < r_L^{-1}$, so positive integers approximate ξ to exponent 1 infinitely often; Corollary 2 says they never do better than exponent 2. The same argument applied to rational points shows:

Corollary 3. ξ is not rational. Equivalently, no element of $\mathbb{Z}_{(2)} = \{a/b : b \text{ odd}\}$ has the critical word **e** as its T-parity vector.

This is a special case of the Periodicity Conjecture proposed by Lagarias [13] and restated by Bernstein and Lagarias [2], which predicts that rational 2-adic integers have eventually periodic

parity vectors (see also [15]). It is not new: it is contained in Corollary 14 of De Jesús [4], which shows, by a drift-sensitive refinement of Dubickas's counting argument, that every parity word of odd-step frequency $\log_3 2$ and linear factor complexity has an irrational preimage. De Jesús has since proved directly that the conjugacy value of every Sturmian parity word, the critical word included, is irrational [51]. We include the short direct proof because it uses only the lemmas needed for Theorem 1.

Between consecutive convergent denominators of α a much sharper bound holds. Let $q_0 < q_1 < \cdots$ (namely 1,2,3,8,19,65,84,485,1054,24727, ...) be the distinct denominators of the convergents of α.

Theorem 4 (Sharper bound away from convergents). Let n have bit length k, that is $2^{k-1} \le n < 2^k$, and suppose $q_m + 4 \le k \le q_{m+1} - 4$ for some m. Then $\sigma_3(n) \le k + q_m + 3$.

When k is large compared with q_m, this is substantially smaller than the bound of Theorem 1, which is about $2k$; at the lower endpoint $k = q_m + 4$ the two bounds essentially coincide. For example, with $q_m = 1054$, every n with $1058 \le k \le 24723$ has $\sigma_3(n) \le k + 1057$: for $k = 1058$ this gives 2115 (Theorem 1 gives 2115 or 2116), while for $k = 10000$ it gives 11057 instead of 19999 or 20000.

The decisive step in the proofs of Theorems 1 and 4 is the following classification, in which *balanced* means that any two cyclic blocks of equal length of the periodic parity word differ in their number of odd terms by at most one.

Theorem 5 (Balanced cycles). A positive integer cycle of T whose parity word is balanced is the cycle $1 \to 2 \to 1$.

Theorem 5 is not new. Halbeisen and Hungerbühler identified the extremal parity word among rational cycles, given by a floor-difference formula; in later terminology it is the Christoffel (balanced) word [9]; Knight proved that the corresponding "high cycles" are never integral [11], a result proved in Lean using only Lean's standard axioms in [12]; Steiner had earlier excluded the "circuits" [22]; and the statement also follows from the classification of mechanical itineraries in [21]. See also [7]. We give a short self-contained proof (Section 4), including the classical reduction from balanced words to Christoffel rotations (Lemma 10), because it is the form used in our formalization.

Why the corridor [1,3]

The proof of Theorem 1 uses that only one infinite word survives in [1,3], so that the surviving words have only $k + 1$ distinct blocks of length k. This fails as soon as the corridor is widened (Section 7): for every $K > 3$ the number $N_K(L)$ of parity words of length L admissible in $[1, K]$ grows exponentially in L, and for $K = 4$ it is an explicit power of two. An extension of this counting argument beyond [1,3] would need an additional restriction, for instance one tied to a single orbit or to the size of the seed.

Formal verification

Theorems 1 and 5 are proved in Lean 4.35.0 with Mathlib; see Section 8. The development was rebuilt on an independent GitHub-hosted machine, and an automatic check confirms that the final theorems depend only on the axioms *propext*, *Classical.choice* and *Quot.sound*. Lean round 2 adds proofs of Corollary 2, Proposition 14 and several results of Sections 6A, 7 and 7A, listed in Section 8.

What is not claimed

Nothing here proves that any orbit reaches 1, excludes divergent orbits, or excludes nontrivial cycles beyond the balanced class. The same exit inequality holds for the shortcut $5x + 1$ map, $T_5(n) = n/2$ for even n and $T_5(n) = (5n + 1)/2$ for odd n, with coefficients $5^{a_j}/2^j$: in fact every coefficient path leaves $[1,3]$ within three steps. Yet that map has the nontrivial cycle $13 \to 33 \to 83 \to 208 \to 104 \to 52 \to 26 \to 13$. So the first-exit bound used here does not, by itself or together with the affine composition identities of the orbit, force descent.

3 The critical word

Throughout, $\alpha = \log_3 2 = 0.6309\ldots$, $Q_j = \lceil j\alpha \rceil$ for $j \geq 0$, and $\delta_t = Q_t - t\alpha$. Thus $\delta_0 = 0$ and $\delta_t \in (0,1)$ for $t \geq 1$, since $t\alpha \notin \mathbb{Z}$.

Lemma 6 (Forcing). For $n \geq 1$ and $L \geq 0$ the following are equivalent: (a) $C_j(n) \in [1,3]$ for $1 \leq j \leq L$; (b) the parities of $x_0, \ldots, x_{L-1}$ are $e_1, \ldots, e_L$; (c) $n \equiv r_L \pmod{2^L}$. In that case $a_j(n) = Q_j$ and $C_j(n) = 3^{\delta_j} \in (1,3)$ for $1 \leq j \leq L$. Consequently $\sigma_3(n) = v_2(n - \xi) + 1$.

Proof. For $j \geq 1$, $1 \leq 3^a/2^j \leq 3$ is equivalent to $j\alpha \leq a \leq j\alpha + 1$; the endpoints are irrational, so the only integer solution is $a = Q_j$. Hence (a) holds iff $a_j = Q_j$ for $j \leq L$, which is (b). The equivalence of (b) and (c) is the standard bijection between parity vectors of length L and residues modulo 2^L [24, 13]. Finally $\sigma_3(n) > L$ iff $n \equiv \xi \pmod{2^L}$, which gives the last statement. ▫

Lemma 7 (Blocks and balance). For $t \geq 0$ and $a \geq 0$, the number of ones in $e_{t+1} \cdots e_{t+a}$ is $Q_{t+a} - Q_t = \lceil a\alpha - \delta_t \rceil \in \{\lfloor a\alpha \rfloor, \lceil a\alpha \rceil\}$. Consequently:

1. **e** is balanced: blocks of equal length differ in weight by at most one;
2. for each $k \geq 1$, **e** has at most $k + 1$ distinct blocks of length k.

Proof. $(t + a)\alpha = Q_t - \delta_t + a\alpha$, so $Q_{t+a} = Q_t + \lceil a\alpha - \delta_t \rceil$, and since $0 \leq \delta_t < 1$ this lies in $\{\lfloor a\alpha \rfloor, \lceil a\alpha \rceil\}$. The block $e_{t+1} \cdots e_{t+k}$ is determined by its prefix weights, i.e. by the vector $(\lceil a\alpha - \delta_t \rceil)_{1 \leq a \leq k}$. As a function of $\delta \in [0,1)$ the a-th coordinate equals $\lceil a\alpha \rceil$ on $[0, \{a\alpha\})$ and $\lfloor a\alpha \rfloor$ on $[\{a\alpha\},1)$. The k cut points $\{a\alpha\}$ are distinct because α is irrational, so the vector takes at most $k + 1$ values. ▫

Lemma 8 (Six-symbol cells). If $e_{i+1} \cdots e_{i+6} = e_{j+1} \cdots e_{j+6}$, then $|C_i - C_j| \le 4/9$, where $C_t = 3^{\delta_t}$.

Proof. By the proof of Lemma 7, the block of length 6 after position t is a function of δ_t that changes exactly at the points $\{a\alpha\}$, $1 \le a \le 6$; in terms of $C_t = 3^{\delta_t}$ these are $3^{\{a\alpha\}} = 2^a/3^{\lfloor a\alpha \rfloor}$, namely $2,\ 4/3\,,\ 8/3\,,\ 16/9\,,\ 32/27\,,\ 64/27$. They cut $[1,3)$ into the seven cells

$$1 < 32/27 < 4/3 < 16/9 < 2 < 64/27 < 8/3 < 3,$$

of widths $5/27\,, 4/27\,, 4/9\,, 2/9\,, 10/27\,, 8/27\,, 1/3$. Each coordinate $\lceil a\alpha - \delta \rceil$ is nonincreasing in δ and drops at exactly one cut, so distinct cells carry distinct blocks. No C_t equals a cut, since $\delta_t = \{a\alpha\}$ would make $(t + a)\alpha$ an integer. Hence equal six-blocks put C_i, C_j in one cell, of width at most $4/9$. ▫

Lemma 9 (Affine form). If n follows $e_1 \cdots e_L$, then for $0 \le t \le L$

$$x_t = C_t\, n + E_t, \qquad 0 \le E_t \le t,$$

and $x_t > n$ for $1 \le t \le L$.

Proof. Put $y_t = x_t/C_t$. An even step leaves y unchanged; an odd step gives $y_{t+1} = \left(\frac{3}{2}x_t + \frac{1}{2}\right) / \left(\frac{3}{2}C_t\right) = y_t + 1/(3C_t)$. Hence $E_t = \sum_{s<t,\, e_{s+1}=1} C_t\,/(3C_s)$, and each term lies in $(0,1]$ because $1 \le C_s$ and $C_t \le 3$. Since $e_1 = 1$, the term $s = 0$ is present for $t \ge 1$, so $E_t \ge C_t/3 > 0$; and $C_t > 1$ there. ▫

4 Balanced cycles

A finite binary word $w = w_1 \cdots w_p$ is *cyclically balanced* if any two cyclic blocks of the same length $a \le p$ (indices modulo p) have weights differing by at most one; it is *primitive* if it is not a proper power. The upper and lower mechanical words of slope q/p are $U_a = \lceil aq/p \rceil - \lceil (a-1)q/p \rceil$ and $V_a = \lfloor aq/p \rfloor - \lfloor (a-1)q/p \rfloor$.

Lemma 10. Let w be primitive and cyclically balanced, of length p and weight q with $0 < q < p$. Then $\gcd(p, q) = 1$, and w has rotations $1u0$ and $0u1$ with the same word u.

Proof. Weights. The p cyclic blocks of length a have total weight aq, so their mean is aq/p; balance puts all of them in some $\{m, m+1\}$ with $m \le aq/p \le m+1$. Hence each weight is $\lfloor aq/p \rfloor$ or $\lceil aq/p \rceil$, and if $aq/p \in \mathbb{Z}$ every weight equals aq/p.

Coprimality. With $g = \gcd(p, q)$ and $a = p/g$, all cyclic a-blocks have equal weight. Comparing the blocks starting at s and $s+1$ gives $w_{s+1} = w_{s+a+1}$ for all s, so w has period $a \mid p$ and $w = (w_1 \cdots w_a)^g$. Primitivity gives $g = 1$; thus $aq/p \notin \mathbb{Z}$ for $0 < a < p$, and $\lceil aq/p \rceil = \lfloor aq/p \rfloor + 1$ there.

Extremal rotations. Let S_j be the number of ones among $w_1, \ldots, w_j$, extended periodically ($S_{j+p} = S_j + q$), and $h(j) = pS_j - qj$, which has period p. If h is minimal at s, the rotation w'

starting after s has prefix weights $S'_a = S_{s+a} - S_s \geq aq/p$, hence $S'_a = \lceil aq/p \rceil$, so $w' = U$. At a maximum of h one obtains V in the same way.

Endpoints. $U_1 = 1$, $U_p = 0$, $V_1 = 0$, $V_p = 1$, and for $2 \leq a \leq p-1$ the relation $\lceil x \rceil = \lfloor x \rfloor + 1$ at $x = aq/p$ and $x = (a-1)q/p$ gives $U_a = V_a$. So $U = 1u0$ and $V = 0u1$. ▫

The conclusion of Lemma 10 is classical (balanced primitive cyclic words are conjugates of Christoffel words [3, 16]); the proof above is the self-contained form used in the formalization.

For a parity word w of length p with q ones, composing the steps gives $2^p T^p(y) = 3^q y + B(w)$ for every y whose next p parities are w, where

$$B(w) = \sum_{s:\, w_s = 1} 3^{|\{r > s:\, w_r = 1\}|}\ 2^{s-1}.$$

Here |S| is the number of elements of a set S. A direct computation gives

$$B(01u) - B(10u) = 2 \cdot 3^{q-1} - 3^{q-1} = 3^{q-1}$$

for every word u of weight $q-1$.

Proof of Theorem 5. Let the cycle have least period p and parity word w with q ones; $0 < q < p$ because a positive orbit cannot consist only of even or only of odd steps. The word w is primitive: if $w = z^g$ with $g > 1$, the states y and $T^{|z|}(y)$ would have the same infinite parity vector z^∞, hence be equal, since equality of their first L parities implies congruence modulo 2^L for every L, by the general parity-to-residue bijection recalled in Section 2 [24, 13], contradicting minimality. By Lemma 10, w has rotations $1u0$ and $0u1$; moving the last letter of each to the front shows that $01u$ and $10u$ are also rotations of w, so the cycle contains states y, y' whose next p parities are $01u$ and $10u$. Each satisfies $(2^p - 3^q)y = B(\cdot)$ with $y > 0$, so $D = 2^p - 3^q > 0$ divides both numerators and, by the identity for B(01u) − B(10u) above, $D \mid 3^{q-1}$. As $D \equiv 2^p \not\equiv 0 \pmod 3$, $D = 1$. For $p \geq 3$, $2^p - 1 \equiv 7 \pmod 8$ while $3^q \bmod 8 \in \{1,3\}$; so $p = 2$, $q = 1$, and the cycle is $\{1,2\}$. ▫

5 Proof of Theorem 1

For $2 \leq n \leq 63$ the theorem is checked directly; equality holds only at $n = 3$ ($C_1, \ldots, C_4 = 3/2, 9/4, 9/8, 9/16$), and $n = 1$ fails because its bound is 1 while $C_1 = 3/2$.

Let $n \geq 64$ and suppose, for a contradiction, that $\sigma_3(n) > L := \lfloor \log_2(n^2) \rfloor + 1$, so n follows $e_1 \cdots e_L$ (Lemma 6). Put $k = \lfloor (L-1)/2 \rfloor$. Then $L \geq 13$, $k \geq 6$, $2k+1 \leq L \leq 2k+2$, and $n < 2^{L/2} \leq 2^{k+1}$.

A repeated block. The $k+2$ blocks $e_{i+1} \cdots e_{i+k}$, $0 \leq i \leq k+1$, lie inside $e_1 \cdots e_L$; by Lemma 7 two coincide, at $0 \leq i < j \leq k+1$.

Equal states. The states x_i and x_j have the same next k parities, so $x_i \equiv x_j \pmod{2^k}$. Since $k \geq 6$, Lemmas 8 and 9 give

$$\left|x_i - x_j\right| \le 4/9\, n + j < 8/9\ 2^k + k + 1 < 2^k,$$

because $9(k+1) < 2^k$ for $k \ge 6$. Hence $x_i = x_j$. Since $x_t > n = x_0$ for $t \ge 1$, $i \ge 1$.

A balanced cycle. Thus x_i lies on a cycle; let $d = j - i$ and $v = e_{i+1} \cdots e_{i+d}$. Because $i + 2d = 2j - i \le 2k + 1 \le L$, the word vv is a block of **e**, so by Lemma 7(1) every cyclic block of v is balanced, and the same holds for the parity word of the least period, which divides d. Theorem 5 forces the cycle to be $\{1,2\}$, but all its states exceed $n \ge 64$. This contradiction proves Theorem 1. ▫

Proof of Corollary 2. If $r_L^2 < 2^L$ for some $L \ge 2$, then $n = r_L \ge 3$ satisfies $\lfloor \log_2 n^2 \rfloor + 1 \le L < \sigma_3(n)$. Conversely, a seed violating Theorem 1 follows the word for $L = \lfloor \log_2 n^2 \rfloor + 1$ steps, so $n \equiv r_L$ and $n < 2^{L/2}$, forcing $n = r_L$ and $r_L^2 < 2^L$. Part (2) is $v_2(n - \xi) = \sigma_3(n) - 1 \le \lfloor \log_2 n^2 \rfloor$. ▫

Proof of Corollary 3. Suppose $\xi = a/b$ with $a \in \mathbb{Z}$ and $b > 0$ odd. The real numbers $z_t = T^t(\xi)$ (computed with the parities e_{t+1}) satisfy $bz_t \in \mathbb{Z}$, and the proof of Lemma 9 gives $z_t = C_t\xi + E_t$ with $0 \le E_t \le t$. Fix $k \ge 6$ large. Equal k-blocks at $i < j \le k + 1$ give $z_i \equiv z_j \pmod{2^k}$ in $\mathbb{Z}_2$, by the parity-to-residue bijection in $\mathbb{Z}_2$; since b is a 2-adic unit, the integer $bz_i - bz_j$ is divisible by 2^k. By Lemma 8, $\left|bz_i - bz_j\right| \le b(4/9\,|\xi| + k + 1) < 2^k$ once k is large enough. So $z_i = z_j$, and the parity vector of ξ would be eventually periodic, which **e** is not because α is irrational. ▫

6 Proof of Theorem 4

Lemma 11 (Convergent periods). Let $q = q_m$. Then $e_t = e_{t+q}$ for $2 \le t \le q_{m+1} - 1$.

Proof. Write $q\alpha = p + \varepsilon$ with $p \in \mathbb{Z}$ and $|\varepsilon| = \| q\alpha \|$. Then $\lceil (s+q)\alpha \rceil = p + \lceil s\alpha + \varepsilon \rceil$, so $e_t = e_{t+q}$ unless $\lceil s\alpha + \varepsilon \rceil \ne \lceil s\alpha \rceil$ for $s = t - 1$ or $s = t$. That would put an integer strictly between $s\alpha$ and $s\alpha + \varepsilon$, giving $\| s\alpha \| < \| q_m\alpha \|$ with $1 \le s \le q_{m+1} - 1$, which contradicts the best-approximation property of convergents [10]. ▫

Proof of Theorem 4. Let $q = q_m$ and $S = k + q + 3$, and suppose $\sigma_3(n) > S$, so n follows $e_1 \cdots e_S$. The parities of $x_1, \ldots, x_{k+2}$ are $e_2, \ldots, e_{k+3}$, and those of $x_{1+q}, \ldots, x_{k+2+q}$ are $e_{2+q}, \ldots, e_{k+3+q}$; they agree by Lemma 11, because $k + 3 \le q_{m+1} - 1$. Hence $x_1 \equiv x_{1+q} \pmod{2^{k+2}}$. Also $x_1 = (3n+1)/2 < 2^{k+2}$ and, by Lemma 9, $x_{1+q} \le 3n + q + 1 < 2^{k+2}$, since $q \le k - 4$. So $x_1 = x_{1+q}$: the orbit enters a cycle whose states exceed $n \ge 16$. Its parity word $v = e_2 \cdots e_{q+1}$ satisfies $vv = e_2 \cdots e_{2q+1}$, a block of **e** because $2q + 1 \le S$. As in Section 5, the cycle is balanced and Theorem 5 gives a contradiction. ▫

Remark 12. Theorem 4 is only partly proved in Lean. Lean round 2 proves the step from the convergent-period property of the critical word to its conclusion (Section 8); Lemma 11, which supplies that property, is not yet proved in Lean. It has been reviewed by AI systems, internally and in two rounds of adversarial review, but not yet by a human specialist. It was checked numerically for all bit lengths $k \le 10^4$ in its range, with no exception, using the exact residues r_L.

6A A second proof of Theorem 1 (new in v10)

Section 18.4 and Appendix F describe a second route to Theorem 1, archived on 21 October as a working record. That record compressed several deductions, and one of its statements (Lemma S, at the endpoint) is false (editorial note in Appendix F). This section gives a complete version of the route. It keeps the archived structure: Theorem 4 (the archived Branch P) away from convergent denominators, a separate estimate near them (Branch L), and a finite computation for small bit lengths. It replaces the archived balanced-word step with the argument already used in Section 6. The proof uses Lemma 6 and the parity-to-residue correspondence behind it, Lemmas 7, 9 and 11, Theorems 4 and 5, and the standard error bound for convergents. It does not use Theorem 1, Corollary 2, Lemma 7(2), Lemma 8 or the pigeonhole argument of Section 5. None of the results it uses depends on them: Lemmas 6, 7 and 9 are proved in Section 3 from the definitions; Lemma 10 and Theorem 5 in Section 4 from the identity for $B(w)$; Lemma 11 in Section 6 from the best-approximation property of convergents; and Theorem 4 in Section 6 from Lemmas 9 and 11 and Theorem 5.

Proposition N7 (near a convergent denominator). Let $q = q_{m+1} \geq 65$ be a convergent denominator of α, and let $q - 3 \leq k \leq q + 3$. Then $\sigma_3(n) \leq 2k - 1$ for every n with $2^{k-1} \leq n < 2^k$.

Proof. Let $Q = q_{m+2}$ be the next denominator. Then $Q \geq q + q_m \geq q + 19$, and $\| q\alpha \| < 1/Q$ by the standard error bound for convergents [10]. Suppose $\sigma_3(n) \geq 2k$, so that n follows $e_1 \cdots e_L$ with $L = 2k - 1 \geq 2q - 7 \geq q + 2$.

Congruence. For $t \leq L - 1$ the parity of x_t is e_{t+1}. Put $\ell = L - 1 - q = 2k - 2 - q \geq q - 8$. The parities of $x_1, \ldots, x_\ell$ are $e_2, \ldots, e_{\ell+1}$, and those of $x_{1+q}, \ldots, x_{q+\ell}$ are $e_{q+2}, \ldots, e_{q+\ell+1}$. They agree by Lemma 11, because $\ell + 1 = 2k - 1 - q \leq q + 5 \leq Q - 1$. Hence $x_1 \equiv x_{1+q} \pmod{2^\ell}$.

Size. By Lemma 9, $x_t = C_t n + E_t$ with $0 \leq E_t \leq t$ for $t \leq L$. Here $C_1 = 3/2$ and $C_{1+q} = C_1 C'$, where $C' = 3^N/2^q$ and N is the number of ones in $e_2 \cdots e_{q+1}$. By Lemma 7 with $t = 1$, $N = Q_{q+1} - Q_1 = \lceil q\alpha - \delta_1 \rceil$ with $\delta_1 = 1 - \alpha$, so N is $\lfloor q\alpha \rfloor$ or $\lceil q\alpha \rceil$. Since $\| q\alpha \| < 1/Q < \min(\delta_1, 1 - \delta_1)$, N is the integer nearest to $q\alpha$, and $|N - q\alpha| = \| q\alpha \| < 1/Q$. Put $u = |N - q\alpha| \ln 3$, so that $C' = e^{\pm u}$. Since $1 - e^{-u} \leq e^u - 1 \leq u e^u$ for $u \geq 0$, $|C' - 1| \leq (\ln 3/Q)\, 3^{1/Q}$ whichever the sign, and

$$\left| x_{1+q} - x_1 \right| \leq 3/2\, |C' - 1|\, n + q + 1 < \left(\frac{3 \ln 3}{2} \cdot \frac{3^{1/Q}}{Q} + \frac{q+1}{2^k} \right) 2^k.$$

For $q \geq 65$ we have $Q \geq 84$ and $k \geq 62$. Since $3^{1/Q}/Q$ decreases in Q, the first term is at most $3 \ln 3/2 \cdot 3^{1/84}/84 < 0.0199$. Since $q + 1 \leq k + 4$ and $(k + 4)/2^k$ decreases, the second is at most $66 \cdot 2^{-62}$. The bracket is therefore less than $2^{-5} \leq 2^{k-2-q}$. So $\left| x_{1+q} - x_1 \right| < 2^\ell$, and with the congruence, $x_1 = x_{1+q}$.

A balanced cycle. Thus x_1 lies on a cycle whose length divides q, and all its states exceed $n \geq 2^{61}$. The parity word of q steps from x_1 is $v = e_2 \cdots e_{q+1}$. By Lemma 11, $e_t = e_{t+q}$ for $2 \leq t \leq$

$q+1$, so $vv = e_2 \cdots e_{2q+1}$ is a block of $\mathbf{e}$. Every cyclic block of v of length at most q is therefore a block of $\mathbf{e}$, so v is cyclically balanced by Lemma 7(1). As in Section 5, the parity word of the least period, whose q/p-th power is v, is then cyclically balanced too. Theorem 5 forces the cycle to be $1 \to 2 \to 1$, a contradiction. ▫

The exact constants at the known windows are better than the uniform estimate. The left side of the size inequality, divided by 2^{ℓ}, is at most about 0.547 at $q = 65$, 0.100 at $q = 84$, 0.049 at $q = 485$, 0.002 at $q = 1054$ and 0.001 at $q = 24727$, the maximum over the seven k of each window. These values were computed in exact rational arithmetic. The archived record gave 0.561 for $q = 65$, with different bookkeeping.

The window. The two ends of the window $q - 3 \le k \le q + 3$ are limited differently. At the lower end the size estimate binds: the uniform bound $2^{-5} \le 2^{k-2-q}$ needs $k \ge q - 3$, and at $q = 65$ the exact ratio is about 1.09 at $k = q - 4$, so the argument fails there. At the upper end nothing binds at $q + 3$: the congruence step needs only $\ell + 1 \le Q - 1$, that is $2k \le q + Q$, and the size ratio decreases as k grows. The upper end is $q + 3$ because the second proof below needs exactly that much to cover the bit lengths between the ranges of Theorem 4.

Second proof of Theorem 1. Let $n \ge 2$ have bit length k; then $\lfloor \log_2 n^2 \rfloor + 1 \ge 2k - 1$.

- *Small bit lengths.* If $k \le 22$, Theorem 1 holds by exact computation for every $2 \le n < 2^{22}$, including equality at $n = 3$.
- *All other bit lengths.* If $k \ge 23$, let m be the largest index with $q_m + 4 \le k$. Then $m \ge 4$, since $q_4 = 19$, and $k \le q_{m+1} + 3$.
  - If $k \le q_{m+1} - 4$, Theorem 4 gives $\sigma_3(n) \le k + q_m + 3 \le 2k - 1$.
  - Otherwise $q_{m+1} - 3 \le k \le q_{m+1} + 3$ with $q_{m+1} \ge q_5 = 65$, and Proposition N7 gives $\sigma_3(n) \le 2k - 1$.

So for $k \ge 23$, $\sigma_3(n) \le 2k - 1 \le \lfloor \log_2 n^2 \rfloor + 1$. Together with the finite base for $k \le 22$, this proves Theorem 1. ▫

The argument shows slightly more for $k \ge 23$: $\sigma_3(n) \le 2b(n) - 1$. This stronger form is restricted to $k \ge 23$; it fails at $n = 3$, where $\sigma_3(3) = 4 > 2 \cdot 2 - 1$, and the exact computation shows that $n = 3$ is the only seed below 2^{22} for which it fails. It uses Theorem 5 exactly as the principal proof does. Otherwise its route is different: convergent periods (Lemma 11) and a one-lap size estimate take the place of the complexity bound, the six-symbol cells and the pigeonhole.

Checks. The finite base was checked by direct simulation of every seed below 2^{22}. Coverage of every bit length $23 \le k \le 50511$ by the ranges of Theorem 4 and the windows of Proposition N7 was checked exactly; the proof covers all k. The data satisfy $\max \sigma_3 \le 2k - 1$ at all 28 window bit lengths below 20000. The finite base is also proved in Lean (Lean round 2, Section 8): a kernel-checked residue certificate covers every $2 \le n < 2^{22}$ without using Theorem 1 or Corollary 2. Proposition N7 and the rest of this proof have not been proved in Lean.

7 The corridor [1,3] is critical for the method

For $K \geq 1$ let $N_K(L)$ be the number of binary words of length L such that $1 \leq 3^{a_j}/2^j \leq K$ for every prefix, where a_j counts the ones among the first j letters. By Lemma 6, $N_3(L) = 1$ for every L.

Proposition 13. For every $K > 3$,

$$\liminf_{L\to\infty} \frac{\log_2 N_K(L)}{L} \;\geq\; \delta_K := \log_3 \frac{\min(3,2K/3)}{2} > 0.$$

On the other hand $N_K(L) \leq 2^{H(\alpha)L}$ for every K, where $H(\alpha) = -\alpha\log_2\alpha - (1-\alpha)\log_2(1-\alpha) = 0.94995\ldots$.

Proof. On the critical path, a coefficient $c \in (2,3)$ is followed by the letters 01: $c \mapsto c/2 \mapsto 3c/4$. The alternative 10 gives $c \mapsto 3c/2 \mapsto 3c/4$, with the same end value, and stays in $[1, K]$ whenever $c < 2K/3$. These blocks do not overlap, so the swaps can be made independently, and each one doubles the count. The coefficients on the critical path are 3^{δ_t} with $\delta_t \equiv -t\alpha \pmod 1$, which is equidistributed in $[0,1)$; the proportion of times with $3^{\delta_t} \in \big(2, \min(3,2K/3)\big)$ is therefore δ_K. For the upper bound, every admissible word has at least αL ones, and $\sum_{a\geq\alpha L}\binom{L}{a} \leq 2^{H(\alpha)L}$ because $\alpha > 1/2$. ▫

Proposition 14. With $f_j = \lfloor j\alpha \rfloor$ (not to be confused with the residues r_L), for every $L \geq 0$

$$N_4(L) = 2^{\,f_{L+1}+f_{L+2}-L-1},$$

so $\lim\limits_{L\to\infty} 1/L \log_2 N_4(L) = 2\alpha - 1 = \delta_4 \approx 0.2619$.

Proof. For $j \geq 1$, $1 \leq 3^a/2^j \leq 4$ means $j\alpha \leq a \leq (j+2)\alpha$, so the admissible odd counts at time j form the set $A_j = \{f_j + 1, \ldots, f_{j+2}\}$, of size $f_{j+2} - f_j \in \{1,2\}$; it always contains the critical value $Q_j = f_j + 1$. A word is a path $a_0 = 0, a_1, \ldots, a_L$ with $a_j \in A_j$ and $a_j - a_{j-1} \in \{0,1\}$. Since $3\alpha < 2$, $f_{j+3} - f_j \leq 2$, so two-element sets are never consecutive. If $|A_j| = 2$, then $A_{j-1} = \{f_{j+1}\}$, $A_j = \{f_{j+1}, f_{j+1}+1\}$ and, when $j < L$, $A_{j+1} = \{f_{j+1}+1\}$: both values at time j are reachable from time $j-1$ and both lead to the value at time $j+1$. At one-element times the path is forced to the critical value. Hence each two-element time doubles the count independently, and $\log_2 N_4(L) = \sum_{j=1}^{L}\big(f_{j+2} - f_j - 1\big) = f_{L+1} + f_{L+2} - f_1 - f_2 - L$, with $f_1 = 0$, $f_2 = 1$. ▫

Remark 15. The formula agrees with direct exact counting for all $L \leq 2000$. Numerically, δ_K appears to be the exact growth rate for $3 < K \leq 9/2$; we have not proved this.

Status note (v10). The observation in Remark 15 is now proved as Proposition N1 below. Its proof was proposed in the Completion Assessment of 24 October 2026; it has since been examined in adversarial reviews, and the corrections they required are incorporated here

(Appendix G.5). The sentence "we have not proved this" records the status at v6 and is kept for the historical record. Labels N1–N7 are provisional until their numbering is settled.

Proposition N1 (exact count for $3 < K \le 9/2$). Put $\beta = \log_3(K/3)$, where $3 < K \le 9/2$, and for $L \ge 0$ let

$$B_K(L) = \#\{j : 2 \le j \le L,\ \{j\alpha\} \ge 1 - \beta\},$$

where $\{x\}$ is the fractional part. Then $N_K(L) = 2^{B_K(L)}$ for every $L \ge 0$, and

$$\lim_{L\to\infty} \frac{\log_2 N_K(L)}{L} = \beta = \log_3 \frac{K}{3} = \delta_K.$$

In particular the lower rate of Proposition 13 is the exact growth rate on $3 < K \le 9/2$, as Remark 15 suggested. At $K = 4$ the formula is Proposition 14, and at $K = 9/2$ it gives $N_{9/2}(L) = 2^{L-\lceil L\alpha\rceil}$.

Proof. For $j \ge 1$ write $j\alpha = f_j + s_j$ with $f_j = \lfloor j\alpha \rfloor$ and $0 < s_j < 1$; put $f_0 = s_0 = 0$. Note that $K \le 9/2$ is exactly $\beta \le 1 - \alpha$, and $1 - \alpha < 1/2 < \alpha$.

Admissible values. At time $j \ge 1$ the condition $1 \le 3^a/2^j \le K$ is $j\alpha \le a \le j\alpha + 1 + \beta$, the endpoints included (for rational K the upper endpoint can be attained, for instance at $K = 27/8$, $j = 3$; the lower one cannot, since $j\alpha \notin \mathbb{Z}$). The integers in this interval are $Q_j = f_j + 1$ and, exactly when $s_j \ge 1 - \beta$, also $Q_j + 1$; no larger value fits because $\beta < 1$. Call j a *choice time* if $s_j \ge 1 - \beta$, and let A_j be the set of admissible values.

Isolation. Let $j \ge 2$ be a choice time. Then $s_j \ge 1 - \beta \ge \alpha$, and $s_j \ne \alpha$, since otherwise $(j-1)\alpha = f_j$ would be an integer with $j - 1 \ge 1$. So $s_{j-1} = s_j - \alpha \in (0, 1 - \alpha)$, which gives $f_{j-1} = f_j$, $Q_{j-1} = Q_j$, and, because $1 - \alpha < 1 - \beta$ (that is, $\beta < \alpha$, as $\beta \le 1 - \alpha < 1/2 < \alpha$), time $j - 1$ is not a choice time. Also $s_j \ge \alpha > 1 - \alpha$, so $s_{j+1} = s_j + \alpha - 1 < \alpha \le 1 - \beta$; hence $Q_{j+1} = Q_j + 1$ and time $j + 1$ is not a choice time.

Counting. Let R_j be the set of odd counts a_j reached by admissible words of length j, and N_j the number of such words. We show by induction on $j \ge 1$ that $R_j = A_j$, except that $R_1 = \{1\}$ when $K = 9/2$, and that $N_j = 2^{B_K(j)}$, each value in R_j being reached by the same number of words. For $j = 1$: the one-letter word 1 is admissible ($C_1 = 3/2$) and 0 is not, and time 1 is a choice time only when $\beta = 1 - \alpha$, that is $K = 9/2$, where the second value 2 cannot be reached from $a_0 = 0$; so $N_1 = 1 = 2^{B_K(1)}$. For the step $j - 1 \to j$ with $j \ge 2$, suppose first that $R_{j-1} = \{Q_{j-1}\}$, which holds when $j - 1$ is not a choice time and also when $j - 1 = 1$. A next letter 0 or 1 gives the count Q_{j-1} or $Q_{j-1} + 1$. If j is a choice time, $Q_{j-1} = Q_j$, so both values of A_j are reached, each by N_{j-1} words, and $N_j = 2N_{j-1}$. If j is not a choice time, $A_j = \{Q_j\}$ and $Q_j - Q_{j-1} \in \{0,1\}$, so exactly one letter is admissible and $N_j = N_{j-1}$. Suppose next that $j - 1 \ge 2$ is a choice time. Then j is not, $A_j = \{Q_{j-1} + 1\}$, and each of the two counts at time $j - 1$ has exactly one admissible continuation (letter 1 from Q_{j-1}, letter 0 from $Q_{j-1} + 1$), so $N_j = N_{j-1}$. This

completes the induction; with $N_K(0) = 1$, it gives $N_K(L) = 2^{B_K(L)}$ for all $L \geq 0$, the case $j = L$ included.

Rate. Since α is irrational, $\{j\alpha\}$ is equidistributed [45], so $B_K(L)/L \to \beta$, the length of $[1 - \beta, 1)$. On $3 < K \leq 9/2$ we have $2K/3 \leq 3$, so $\delta_K = \log_3\big((2K/3)/2\big) = \log_3(K/3) = \beta$.

Special cases. At $K = 4$, $1 - \beta = 2 - 2\alpha$. For $j \geq 0$, $f_{j+2} - f_j = \lfloor s_j + 2\alpha \rfloor = 1 + \big[\, s_j \geq 2 - 2\alpha \,\big]$. Summing over $0 \leq j \leq L$, and noting that $j = 0,1$ contribute nothing to the bracket ($s_0 = 0$ and $s_1 = \alpha < 2 - 2\alpha$) and $f_0 = f_1 = 0$, gives $f_{L+1} + f_{L+2} = L + 1 + B_4(L)$, which is Proposition 14. At $K = 9/2$, $1 - \beta = \alpha$, and for $j \geq 2$ the condition $s_j \geq \alpha$ (equality excluded as above) is $s_{j-1} = s_j - \alpha > 0$ without carry, that is $f_j = f_{j-1}$, $Q_j = Q_{j-1}$, $e_j = 0$. Since $e_1 = 1$, $B_{9/2}(L)$ is the number of zeros among $e_1, \ldots, e_L$, which is $L - \lfloor L\alpha \rfloor$. ▫

The bound $K \leq 9/2$ is needed: the formula fails for every $9/2 < K < 9$. There $\beta < 1$, so at each time $j \geq 1$ the admissible counts are Q_j and, at a choice time, $Q_j + 1$; at time 1 only the count 1 is reachable. An admissible word is determined by its count path, so the words of length L inject into the assignments of one of two values to each choice time $2 \leq j \leq L$, and $N_K(L) \leq 2^{B_K(L)}$. Since $K > 9/2$ means $\beta > 1 - \alpha$, the interval $(2 - \alpha - \beta, 1)$ is nonempty, and by equidistribution some $j \geq 2$ has s_j in it. Then $s_j > 1 - \beta$ and, because $s_j > 1 - \alpha$, $s_{j+1} = s_j + \alpha - 1 > 1 - \beta$: both j and $j + 1$ are choice times, and $Q_{j+1} = Q_j + 1$. The assignment choosing Q_j at j and $Q_{j+1} + 1$ at $j + 1$ would need two odd steps in one step, so no word realizes it. Hence $N_K(L) < 2^{B_K(L)}$ for every $L \geq j + 1$. (This argument was supplied in GPT's review of the completion round's draft; it replaces an earlier argument that failed for $27/4 \leq K < 9$, where already $N_K(2) = 1 < 2 = 2^{B_K(2)}$.) In the examples, $K = 5$ first fails at $L = 12$, where $N_5(12) = 24 < 32$, and $K = 9/2 + 1/1000$ first fails at $L = 1055$.

Checks. The finite identity was compared with an independent dynamic count, in exact integer arithmetic, for every $L \leq 2000$ at nine widths (Completion Assessment); for every $L \leq 3000$ at 22 widths, including 13 widths lying exactly on a choice boundary $K = 9 \cdot 3^{f_j}/2^j$ (the completion round); and by the reviewer for every $L \leq 2000$ at 51 widths, including points within 10^{-30} of eight boundary values, and for the two special cases up to $L = 20000$. There were no disagreements. These checks test boundary handling; the proof supplies the universal statement. Proposition N1's exact count is proved in Lean (Lean round 2, Section 8); its growth rate is not. Its originality has not been assessed.

With $N_K(L)$ exponential, blocks of length proportional to $\log n$ can take more values than there are positions in a window of length $O(\log n)$, so widening the corridor alone does not preserve the low-complexity estimate on which the pigeonhole step of Section 5 relies. This does not exclude other counting arguments for wider corridors.

7A The maximum exit time $M(k)$ (new in v10)

This section collects what is now proved about the largest exit time among all seeds below 2^k, and reports the investigation of the two asymptotic questions left open in v6. Its labels N2–N6 are provisional. Throughout, $b(n) = \lfloor \log_2 n \rfloor + 1$ is the bit length of one seed, while k is a cutoff: $M(k)$ below includes every seed of every bit length up to k. Write $\xi = \sum_{i \geq 0} \xi_i \, 2^i$ for the

binary digits of ξ, so that $r_L = \sum_{i<L} \xi_i\, 2^i$. In the notation of Bernstein and Lagarias [2], $\xi = \Phi(\mathbf{e}) = \Phi(1c_\alpha)$. López and Stoll give a 2-adic series for $\Phi(1c_\alpha)$ for every irrational slope (their Theorem 1), and for $\alpha = \log_3 2$ they print the first binary digits of ξ (their Example 8) [15]. They do not consider runs of zeros in these digits. De Jesús does [50]. His constant $\Xi_{\log_2 3,0}$ is exactly ξ: numbering parity positions from 0, the ones of $\mathbf{e}$ occur at the positions $\lfloor i\log_2 3\rfloor$, $i \geq 0$, so the inverse parity series for ξ is, term by term, his defining series $-\sum_{i\geq 0} 2^{\lfloor i\log_2 3\rfloor}/3^{i+1}$ [50, Theorem 4.1 and addendum]. His Proposition 7.2 shows that an effective measure $|\xi - y|_2 \geq Cy^{-\mu}$ for all integers $y \geq 1$ is equivalent, up to constants, to a bound $(1 - 1/\mu)d + C'$ on the length of every zero run of ξ ending at depth d (that is, with last zero at digit $d-1$). A run beginning at digit t with length z ends at depth $t + z$, so in the indexing of Corollary N3 the bound reads $z \leq (\mu - 1)t + O(1)$. Corollary 2 gives $\mu = 2$ with $C = 1/2$, by a different method from the Mahler-type construction proposed in his Open Problem 2, and only at intercept 0. Its explicit constants give the exact run bound $z_k \leq k - 1$ of Corollary N3, which the equivalence up to constants does not give by itself. In the same terms, Target A is equivalent to such a measure with some $\mu < 2$, and De Jesús's Open Problem 3 (zero runs of length $O(\log k)$, or a fixed power below 1) implies Target B. The exact formula for the cumulative maximum $M(k)$ in Corollaries N2–N3 is not in [50].

7A.1 Definitions and the two targets
For $k \geq 2$ put

$$M(k) = \max\{\sigma_3(n) : 2 \leq n < 2^k\}.$$

Two questions about $M(k)$ were fixed, with their quantifiers, before the principal computations of the completion round (preregistration record, Appendix G.3).

Target A (uniform improvement of the constant 2): there are constants $0 < \varepsilon < 1$ and C, independent of k, with $M(k) \leq (2 - \varepsilon)k + C$ for **every** $k \geq 2$.

Target B (optimal leading constant): for every $\eta > 0$ there is $k_0(\eta)$ with $M(k) \leq (1 + \eta)k$ for **every** $k \geq k_0(\eta)$; equivalently, by Corollary N2, $M(k)/k \to 1$.

A statement for infinitely many k, for k in selected intervals, or for a finite range of k is not either target and is labelled accordingly below.

7A.2 Reduction to the residues and to the digits of ξ
Corollary N2. For $k \geq 2$ the set $S_k = \{L \geq 2 : r_L < 2^k\}$ is finite and nonempty, and

$$M(k) = 1 + \max S_k, \qquad k + 1 \leq M(k) \leq 2k.$$

Proof. $M(k)$ is a maximum over the finite nonempty set $\{2, \ldots, 2^k - 1\}$. By Lemma 6, $\sigma_3(n) = v_2(n - \xi) + 1$, where $v_2(n - \xi) = \max\{L : n \equiv r_L \pmod{2^L}\}$ is finite by Corollary 3. Since $r_2 = 3 < 2^k$, $2 \in S_k$. If $L \in S_k$, then $L \geq 2$ and Corollary 2 gives $2^L \leq r_L^2 < 2^{2k}$, so $\max S_k \leq 2k - 1$. For $L = \max S_k$, the seed $n = r_L$ satisfies $r_L^2 \geq 2^L \geq 4$, so $2 \leq n < 2^k$, and it follows $e_1 \cdots e_L$, so $M(k) \geq L + 1$. Conversely, if $2 \leq n < 2^k$ and $L = \sigma_3(n) - 1$, then $n \equiv r_L \pmod{2^L}$ with $0 \leq r_L \leq n < 2^k$; either $L \geq 2$ and $L \in S_k$, or $\sigma_3(n) \leq 2 < 1 + \max S_k$. Finally $k \in S_k$ because $r_k < 2^k$, so $M(k) \geq k + 1$, and $\max S_k \leq 2k - 1$ gives $M(k) \leq 2k$. ▫

The upper bound is Theorem 1 restated for the cumulative maximum, and the lower bound is the one discussed under Figure 8. Neither shows that the constant 2 is needed asymptotically.

Corollary N3 (zero runs). For $k \geq 2$ let z_k be the number of consecutive zero digits of ξ beginning at ξ_k (so $z_k = 0$ if $\xi_k = 1$). Then z_k is finite, $z_k \leq k - 1$, and

$$M(k) = k + 1 + z_k.$$

For seeds of exact bit length $b \geq 2$, the largest exit time is $b + 1 + z_b$ if $\xi_{b-1} = 1$ and b if $\xi_{b-1} = 0$.

Proof. For $2 \leq L \leq k$, $r_L < 2^L \leq 2^k$; for $L > k$, $r_L < 2^k$ exactly when $\xi_k = \cdots = \xi_{L-1} = 0$, and once some $\xi_i = 1$ with $i \geq k$, every $L > i$ has $r_L \geq 2^i \geq 2^k$. The run is finite, since otherwise every $L > k$ would lie in S_k, contrary to Corollary N2. Hence $S_k = \{2, \ldots, k + z_k\}$, and Corollary N2 gives the formula and $z_k \leq k - 1$. For exact bit length $b \geq 2$, let $2^{b-1} \leq n < 2^b$ and $L = \sigma_3(n) - 1$. If $L \leq b - 1$, then $\sigma_3(n) \leq b$. If $L \geq b$, then $n \equiv r_L$ with $0 \leq n, r_L < 2^L$, so $n = r_L$; this forces $\xi_{b-1} = 1$ and $\xi_b = \cdots = \xi_{L-1} = 0$, so $L \leq b + z_b$. Conversely, if $\xi_{b-1} = 1$, the seed r_{b+z_b} has bit length b (so it is at least $2^{b-1} \geq 2$) and exit time at least $b + 1 + z_b$. If $\xi_{b-1} = 0$ (which needs $b \geq 3$, since $\xi_0 = \xi_1 = 1$), no seed of bit length b has $L \geq b$, while $r_{b-1} + 2^{b-1}$ has bit length b and follows $e_1 \cdots e_{b-1}$, so its exit time is at least b. ▫

In this form, Theorem 1 says that every run of zeros in the binary expansion of ξ that begins at digit k is shorter than k. Target A asks for runs of length at most $(1 - \varepsilon)k + C$, and Target B for runs of length $o(k)$. These are reformulations, not progress on either target.

7A.3 What Theorems 1 and 4 give for $M(k)$

Theorem 4 bounds seeds of one bit length. To bound $M(k)$, every bit length $b \leq k$ must be covered.

Corollary N4 (selected intervals). If $q_m + 4 \leq k \leq q_{m+1} - 4$ for some m, then $M(k) \leq k + q_m + 3$. (The range is empty for $m \leq 2$.)

Proof. Fix such m and k. Let $2 \leq n < 2^k$ and $b = b(n) \leq k$. If $b \geq q_m + 4$, then $q_m + 4 \leq b \leq q_{m+1} - 4$, and Theorem 4 with this same m gives $\sigma_3(n) \leq b + q_m + 3 \leq k + q_m + 3$. If $b \leq q_m + 3$, then $n^2 < 2^{2b}$, so Theorem 1 gives $\sigma_3(n) \leq \lfloor \log_2 n^2 \rfloor + 1 \leq 2b \leq 2q_m + 6 \leq k + q_m + 3$. ▫

For example, $M(k) \leq k + 1057$ for $1058 \leq k \leq 24723$, so $M(k) \leq 1.5\,k$ for $2114 \leq k \leq 24723$.

Proposition N5 (lim inf). $\liminf_{k\to\infty} \frac{M(k)}{k} \leq \sqrt{2}$.

Proof. The continued fraction of α is $[0; a_1, a_2, \ldots] = [0; 1,1,1,2,2,3,1,5,2,23, \ldots]$. Its convergent denominators D_n satisfy $D_{n+1} = a_{n+1}D_n + D_{n-1}$ with $D_0 = D_1 = 1$, so the distinct list is $q_m = D_{m+1}$, and $q_{m+1} = a_{m+2}q_m + q_{m-1}$ for $m \geq 1$ (for instance $8 = 2 \cdot 3 + 2$).

Partial quotients. Since $3^{\alpha} = 2$ and α is irrational (no power of 2 equals a power of 3), the Gelfond–Schneider theorem [49] shows that α is transcendental. An eventually periodic continued fraction represents a quadratic irrational [10], so the continued fraction of α is not eventually periodic, and in particular not eventually constant equal to 1. Hence $a_{m+2} \geq 2$ for infinitely many m.

Ratios of denominators. Let $R = \limsup_m q_{m+1}/q_m \in [1, \infty]$. Suppose R is finite, and let $\delta > 0$. For all large m, $q_m/q_{m-1} \leq R + \delta$. At each of the infinitely many m with $a_{m+2} \geq 2$,

$$\frac{q_{m+1}}{q_m} \geq 2 + \frac{q_{m-1}}{q_m} \geq 2 + \frac{1}{R+\delta}.$$

Hence $R \geq 2 + 1/(R+\delta)$ for every $\delta > 0$, so $R \geq 2 + 1/R$, that is, $R \geq 1 + \sqrt{2}$.

Conclusion. Choose infinitely many m along which q_{m+1}/q_m tends to R (or to infinity if $R = \infty$). Along these m the gaps $q_{m+1} - q_m$ eventually exceed 8, so the cutoff $\kappa_m = q_{m+1} - 4$ satisfies $q_m + 4 \leq \kappa_m \leq q_{m+1} - 4$, and Corollary N4 gives

$$\frac{M(\kappa_m)}{\kappa_m} \leq 1 + \frac{q_m + 3}{q_{m+1} - 4} \to 1 + \frac{1}{R} \leq 1 + \frac{1}{1+\sqrt{2}} = \sqrt{2},$$

with $1/R = 0$ when $R = \infty$. Hence $\liminf_k M(k)/k \leq \sqrt{2}$. ▫

The constant has a short history. The argument was first written with the golden ratio $\varphi \approx 1.618$, which needs only $q_{m+2} \geq q_{m+1} + q_m$. The second-round reviewer improved it to $3/2$ using Gelfond–Schneider. GPT's review then improved it to $\sqrt{2} \approx 1.414$ by the lim sup argument above, which Claude checked (Appendix G.5). A related route follows from Ren's rational-height staircase theorem [52, Theorem 4.1]. Its lower estimate, restricted to positive integer realizers and converted to parity depth, gives $\liminf M(k)/k \leq 1 + 1/R$, and the denominator-ratio argument above then gives $\sqrt{2}$ (GPT's deduction, checked by Claude against Ren's text; Appendix G.5, Rounds 9–10). Ren's lower estimate rests on his Proposition 3.2, an exact-reach law for the convergent approximants, which his paper states without proof; Claude checked it by exact computation in six cases. The self-contained derivation through Corollary N4 is retained.

Where a partial quotient is large, Corollary N4 gives much more. After $a = 23$, for instance, $M(k) \leq 1.05\,k$ at $k = 24723$. Proposition N5 is a statement about infinitely many k; it is not Target A, which concerns every k. It is also far from what the data suggest: the exact values below give $M(k) \leq k + 16$ for $2 \leq k < 20000$.

Remark N6 (the limit of Theorems 1 and 4). The ranges $R_m = [q_m + 4, q_{m+1} - 4]$ of Theorem 4 are pairwise disjoint because the q_m increase. Let $F(b) = b + q_m + 3$ for $b \in R_m$ and $F(b) = 2b$ otherwise. This is the bound the two theorems give for seeds of bit length b: Theorem 1's bound $\lfloor \log_2 n^2 \rfloor + 1$ equals $2b$ at $n = 2^b - 1$ ($b \geq 2$), and on R_m we have $b + q_m + 3 \leq 2b - 1$. So $G(k) = \max_{2 \leq b \leq k} F(b) \leq 2k$ is the best bound for $M(k)$ that they give together. If $q_m - 3 \leq k \leq q_m + 3$, then k lies in no range: R_{m-1} ends at $q_m - 4$, R_m begins at $q_m + 4$, earlier ranges end below q_{m-1} and later ones begin above q_{m+1}. So $F(k) = 2k = G(k)$. Since $q_m \to$

∞, $G(k) = 2k$ for infinitely many k, and $2k > (2-\varepsilon)k + C$ once $k > C/\varepsilon$. Hence Theorems 1 and 4, combined over bit lengths, give no constants for Target A. In fact $G(k) = 2k$ exactly when $2 \le k \le 11$ or $q_m - 3 \le k \le q_m + 3$ with $m \ge 4$. For $k \in R_m$, $G(k) = \max(2q_m + 6, k + q_m + 3) < 2k$; and every k outside all ranges lies in one of those sets, because $R_3 = [12,15]$ is the first nonempty range. This is a property of the two bounds, not of $M(k)$: for example $M(1054) = 1055$, while $G(1054) = 2108$.

7A.4 What exact computation shows

The residues r_L were computed for $L \le 20000$ in two independently written ways (a closed form through $3^{-Q_L} \bmod 2^L$, and bit-by-bit lifting of an orbit), with identical results. Corollary N2 was compared with direct enumeration of every seed below 2^{22}, and Corollary N3 with the literal definition of S_k for every $k < 20000$; there were no disagreements. For $2 \le k < 20000$, $M(k) - k \le 16$, with maximum at $k = 1131$, and the successive record values of $M(k) - k$ occur at $k = 2,15,40,49,196,415,1131$ with values 2,3,7,8,9,14,16. The computed values satisfy the bounds of Theorems 1 and 4 for every $k < 20000$, both for $M(k)$ and for each exact bit length. (The largest index of ξ needed for $k < 20000$ is 19999, inside the computed range; a reviewer extended the computation to $L = 40000$ with the same results.) Figure 3 shows these data. They are consistent with $M(k) = k + O(\log k)$, but a finite computation cannot establish Target A or B.

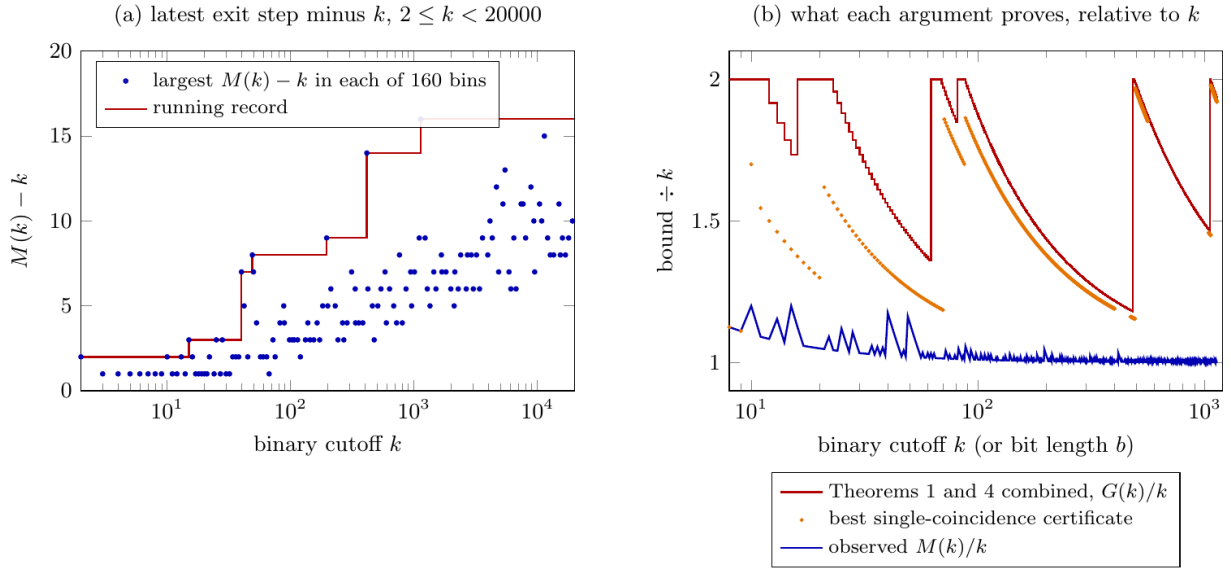


Figure 3. The maximum exit time and what each argument proves. (a) For mathematicians: the largest value of $M(k) - k$ in each of 160 logarithmically spaced bins of $2 \le k < 20000$, and the running record, from the exact residues r_L (Corollary N3). (b) The best bound for $M(k)$ that Theorems 1 and 4 give together, $G(k)/k$, which returns to 2 just above every convergent denominator (Remark N6); the best single-coincidence certificate divided by the bit length (Section 7A.5), computed exhaustively for $8 \le b \le 400$ (shown) and near 485 and 1054, using $j + \ell$ for $i = 0$ and $\max(j + \ell, 2j - i)$ for $i \ge 1$, as defined in Section 7A.5; the preregistered value $j + \ell$ differs at only two of these bit lengths; and the observed $M(k)/k$. For the general reader: the left panel shows that, among all numbers with at most k binary digits, the latest exit

comes only a few steps after step k. The right panel shows that the present proofs, by contrast, guarantee nothing better than about twice k just above certain special sizes. The data are computations, not proofs.

7A.5 Evidence that the present method stops at coefficient 2

Both proofs of the upper bounds rest on one mechanism: two positions $i < j$ at which the critical word has the same next ℓ letters give $x_i \equiv x_j \pmod{2^\ell}$; a size estimate $|x_i - x_j| \le |C_i - C_j| n + j < 2^\ell$ forces $x_i = x_j$, and Theorem 5 excludes the resulting cycle. For seeds below 2^b, positions $0 \le i < j$ with equal next ℓ letters and $|C_i - C_j| 2^b + j < 2^\ell$ certify $\sigma_3(n) \le j + \ell$ when $i = 0$, and $\sigma_3(n) \le \max(j + \ell, 2j - i)$ when $i \ge 1$ (the second term makes room for two copies of the cycle's word). We call such a triple a *single-coincidence certificate*. The best such certificate was computed exactly, by an exhaustive search, for every $2 \le b \le 400$ and for b in $[470,560]$ and $[1040,1130]$. The preregistered test (Appendix G.3) asked for $b \le 400$ with the value $j + \ell$. The first run covered only $8 \le b \le 300$ and used the value $\max(j + \ell, 2j - i)$ when $i \ge 1$. The two windows were added after the first results were seen. These departures were disclosed late, in a retrospective amendment, and the exhaustive recomputation under both definitions was added after review. The two definitions differ at only two bit lengths, by one step each. Under either, the ratio to b falls between denominators but peaks just above each denominator in the computed range: about 1.86 near $q = 65$ and $q = 84$, 1.967–1.970 at $b = 493$ just above $q = 485$, and 1.979 at $b = 1062$ just above $q = 1054$.

The reason appears to be structural. Just above a denominator q, blocks long enough to make the size estimate work (length about b less a logarithmic gain from the closeness of C_i and C_j) recur first at a lag of at least about q, so the certificate costs about $q + b \approx 2b$. A proof that every single-coincidence certificate has $j + \ell \ge (2 - o(1))b$ on such windows is a candidate result, not yet written in full; the computation above is evidence for it. If proved, it would show that Target A needs an ingredient beyond one coincidence and one size estimate, for instance several coincidences used together, or arithmetic information about the periodic 2-adic points $B(w)/(2^q - 3^p)$ that the critical word follows for long stretches. It would not show that Target A is false. The quantity involved is close to the complexity function $r(n, x)$ of Bugeaud and Kim [44], the length of the shortest prefix of a word x that contains two occurrences of some factor of length n. The correspondence is through the block length. A certificate (i, j, ℓ) exhibits the factor $e_{i+1} \cdots e_{i+\ell} = e_{j+1} \cdots e_{j+\ell}$ twice in the prefix $e_1 \cdots e_{j+\ell}$, so its value is at least $j + \ell \ge r(\ell, \mathbf{e})$. The size condition sets how small ℓ may be: about b, less a logarithmic gain. That coefficient-sensitive lower bound on ℓ is what a proof would have to add to the results of [44] on Sturmian words, which are the natural starting point.

7A.6 Status of these results

Corollaries N2–N4, Proposition N5 and Remark N6 have written proofs constructed in the completion round. Proposition N1 completes the proof proposed in GPT's Completion Assessment.

Review history (Appendix G.5). The review history differs by result.

- *N2–N4.* Reviewed twice by Claude sub-agents that had not constructed them, then by GPT, then by Grok.

- *N6.* Reviewed twice by such sub-agents, and then by GPT.

- *N1.* Reviewed twice by such sub-agents. GPT, the original author of its proof, reviewed the corrections. The argument that the formula fails above $9/2$ was supplied by GPT in that review, checked by Claude and reviewed by Grok.

- *N5.* The final $\sqrt{2}$ form was proposed in GPT's review, checked by Claude and reviewed by Grok; its $3/2$ predecessor was reviewed by GPT.

Other status. Proposition N1's exact count (for every real K), Corollary N2 and Corollary N3 are proved in Lean (Section 8). Corollary N4 and Proposition N5 are only partly proved in Lean, and Remark N6 and N1's growth rate are not proved in Lean. A documented literature search (Appendix G.5, Round 8) did not locate any of them as stated. The equivalence between irrationality measures of ξ and its zero runs, which underlies N3, is in De Jesús [50], and the bound of N5 can be deduced from Ren's Theorem 4.1 [52] (Appendix G.5, Round 9).

Targets A and B remain open. The investigation established:

- the reductions N2–N3;

- the interval bound N4;

- the lim inf bound N5;

- the fact that Theorems 1 and 4 cannot give Target A (N6).

It also gave computational evidence, with a candidate explanation that is not proved, that the single-coincidence method on which both proofs rest cannot go below coefficient 2 (Section 7A.5). The next steps are to prove or refute that candidate obstruction and to investigate stronger arguments for Targets A and B. Neither target needs to be resolved for the paper's narrower proved claims to be complete. Figure 4 shows how the new results depend on the old.

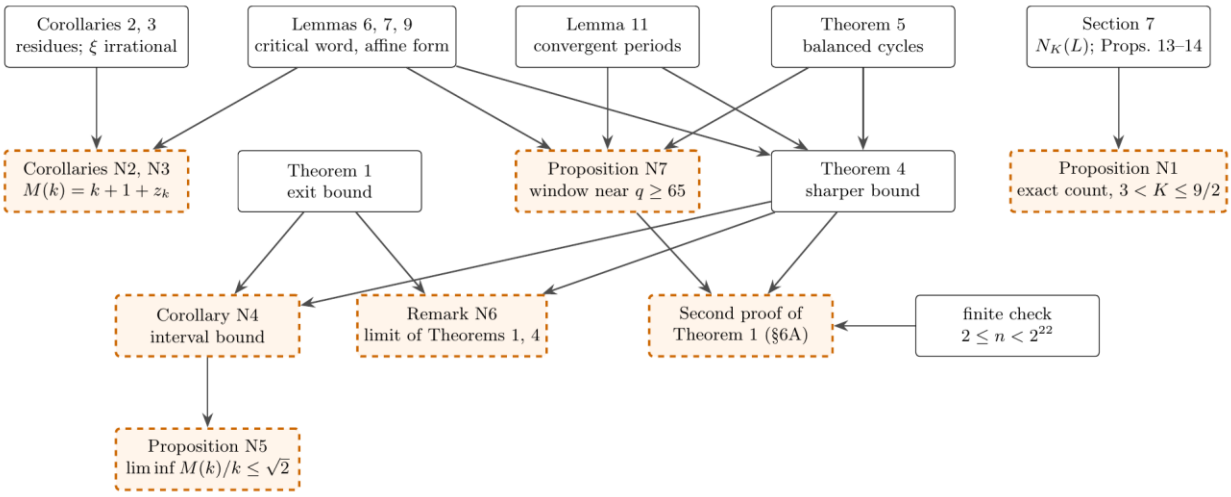


Figure 4. How the results added in v10 depend on earlier ones. An arrow means "is used to prove". Dashed boxes are new in v10; in Lean, N1's exact count, N2, N3 and the finite check are proved, Theorem 4, N4 and N5 are partly proved, and the others are not proved in Lean (Section 8). Proposition N5 also uses two results outside the paper, the Gelfond–Schneider theorem and the periodicity of continued fractions of quadratic irrationals, which are not drawn. Proposition N7, Theorem 4 and a finite check together give the second proof of Theorem 1 (Section 6A), which uses neither Lemma 8 nor the pigeonhole argument of Section 5. For the general reader: each box is a step of an argument, and the dashed boxes are the steps added in v10.

8 Formal verification

The proof assistant is Lean 4 [18] with the Mathlib library [17]. The statement formalized is the exact seed form of Theorem 1, with no real numbers: for all $n \geq 2$ there is j with $1 \leq j \leq$ Nat.log2$(n \cdot n) + 1$ and $3^{a_j(n)} < 2^j$ or $3 \cdot 2^j < 3^{a_j(n)}$, where T, the orbit and a_j are defined in a frozen file *MStatement.lean*. The final declarations are

HypothesisMAssembly.hypothesis_M : HypothesisMTarget.MStatement
HypothesisMAssembly.balanced_return_trivial(Theorem 5).

The formal proof follows Sections 3–5 with one change: instead of the irrational rotation by α, it identifies the confined parity sequence with the orbit of the exact rational map $c \mapsto 3c/2$ $(1 \leq c < 2)$, $c \mapsto c/2$ $(2 \leq c < 3)$, and derives balance and the $k + 1$ block bound by induction over rationals. Seeds $2 \leq n \leq 63$ are handled by kernel evaluation (*decide*). From the formalization of Knight's theorem [12] it imports only axiom-free arithmetic lemmas (the cancellation identity and the oddness of $2^p - 3^q$); that library declares four auxiliary axioms, which neither its own main theorem nor our development uses.

The development (Lean 4.35.0, Mathlib v4.35.0, 13 modules) was rebuilt from source on a GitHub-hosted runner from commit *449929d* of the public repository

https://github.com/whouxe82/hypothesis-m-lean.

The run (i) checks the SHA-256 hashes of all 33 frozen sources, (ii) compiles the development, (iii) compiles a separately written file that checks the theorem against a hand-written expansion of the statement and evaluates the definitions on sample orbits, and (iv) fails unless *#print axioms* reports only *propext*, *Classical.choice* and *Quot.sound*. All four steps passed (run of 22 October 2026, 15:45–15:47 UTC; Lean 4.35.0, Mathlib commit *56a00585*). The run record is public at

https://github.com/whouxe82/hypothesis-m-lean/actions/runs/39807720548/job/119925315584

(full commit *449929d2b453f24e97df37672f77031c1c75ae30*).

Formal proofs of the new results. The completion round (Appendix G) added N1–N7 and the second proof of Section 6A without formal proofs. Lean round 1 proved the first two items of the order proposed then. Three new modules, written by Claude, use only core Lean 4.35.0 and no Mathlib, and are kept outside the frozen sources, whose hashes are unchanged.

- *Rev8Bounds.lean* repeats the definitions of *MStatement.lean* and proves Corollary N2. Its theorem *n2_lower* is unconditional: for every $k \geq 2$ some seed $2 \leq n < 2^k$ makes no coefficient exit within k steps, so $M(k) \geq k + 1$. Its proof constructs the seed by lifting the forced parity pattern one binary digit at a time. Its theorem *n2_upper* derives $M(k) \leq 2k$ from the seed form of Theorem 1.

- *Rev8Bridge.lean* proves that these definitions agree with the frozen ones, and restates both bounds for the frozen definitions. The upper bound uses *HypothesisMAssembly.hypothesis_M*.

- *N1.lean* proves Proposition N1 in integer form. A word of length L is admissible when $2^j \leq 3^{a_j}$ and $Q \cdot 3^{a_j} \leq P \cdot 2^j$ for $1 \leq j \leq L$. A time $j \geq 2$ is a choice time when $Q \cdot 3^{Q_j+1} \leq P \cdot 2^j$, which is the condition $\{j\alpha\} \geq 1 - \beta$ of Section 7 written without real numbers. The theorem *N1.N1* states that the number of admissible words is $2^{B(L)}$ whenever $Q > 0$, $3Q < P$ and $2P \leq 9Q$. This covers every rational K in $(3, 9/2]$; irrational K is not covered in Lean round 1; Lean round 2 covers it (below).

Each theorem's *#print axioms* report lists only *propext*, *Classical.choice* and *Quot.sound*, and the files contain no *sorry*, *native_decide* or added axioms. The modules were first compiled in Claude's working environment. Mathlib's binary cache was not reachable there, so the bridge was checked against a copy of the frozen definitions. On 24 October 2026 the author ran the GitHub Actions workflow "Hypothesis M replay with revision 8" in his repository, on a GitHub-hosted machine. It repeated the full replay of the frozen development, then compiled the three modules against it and applied an axiom gate. The run passed:

https://github.com/whouxe82/hypothesis-m-lean/actions/runs/40109923980

In that run *n2_upper_frozen* reports only the standard axioms, so the bridge was checked against the formal Theorem 1 itself. The same day, four further runs of the original replay workflow rebuilt the frozen development from source, and all passed.

Lean round 2. GPT wrote thirteen further modules against the frozen development and the Lean round 1 modules, under a brief that forbade *sorry*, *native_decide* and added axioms and required every statement to use the frozen definitions. It delivered them with a correspondence table giving each paper result, its Lean statement, its axiom report and its status. Claude checked the SHA-256 hashes, the absence of forbidden constructs, and each statement against the paper. The modules prove the following.

- *HypMRev9.corollary2_residue*, *corollary2_integer* and *no_exit_iff_mod*: Corollary 2, with ξ represented by its canonical residues r_L, and the absence of an exit within L steps proved equivalent to n ≡ r_L (mod 2^L).

- *HypMRev9Four.proposition14_real_formula* and *proposition14_rate*: the exact formula for $N_4(L)$ and its growth rate $2\alpha - 1$, with $\alpha = \log 2/\log 3$, by elementary bounds and without equidistribution.

- *HypMRev9Real.propositionN1_real_count*: N_K(L) = 2^(B_K(L)) for every real K with $3 < K \leq 9/2$; *admissible_frozen_orbit* ties the counted words to the frozen orbit words.

- *HypMRev9.corollaryN2_identity*, *maximum_lower* and *maximum_upper*, with *residueDepths_finite* and *residueDepths_nonempty*: Corollary N2 for the actual maximum M(k).

- *HypMRev9.residue_binary_expansion*, *zeroRun_spec*, *zeroRun_unique*, *corollaryN3_identity*, *corollaryN3_run_bound* and *corollaryN3_exact*: Corollary N3, including both exact-bit-length cases.

- *HypMRev9Base.second_proof_finite_base*: Theorem 1 for every $2 \leq n < 2^{22}$, from 45 residue certificates checked by the kernel with *decide* (the certificate's axiom report is *propext* alone). A dependency check, *Rev9IndependenceAudit.lean*, confirms that neither the original Theorem 1 nor Corollary 2 occurs in its proof or statement.

Two modules are partial. *HypMRev9.period_implies_exit* and *period_implies_maximum_bound* derive the conclusions of Theorem 4 and Corollary N4 from an explicit hypothesis that the critical word is q-periodic on the relevant range, which is the content of Lemma 11. The three lemmas of *HypMRev9Ratio* prove the arithmetic of Proposition N5 under stated hypotheses about the denominators. So Theorem 4, N4 and N5 are partly proved in Lean, not proved in Lean: an axiom report shows how a theorem is proved, not that its hypotheses hold. The proofs of N1, N2 and N3 use the formal Theorem 1; the proof of the finite base does not.

The author added a third workflow, “Hypothesis M replay with revision 9”. It repeats the full replay, compiles the round 1 and round 2 modules in dependency order, and fails unless all 27 theorems named in *Rev9Audit.lean* report only *propext*, *Classical.choice* and *Quot.sound*, no log shows an error, *sorryAx* or *ofReduceBool*, and the dependency check passes. He then pinned all three workflows to the Ubuntu 24.04 build image. The final runs, on commit *13bd411*, all passed:

Round 2 workflow: https://github.com/whouxe82/hypothesis-m-lean/actions/runs/40131960252

Round 1 workflow: https://github.com/whouxe82/hypothesis-m-lean/actions/runs/40131960287

Independent replay: https://github.com/whouxe82/hypothesis-m-lean/actions/runs/40131960242

(full commit *13bd41132b25f678d5bb7bed74b301b5e1556c48*). Appendix G.7 gives the record.

Not yet proved in Lean:

1. Corollary 3, which needs rational 2-adic orbit arithmetic and the irrationality of α.
2. Lemma 11, and with it Theorem 4 and Corollary N4, with their exact endpoint ranges.
3. Proposition N7, which needs a certified bound for $\| q\alpha \|$ at convergents, and with it the second proof of Section 6A beyond its finite base.
4. Proposition N5, which uses the Gelfond–Schneider theorem, not yet available in Mathlib.
5. Proposition 13, Remark N6 and the growth rate of Proposition N1, which need a theorem on the frequency of an irrational rotation.

Part II. The Same Result for the General Reader

This part retells Part I without formulas beyond very basic arithmetic. Nothing in it depends on Part I, and nothing in it claims more about the result than Part I proves.

9 A game with numbers

Pick any positive whole number and apply this rule:

- if the number is **even**, cut it in half;
- if the number is **odd**, triple it, add 1, and then cut the result in half.

Apply the rule again to the result, and keep going. Starting from 7, for example, you get

$$7 \to 11 \to 17 \to 26 \to 13 \to 20 \to 10 \to 5 \to 8 \to 4 \to 2 \to 1 \to 2 \to 1 \to \cdots$$

Once a number reaches 1 it is stuck in the little loop $1 \to 2 \to 1$.

The German mathematician Lothar Collatz studied questions of this kind in the 1930s. The best known is this: does *every* starting number eventually reach 1? Computers have checked every starting number below about 2.4×10^{21} [1], and all of them do. It is also proved that almost all numbers eventually drop below their starting value [24, 6], and Tao proved that, for every function $f(n)$ tending to infinity, the smallest value the chain reaches is at most $f(n)$ for almost all starting numbers n, where "almost all" has the technical meaning of logarithmic density [23]. Such "almost all" results measure how sparse the exceptions are; they do not show that there are no exceptions. But no one has proved that every number reaches 1. Checking many cases is not a proof, because the next unchecked number could always be the exception. The question remains one of the best-known unsolved problems in mathematics. A good guide to it is the collection edited by Lagarias [14].

This paper does **not** solve the problem. It proves one precise fact about a narrow corner of it.

10 Keeping score: the hallway

Each step of the rule either pushes the number up or pulls it down. An odd step multiplies the number by about one and a half (triple, then halve). An even step multiplies it by one half. Ignore the small "add 1" and keep a running **score** of how much the pushes and pulls have multiplied the number so far:

- start the score at 1;
- on an odd step, add half the score to itself (for example $1 + 0.5 = 1.5$);
- on an even step, halve the score (for example $1.5 \div 2 = 0.75$).

Now picture a hallway with a floor and a ceiling. The **floor** is a score of 1: the multiplications alone would bring the number back to its starting value (the small additions are not counted in

the score). The **ceiling** is a score of 3: the multiplications alone would triple it. A number is *in the hallway* as long as its score stays between 1 and 3. Once the score drops below 1 or rises above 3, the number has *left the hallway*.

Only one rhythm of odd and even steps keeps the score in the hallway forever. Its rule is simple: halve whenever halving keeps the score at 1 or above, and take an odd step otherwise. An odd step happens only when the score is below 2, so adding half never pushes it past 3. The rhythm begins

odd, odd, even, odd, odd, even, odd, odd, even, odd, even, odd, odd, even, …

and continues forever without ever settling into an exact repeat. We call it the **drumbeat**. Mathematicians call it the critical word. Figure 5 shows the score following the drumbeat, together with two numbers that fall out of step.

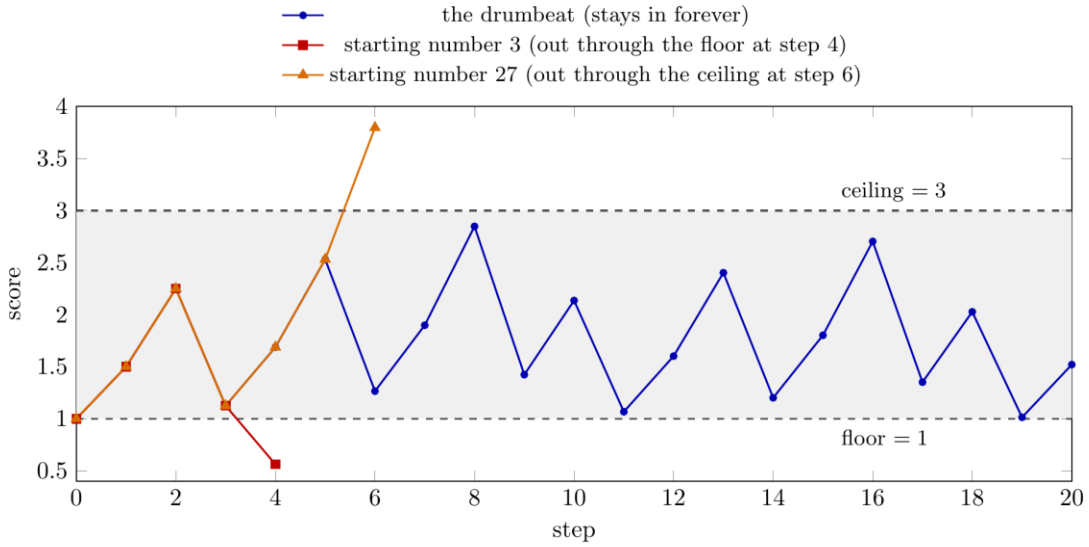


Figure 5. The hallway (shaded) lies between a score of 1 and a score of 3. The drumbeat keeps the score inside forever. A real starting number keeps step only for a while: 3 leaves through the floor at step 4, and 27 leaves through the ceiling at step 6.

11 What the theorem says

Numbers can follow the drumbeat for a while. The main theorem asks how long, and answers: **not long, and here is the exact deadline**. To find the deadline for a starting number:

1. multiply the number by itself;
2. count how many times you can halve the result and still have 1 or more;
3. add 1.

Keep any fractions when halving, as in the examples below. Mathematicians write the same deadline as $\lfloor \log_2(n^2) \rfloor + 1$, the number of binary digits of $n \times n$.

Every starting number from 2 upward leaves the hallway no later than that step. The deadline is roughly twice the number of digits the number has when written in binary. (Binary writes numbers with only the digits 0 and 1: for example, 7 is 111 and 26 is 11010.) A number twice as large gets a deadline just two steps later.

The number 3 uses all of its allowance, as the next section shows, so the deadline cannot be reduced by even one step as a rule for every starting number. Mathematicians call a bound like this *sharp*, meaning that some case lands exactly on it. That does not make it the best possible deadline for larger numbers; for many sizes Part I proves a tighter one.

In the project's records the claim was called *Hypothesis M* while it was still unproved. Now that it is proved, it appears in Part I as Theorem 1, also called Theorem M.

Table 3 works out the deadline for some starting numbers and shows when each one actually leaves.

Table 3. Deadlines and actual exits. Every even number leaves at step 1. Of these numbers, only 3 reaches its deadline exactly.

Starting number	Number times itself	Halvings that stay at 1 or more	Deadline (halvings +1)	Actually leaves at step	Exit
2	4	2	3	1	floor
3	9	3	4	4	floor
4	16	4	5	1	floor
5	25	4	5	2	floor
6	36	5	6	1	floor
7	49	5	6	3	ceiling
8	64	6	7	1	floor
9	81	6	7	2	floor
10	100	6	7	1	floor
11	121	6	7	5	floor
12	144	7	8	1	floor
27	729	9	10	6	ceiling
100	10,000	13	14	1	floor
1,000	1,000,000	19	20	1	floor

A useful image is a dance floor with one perfectly precise choreography. The theorem says that every dancer, however skilled, loses the step within a known number of beats, and it gives the number of beats for each dancer.

12 The hallway in plain arithmetic

Example 1: the number 3 uses every step it is allowed

Deadline: $3 \times 3 = 9$. Halving gives $4.5,\ 2.25,\ 1.125$, which is three halvings (the next would be 0.5625, below 1). Then $3 + 1 = 4$, so the deadline is **4 steps**.

Table 4. The number 3, step by step: the score falls through the floor at step 4, its deadline.

Step	Number	Odd/even	Score arithmetic	Score	In the hallway?
1	$3 \to 5$	odd	$1 + 0.5$	1.5	yes
2	$5 \to 8$	odd	$1.5 + 0.75$	2.25	yes
3	$8 \to 4$	even	$2.25 \div 2$	1.125	yes, only $1.125 - 1 = 0.125$ above the floor
4	$4 \to 2$	even	$1.125 \div 2$	0.5625	no: $1 - 0.5625 = 0.4375$ below the floor

The number 3 leaves on step 4, exactly at its deadline.

Example 2: the number 7 leaves through the ceiling while growing

Deadline: $7 \times 7 = 49$. Halving gives $24.5,\ 12.25,\ 6.125,\ 3.0625,\ 1.53125$, which is five halvings. Then $5 + 1 = 6$, so the deadline is **6 steps**.

Table 5. The number 7, step by step: the score goes through the ceiling at step 3 while the number is still climbing.

Step	Number	Odd/even	Score arithmetic	Score	In the hallway?
1	$7 \to 11$	odd	$1 + 0.5$	1.5	yes
2	$11 \to 17$	odd	$1.5 + 0.75$	2.25	yes
3	$17 \to 26$	odd	$2.25 + 1.125$	3.375	no: $3.375 - 3 = 0.375$ above the ceiling

The number 7 leaves on step 3, well inside its deadline of 6. But it has climbed from 7 to 26. Leaving the hallway is not the same as coming down. This is why the theorem says nothing about reaching 1.

Why staying in is so hard

A number stays in the hallway only while its own odd and even steps match the drumbeat exactly. The first step that departs from the drumbeat puts it out. If the drumbeat calls for an odd step and the number is even, the score is halved below the floor. If the drumbeat calls for an even

step and the number is odd, the score is pushed above the ceiling. Every even starting number mismatches at once, because $1 \div 2 = 0.5$ is below the floor.

How well do real numbers do?

A computer search over every starting number from 2 to just under two million finds 16 successive record holders, counting the first, 2. A record holder leaves the hallway later than every smaller starting number in the search. Table 6 shows a selection.

Table 6. A selection of the record holders below two million, with the step at which each leaves the hallway and its deadline from Theorem 1.

Starting number	Binary digits	Leaves the hallway at step	Deadline from the theorem
3	2	4	4
27	5	6	10
1,019	10	12	20
23,547	15	18	30
1,989,627	21	22	42

Apart from 3, even the best performers leave well before their deadlines. The search only illustrates the theorem. The proof covers every number, however large, and no computer search could do that.

13 Why it is true: the idea of the proof

The proof is an argument by contradiction. Suppose some large starting number kept to the drumbeat past its deadline. Then the following would happen.

1. **The drumbeat has very little variety.** Cut any stretch of, say, 20 beats out of the drumbeat. There are over a million conceivable patterns of 20 odd-or-even steps, but no more than 21 of them ever occur in the drumbeat. In general, stretches of a given length come in at most one more variety than that length.

2. **So two stretches must match.** Take stretches of the same length starting at consecutive positions, using one more starting position than there are varieties. Two of these stretches must be identical. This is the *pigeonhole principle*: with more pigeons than holes, two pigeons share a hole.

3. **Matching futures mean matching final digits.** Which odd and even steps a number takes next is decided by its last binary digits. The link also works the other way: different endings always lead to different step patterns. So if two numbers in the chain face the same next 20 steps, they end in the same 20 binary digits.

4. **The two numbers are too close to differ.** Two different numbers that share their last 20 binary digits must differ by at least 2^{20}, about a million. But when two numbers face the same stretch of at least six beats, their scores differ by at most $4/9$. And each

number in the chain equals its score times the starting number, plus a small correction no larger than the number of steps taken so far. With the stretch length matched to the size of the starting number, these two facts put the two numbers less than 2^{20} apart. Their difference is a multiple of 2^{20} and smaller than 2^{20}, so it is zero. Picture an odometer. Two readings that end in the same six digits, yet are less than a million miles apart, must be the same reading. So the two numbers are equal.

5. **Equal numbers mean a loop.** If the chain returns to a number it has already visited, it repeats forever. There are enough beats left in the assumed run to hold two copies of the loop's pattern back to back. So every stretch taken around the loop, even one that wraps past its end, occurs in the drumbeat, and the loop's odd steps are spread as evenly as possible. Mathematicians call such a pattern *balanced*.

6. **The only balanced loop is $1 \to 2 \to 1$.** This is known from earlier work (Knight, with earlier partial results by Steiner and by Halbeisen and Hungerbühler), and the paper also gives a short proof. But the loop found above consists of numbers larger than the starting number, which is at least 64, so it cannot be the loop of 1 and 2. That is impossible.

The supposition has led to an impossibility, so it is false. Every starting number from 2 upward leaves the hallway by its deadline. Small starting numbers, from 2 to 63, are checked directly.

Part I also proves a second, sharper deadline for numbers whose length in binary digits falls inside certain wide ranges. It shows as well that the hallway must be exactly this narrow for the argument to work. Raise the ceiling even slightly above 3 and the number of possible rhythms starts to grow exponentially, that is, by repeated multiplication as the rhythm gets longer. Then the low-variety estimate in step 1 no longer holds. That does not rule out every other argument for a wider hallway.

Version 10 adds three things for the general reader.

- *A second check of the main theorem.* It completes a second proof of the main theorem by a different route. Instead of counting the varieties of drumbeat stretches, that proof uses the near-repeats of the drumbeat at special distances, the convergent denominators, and checks small numbers by computer.

- *An exact count for slightly wider hallways.* It counts exactly how many rhythms survive when the ceiling of the hallway is between 3 and 4½. In that range each extra choice happens alone: a rhythm can go "down then up" or "up then down" and return to the same score, so every permitted choice doubles the number of rhythms.

- *The latest exit among numbers of a given size.* Among all starting numbers from 2 with at most k binary digits, the latest first exit from the hallway comes at a step between $k + 1$ and, by the main theorem, $2k$. For every k below $20{,}000$, computer checks find it no later than step $k + 16$.

Whether that latest exit stays close to k for every k is still open; computer checks suggest, without proof, where the present method stops.

14 How the result was checked

- **A written proof.** Part I gives the traditional argument, meant to be read and checked by people.

- A Lean proof. The main theorem was also proved in Lean, a computer program that checks proofs. Lean accepts a proof only if every step is justified down to basic logic. It is like a proofreader who checks every comma and does not care whether the conclusion sounds plausible. The check used only Lean's three standard axioms (an axiom is a foundational assumption that is not itself proved), the usual foundations of mathematics, with nothing extra assumed.

- **An independent rebuild.** The sources and the checking script were run again from scratch on fresh computers hosted by GitHub, and the public log shows that the build passed. This shows that the recorded check is reproducible. It is a little like a notary, who confirms that a document was signed as stated but does not judge what it says: the rebuild does not replace independent assessment of the formal statement or of the written mathematics.

- **Exact computer checks.** Separate programs tested the claims on very large numbers of cases, with exact arithmetic and no rounding. Different machines produced matching results, confirmed by matching digital fingerprints (hashes). These checks are supporting evidence, not proof.

- **New in v10: checks of the new results.** The new results were tested with exact arithmetic by two separately written programs and replicated by the author in Google Colab. The new proofs were reviewed by AI systems told to try to break them, in two rounds of Claude sub-agents, a later GPT review and a Grok review of selected results, with different coverage for different results. GPT reviewed some arguments it had itself written or supplied. Appendix G records which reviews each result received.

- **New in v10: more Lean proofs.** Several of the newer results are now also proved in Lean: the exact count of rhythms, the facts about the latest exit, and the computer check of every number below about four million that begins the second proof. A few others are checked only in part: Lean accepts their final steps, but not yet the facts about the number $\log_3 2$ on which those steps rest. The author ran all of it again on GitHub's computers, and it passed.

- **Reviews so far.** Four AI systems reviewed the work. Claude and GPT reviewed it internally. Grok was asked to try to break the mathematics, and Gemini was asked to search for prior publication. The reviews of the mathematics (by Claude, GPT and Grok) found errors in intermediate work, which were corrected, and reported no unresolved error in the final version. Gemini's review searched for earlier publication and did not check the proofs. AI systems agreeing is not the same as review by human experts, and

that review has not yet taken place. This two-audience edition was itself reviewed again by GPT, Grok and Gemini. Table 2 in the Introduction lists every layer of checking, with a plain-language column.

15 What has been accomplished, and what has not

Accomplished

- **A proved, exact deadline.** The deadline is stated exactly and proved. The number 3 meets it exactly, so it cannot be reduced by even one step for every starting number.

- A checked proof. The main theorem has a written proof, a Lean proof and an independent rebuild.

- **A sharper companion result.** For numbers whose length in binary digits falls inside certain wide ranges, a second theorem gives a much tighter deadline. It has a written proof and exact computer checks, but it is only partly proved in Lean.

- **A map of the method's limits.** The paper shows where the uniform low-variety estimate used in this proof ceases to hold: in any hallway even slightly wider than this one.

- **A record of method.** The exact statement was fixed first and only then proved, attacked, checked and rebuilt, in about two days. Mistakes made by both AI systems were caught and corrected, and they are kept in the project record.

- **Results nobody asked for.** While working toward the main theorem, one of the AI systems found the sharper companion result and a second proof of the main theorem, although no one had proposed either. Part III tells how this happened.

- **New in v10: three further results.** The exact count of rhythms for ceilings up to 4½ is proved. A second proof of the main theorem, by a different route, is now complete. Among all starting numbers from 2 with at most k binary digits, the paper proves that the latest exit occurs between steps $k + 1$ and $2k$, and well before $2k$ for many k.

- **New in v10: more Lean proofs.** Most of the new results, and two older ones, now also have proofs checked by Lean, rebuilt on GitHub's computers.

Not accomplished

- **It says nothing about whether numbers come down to 1.** A number can leave the hallway through the ceiling while it is still climbing, as 7 does. Proving that no car can stay exactly centered in its lane for more than a mile tells you nothing about whether the car gets home.

- **It cannot be stretched into a solution by itself.** In a sibling game, odd numbers are multiplied by 5 instead of 3, and some numbers are known to loop forever without reaching 1. That game passes the same exit test. If a lock known to be broken passes your

safety test, the test alone cannot prove that a lock is safe. Table 7 sets the two games side by side.

Table 7. The broken-lock test. The $5x + 1$ game passes the same exit test yet has a loop that never reaches 1, so passing the test cannot by itself guarantee that numbers reach 1.

	Collatz game ($3x + 1$)	**Sibling game ($5x + 1$)**
Rule for odd numbers	triple, add 1, halve	multiply by 5, add 1, halve
Does every number leave the 1-to-3 hallway?	yes: by the deadline of Theorem 1	yes: within 3 steps (score uses $\times 5/2$)
Loops other than through 1	none known; balanced ones ruled out (Theorem 5)	yes: $13 \to 33 \to 83 \to 208 \to 104 \to 52 \to 26 \to 13$

- **It does not rule out runaway numbers or hidden loops.** The only loops it rules out are the perfectly balanced ones, and that part was already known.

- **The basic idea is not new.** It was already known that no number can follow the drumbeat forever (Dubickas, 2009). The new contribution is turning "eventually" into an exact deadline with a proof. Everyone knew that every runner eventually tires. This result says how many laps each runner gets.

- **Human experts have not yet reviewed it.** The checks so far are by machines and AI systems.

- **Its originality is not fully settled.** No source examined so far contains the result, but a few relevant papers have still to be read in full.

- **New in v10: the best deadline for large numbers is still unknown.** For every k below $20{,}000$, checks find that the latest exit among numbers with at most k binary digits comes no later than step $k + 16$, far before $2k$. Neither "close to k" nor even "clearly less than $2k$" has been proved for every k. Computer checks suggest where the present method stops, but that limit is not proved. Several of the new results are now also proved in Lean, but some are not, and none has been checked by human experts.

Part III. The Result in Context: History, Method and Findings

This part is written for both audiences. Section 16 places the result in the history of the Collatz problem. Sections 17 and 18 describe how the result was directed, found and checked. The part uses no mathematics beyond the statements already given. It refers to results by their numbers in Part I and, where the project records use them, by their project names: Hypothesis M or Theorem M for Theorem 1, Theorem P for Theorem 4, Lemma B for Lemma 10 and Lemma S for Lemma 11. Two different things in the record carry the label M1. One is the *M1 manifold wave* of 17 October, a set of seven Claude workers and two reviewers. The other is *agent M1*, one of GPT's three sub-agents on 21 October. The text always says which is meant.

16 The Collatz problem: history, difficulty and the tree of results

16.1 A short history

The problem is named after Lothar Collatz, who studied iterations of this kind in his notebooks in the 1930s; Lagarias reports a notebook entry dated 1 July 1932 in which he considered a related function [13]. Collatz is said to have circulated it at the International Congress of Mathematicians in Cambridge, Massachusetts, in 1950 and to have described it to Helmut Hasse in 1952. It has travelled under many names: the $3x + 1$ problem, the Syracuse problem, Hasse's algorithm, Kakutani's problem and Ulam's problem. Bryan Thwaites claimed to have found it independently in 1952 [14]. Its systematic study began in the 1970s with the work of Terras and Everett [24, 6]. Lagarias's survey of 1985 and the volume he edited in 2010 remain the standard guides [13, 14].

16.2 How hard is the problem?

There is no objective scale on which open problems can be ranked. Three kinds of evidence nonetheless place the Collatz problem among the hardest problems in mathematics that are easy to state.

Expert judgement. Paul Erdős is reported to have said that "mathematics is not yet ready for such problems" [13]. Lagarias, the problem's leading historian, has written that "this is an extraordinarily difficult problem, completely out of reach of present day mathematics" [14].

A structural warning. John Conway showed in 1972 that a natural generalization of the problem is algorithmically unsolvable [28]. Kurtz and Simon showed in 2007 that the generalized question "does every starting value reach 1?" is undecidable as well [35]. These results do not show that the Collatz problem itself is undecidable. They show only that no single algorithm can settle the question for every map in the generalized family. A proof for the Collatz map therefore cannot be a uniform procedure for the whole family, but it may still be a general theorem covering a large class of maps that includes this one.

Its record. Fermat's Last Theorem was stated around 1637 and proved in 1995; Goldbach's conjecture was stated in 1742 and is still open. Like them, the Collatz problem can be explained to a child, yet it has resisted nearly ninety years of work. It is not one of the Clay Mathematics Institute's Millennium Prize Problems, a list that includes the Riemann Hypothesis and favours

problems central to large areas of mathematics. Its standing rests less on its connections to the rest of mathematics than on its resistance to every method tried. In 2021 a Japanese company offered a prize of 120 million yen for its resolution [26].

16.3 What has been proved

No proof of the conjecture is known, and no counterexample has been found. The proved results fall along a few lines of attack. Table 8 lists the principal ones, and Figure 6 draws them as the branches of a tree.

Statistical results. These show that almost all starting values behave well. Terras and Everett showed that almost every starting value eventually falls below itself [24, 6]. Korec showed that almost every orbit falls below n^{θ} for any $\theta > \log 3/\log 4 \approx 0.7925$ [33]. Tao showed that almost every orbit falls below any function of the starting value that tends to infinity, however slowly [23]. These results cover almost all of the problem but say nothing about any particular starting value, and a counterexample would be exactly such an exception.

Counting results. These give lower bounds on how many numbers are known to reach 1. Krasikov and Lagarias showed that at least $x^{0.84}$ of the numbers up to x do, for all large x [34].

Cycle exclusions. These rule out particular shapes of a hypothetical loop. Steiner excluded loops made of a single climb and a single descent [22]. Simons and de Weger excluded loops with up to 68 such climb-and-descent pairs [40], and Hercher extended this to 91 [32]. Eliahou tied the possible lengths of a loop to the continued fraction of $\log_2 3$ [29], and Knight excluded the so-called high cycles, including the balanced ones [11]. Any nontrivial loop is now known to contain more than 10^{11} odd steps; this combines Hercher's bound with Barina's verification [32, 1].

Parity words. These results study the pattern of odd and even steps itself. Bernstein and Lagarias studied the map that turns a starting value into its parity pattern [2]. Dubickas showed that an orbit that grows without bound must have a complicated parity pattern [5].

Computation. Every starting value below 2^{71}, about 2.4×10^{21}, has been checked by computer [1].

Table 8. Principal proved results on the Collatz problem, by line of attack.

Year	Result	Line	Source
1972	A natural generalization is algorithmically unsolvable (Conway)	Limits	[28]
1976–77	Almost every start eventually falls below itself (Terras; Everett)	Statistical	[24, 6]
1977	No 1-cycles other than $1 \to 2 \to 1$ (Steiner)	Cycles	[22]
1993	Cycle lengths tied to the continued fraction of $\log_2 3$ (Eliahou)	Cycles	[29]
1994	Almost every orbit falls below $n^{0.7925}$ (Korec)	Statistical	[33]
1996	The conjugacy map between starting values and parity patterns (Bernstein–Lagarias)	Parity words	[2]
2003	At least $x^{0.84}$ numbers up to x reach 1 (Krasikov–Lagarias)	Counting	[34]
2005	No m-cycles for $m \leq 68$ (Simons–de Weger)	Cycles	[40]
2007	The generalized question is undecidable (Kurtz–Simon)	Limits	[35]
2009	Unbounded orbits need complex parity patterns (Dubickas)	Parity words	[5]
2019	Almost every orbit attains almost bounded values (Tao; published 2022)	Statistical	[23]
2023	No m-cycles for $m \leq 91$ (Hercher)	Cycles	[32]
2025	Every start below 2^{71} checked by computer (Barina)	Computation	[1]
2026	No high cycles, balanced ones included (Knight)	Cycles	[11]
2026	Exact exit deadline for the critical parity word (this paper)	Parity words	Theorems 1, 4

16.4 Where this paper falls

Figure 6 places the present results on that tree. The trunk is the conjecture itself. Its crown, a proof that every orbit reaches 1, has not been reached by anyone. The results of this paper sit on the parity-word branch, close to where it meets the cycle branch through Knight's theorem, which the proof of Theorem 1 uses.

An important location. The critical word is the boundary between orbits whose coefficient shrinks and orbits whose coefficient grows. Its slope, $\log_3 2$, is the constant that governs the whole problem: the average drift of orbits, the length of possible cycles, and the continued fractions in Eliahou's bounds. Results there expose the geometry of the fork.

Its scope. The starting values that follow the critical word for L steps make up one residue class in 2^L, a share that tends to zero. Theorem 1 says nothing about convergence, divergence or general cycles, and Section 7 shows that its method does not extend to wider corridors. In size it ranks with specialized results such as Dubickas's, and well below Tao's theorem or the cycle exclusions of Simons and de Weger and of Hercher.

An unusual origin. What distinguishes the result is not its size but how it came about. Unlike the other results on the tree, its statement was proposed and its proof found by AI systems under human direction, and its main theorem is proved in Lean. Sections 17 and 18 describe how.

For the general reader. Think of the Collatz problem as a tree whose top no one has climbed. Mathematicians have climbed some way up several of its main branches. They know that almost every number behaves, that many shapes of loop are impossible, and that every number below about two thousand billion billion reaches 1. This paper adds a twig near the fork where the branches divide. It is a real, checked result about one narrow path through the problem, and the top of the tree is as far away as before.

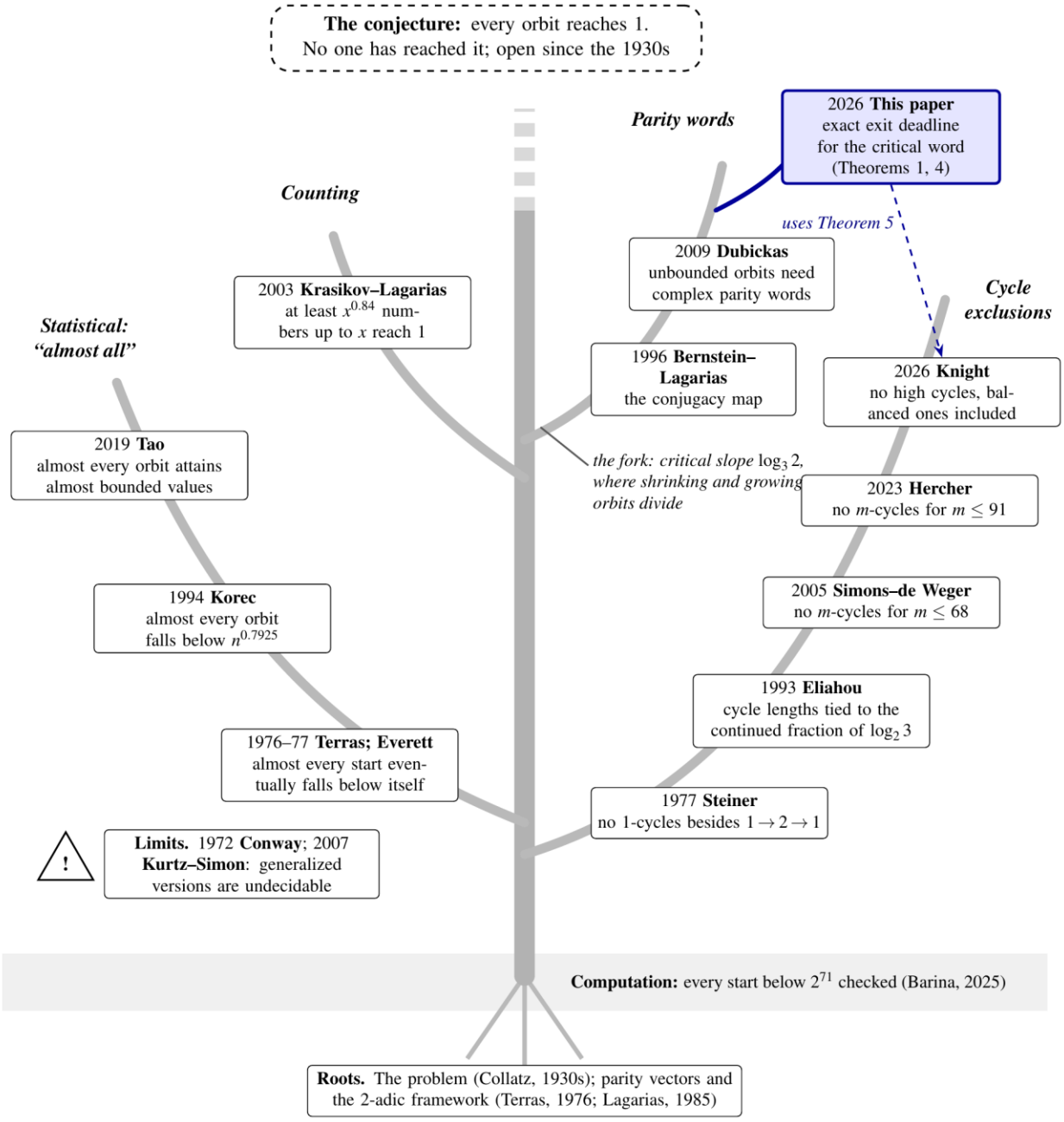


Figure 6. The tree of proved results on the Collatz problem. The trunk is the conjecture; its dashed crown, a proof that every orbit reaches 1, has not been reached. Each branch is a line of attack, and each box on it is a proved result (Table 8). The shaded band is the range checked by

computer, and the warning sign at lower left marks results showing that generalized versions of the problem are undecidable. The blue twig is this paper, on the parity-word branch near the fork, where the critical slope $\log_3 2$ divides shrinking from growing orbits; the dashed arrow marks its use of Knight's balanced-cycle theorem. Branch lengths and positions show lines of attack, not size or difficulty. For the general reader: the top of the tree is the full solution, which no one has reached, and this paper is the blue twig.

17 Methodology

17.1 Participants and roles

The author, an independent scholar whose doctorate is in the humanities, directed the work. He set each task and approved each charter before any computation. He ran the replication scripts himself in Google Colab, a hosted runtime separate from the systems that wrote them, uploaded the formal proof to a public GitHub repository in his own name, and routed every result between the systems. He did not write the proofs or the code, and he does not claim the expertise to referee the mathematics.

Two AI systems did the mathematical work: GPT (OpenAI) and Claude (Anthropic). Each ran several sub-agents in parallel, and each reported to the author through a coordinating thread. Two further systems, Grok (xAI) and Gemini (Google), reviewed the finished work. The instruments were Google Colab for replication, the Lean 4 proof assistant with Mathlib for formal verification, and GitHub Actions for an independent rebuild of the formal proof.

The Maestro role. The role is defined by observable responsibilities in the research process. The author chose the problem and its scope, redirected attention from the wider investigation toward a publishable Collatz result, requested an explicit conjecture, routed arguments between systems, and required challenges to the steps on which the proof depended. He ran replication scripts in Colab and required the preservation of claims, corrections, and failed approaches. These activities provided direction and an evidentiary record while the systems supplied the mathematical formulations, arguments, code, and exposition.

Verification was especially important because the author could not independently evaluate every mathematical step. Agreement between systems was treated as one layer of scrutiny. Exact computation tested specified cases, formal proof checked the encoded main statement, and a separate rebuild tested reproducibility. Assessing the correspondence between the formal statement and the intended mathematics, completing the literature comparison, and obtaining specialist review remain distinct tasks. The methodological proposition examined here is that a non-specialist can organize these forms of scrutiny sufficiently to produce a defensible research preprint. This case study records an instance of that process; it does not measure its general success rate or isolate the causal effect of any particular instruction.

17.2 The project and its X/Y/Z construct

The work grew out of the author's Computational Human–Machine Conjecture (CHMC) project. In rounds R1 to R15, completed on 10–11 October 2026, he alternated tasks between Claude and

GPT. Each system acted in turn as producer and as referee on restricted questions about the Erdős–Straus and Collatz conjectures.

During those rounds the author introduced what he called the X/Y/Z principle. If a method works under condition X and fails under condition Y, then X and Y differ in some respect that the method does not address, and that respect must be named. On 12 October he defined a further component, Z, as "the intuitive element, or the supra-logical sensitivity element". Z is the frame-changing step that breakthrough mathematicians supply and that, he argued, frontier AI systems had not yet shown.

From 14 October a second phase of the project made Z its driving concern. Its rounds, Gate 0 and Z01 to Z10, were each governed by a charter that was frozen and hashed before any computation. Every claim was entered in a claim ledger with a fixed status label (Appendix C). By 16 October the Collatz track had proved, conditional on a stated verification range, that any hypothetical "first-drop paradox" must contain at least 47 odd runs (Appendix C, entry C-U″).

17.3 The Z10 charter

The problem behind Theorem 1 came from a project charter, not from the author's later charge alone. The Z10 charter was frozen on 16 October 2026 at the author's instruction, "Charter it". Its SHA-256 hash is 4876b05c…fc87d (Appendix A). Its protocol required:

A hash before any run. The charter's hash was sent before any computation, and scripts were seeded.

Two machines. Each computation was run locally and replicated by the author in Colab.

Independent review. Every claimed result had at least one independent reviewer.

Fixed status labels. Each claim was labelled proved; proved on a specified infinite family; conditional; finite verification; refuted with witness; or inconclusive.

The charter had four branches. Its fourth and lowest-priority branch, Z10-C2, was labelled exploratory and carried no falsifier. It asked whether the least positive representatives r_{w_j} of parity words that keep the coefficient at or above 1 must grow like $2^{\theta j}$ for some $\theta > 0$. The idea came from GPT's assessment of the previous round. Claude wrote the charter and an accompanying memo to GPT (Appendix B).

17.4 The theorem-first charge

On 21 October 2026 GPT assessed the Z10 charter (Appendix D). The author then asked which of the project's two problems was more promising for a partial proof, and gave this charge:

> Proceed with the Collatz effort, table the E-S for the time being, work to produce a publishable result on Collatz. One consideration is what can be loosely called the Ramanujan approach: Propose the theorem first, then derive the actual proof post-theorem. Thus call on using as many sub-agents as possible to thread together a working hypothesis that I can review next. The hypothesis is the theorem in all

> respects, lacking a definitive proof. What about this approach, GPT? Use the sub-agents to produce the theorem.

Stating a result before proving it is ordinary mathematical practice. Every proof is a proof of something already stated, and Pólya made "guess first, prove afterward" a principle of mathematical method [38]. What the author took from Srinivasa Ramanujan and G. H. Hardy is a division of labor. Ramanujan's 1913 letter to Hardy stated many results without proof. Hardy and J. E. Littlewood then tested them, proving some and finding others false [31]. The work of proving the claims in Ramanujan's notebooks continued for most of a century [27]. The Ramanujan Machine automated the first half of that division, generating conjectures by computer for later proof [39].

In the present work one party, GPT's sub-agents, committed to an exact statement. Both systems then sought, attacked and corrected proofs, and Lean served as the final arbiter. Statements are often reshaped while a proof is sought [36]. The protocol was designed to prevent that for the main theorem.

17.5 The verification regime

Every substantive claim passed through the same controls.

Freezing. Statements and test ranges were frozen, and their hashes recorded, before the computations that tested them.

Status labels. Each claim carried one of the fixed labels of Section 17.3, and no claim could be promoted on data alone.

Cross-review. Each system reviewed the other's load-bearing steps. Fresh-context sub-agents re-derived key lemmas without seeing the existing proof.

Replication. The author re-ran the decisive computations himself, in his own Colab sessions, a hosted runtime separate from the systems' environments. Matching SHA-256 hashes confirmed that the separate runtimes had produced identical outputs. For this edition he also re-ran checks of the values shown in Figure 1, the constants quoted in Section 16 and the hashes of seven of the files cited in the appendices; all passed (Appendix E).

Formal proof. The main theorems were proved in Lean and rebuilt from source by GitHub Actions under the author's account, with a public log.

A permanent record. Refuted claims and corrections were kept in the record, not deleted (Appendix C).

17.6 How attribution was reconstructed

Attribution in Part III is reconstructed from the project's archived records. These are the transcripts of the exchanges, the charters and memos, the claim ledger, the systems' work packages and sub-agent reports, and the SHA-256 hash of each. On 23 October the author asked GPT to state, from its own records, what it had received before proposing Theorem 1, how the

statement was chosen and when it was frozen. Each answer was to be marked as documented or as recalled. GPT's answer and its evidence package (Appendix E) were then checked against the archive by Claude. Where the records do not settle a question, this paper says so.

17.7 Classification of results

Each result is classified in two ways. By origin, it is *directed* (the outcome of a task the author set), *unrequested* (emerging from directed work although no one had proposed it), or a *limit* (established when the author pressed for an extension). By standing, it is not located in the sources examined, a known fact for which the paper gives a proof, or known. Table 9 applies both classifications.

Table 9. The results of the paper by origin and standing.

Result	**Origin**	**Originated by**	**Standing**
Theorem 1 (Hypothesis M)	Directed	Statement: GPT's sub-agents, under the charge. Proof: GPT's sub-agents. Cross-check: Claude	Not located in the sources examined
Corollary 2 (residue and 2-adic forms)	Directed	GPT, with the statement of Theorem 1	Equivalent form of Theorem 1
Theorem 5 and Lemma 10	Directed	GPT (Theorem 5, following Knight's mechanism); Claude (written proof of Lemma 10)	Known in substance [3, 11, 16]
Theorem 4 (Theorem P)	Unrequested	Claude, from a retired proof route, completed with Theorem 5	Not located in the sources examined
Second proof of Theorem 1	Unrequested	Claude, completed with Theorem 5	Proof not located in the sources examined; complete proof in §6A; archived account in Appendix F
Corollary 3 (ξ irrational)	Unrequested	Claude	Known [4, Corollary 14]; the paper gives a short proof
Lemma 11 (Lemma S)	Unrequested	Claude	Classical property of Sturmian words [16], stated and checked for this use
Propositions 13 and 14	Limit	GPT	Proposition 13 not located earlier; Proposition 14 undetermined
Proposition N1 (proof of Remark 15)	Unrequested	GPT (Completion Assessment); gaps filled after Claude's review	Not located in the sources examined (limited search)
Corollaries N2–N4, Prop. N5, Remark N6	Directed (asymptotic investigation)	Claude, under the completion protocol; reviewed	N4 not located; N2–N3 are reformulations, and the measure/zero-run equivalence behind N3 is in De Jesús [50]; N5's bound follows from Ren [52, Thm 4.1] (Appendix G.5, Rounds 8–10)

Result	Origin	Originated by	Standing
Proposition N7; second proof completed (§6A)	Directed (completion protocol)	Claude, completing its archived route; reviewed	Not located in the sources examined (limited search)

Table 9 classifies origin and standing in the literature; it is not a list of unproved claims. Every result in it has a written proof:

- Corollaries 2 and 3 and Theorem 1 in Section 5;
- Theorem 4 and Lemma 11 in Section 6;
- Theorem 5 and Lemma 10 in Section 4;
- Propositions 13 and 14 in Section 7.

Their verification status is given in Tables 1 and 2. “Known in substance” concerns originality, not proof.

The rows added in v10 use the same classification. The proof of Remark 15 was proposed unasked in GPT’s Completion Assessment. The results of Section 7A came from the investigation the author directed through the completion protocol. The completed second proof answers that protocol’s request to repair and complete the archived route. A limited search located none of them in the sources examined; it was not a complete originality review (Appendix G.6).

17.8 Limitations

The account has four limits.

AI review only. All review so far is by AI systems. No human specialist has read the proofs, and the author cannot referee them himself. Confidence rests on the Lean proof, the independent rebuild, the replications and the cross-system reviews.

No controlled comparison. The record cannot show whether a neutral request would have led to the same theorem.

The freeze is a recorded sequence. The freezing of the statement of Theorem 1 is supported by the recorded sequence of files and logs. It is not supported by an independently timestamped publication of its hash.

Partial visibility. How many agents ran, for how long and on what hardware is only partly visible to the user of an AI system. The author has elsewhere called this condition the Data Center Veil.[1]

[1] W. C. Houze, *The Hyperscale Datacenter Corpse and the Forensic Autopsy Industrial Substrate, Epistemic Ground, and the Burden of Proof for Machine Scholarship*, SSRN, 23 June 2026 (posted 30 June 2026), https://ssrn.com/abstract=6984540, doi:10.2139/ssrn.6984540.

The method also carries a bias. A theorem-first protocol tends to select statements that turn out to be provable, because a target that resists tends to be replaced by one that yields.

17.9 The completion round

On 24 October 2026 GPT assessed v6 at the author's request. Its Completion Assessment identified the shared numbering as a source of confusion and found an exact error in Appendix F. It proposed a proof of the conjecture in Remark 15 and recommended investigating two asymptotic questions within the present paper. GPT and the author then prepared a ten-step research protocol for Claude (Appendix G.1). The protocol fixed the order of work:

1. Establish and freeze the baseline.
2. Check the foundations.
3. Fix the targets and their quantifiers before testing them.
4. Account explicitly for the passage from single bit lengths to the cumulative maximum.
5. Complete the archived second proof before relying on it.
6. Use exact computation with a separately written checker.
7. Require review by a reviewer that had not constructed the proof.
8. Keep proof status, Lean status and originality separate.
9. Return an auditable package.
10. Apply release criteria before changing the paper's claims.

Claude carried out the round in a single session.

- *Baseline.* The baseline files were hashed and the public Lean repository was retrieved.
- *Preregistration.* The two targets were recorded with their quantifiers before the principal computations (preregistration SHA-256 *ef183770…55ec*). One later extension of a test was recorded as a dated amendment and labelled exploratory.

There he writes: "The datacenter veil is that thick shroud woven from industrial excellence, flawless operational metrics, and the sheer, staggering scale of hyperscale infrastructure. It encourages a dangerous epistemic slippage: because the physical machinery operates with near-perfect integrity—routing bits, correcting errors, and cooling silicon without a single dropped packet—we mistakenly assume the semantic output inherits that same perfection. The veil blinds us to the absolute fracture between operational execution and epistemic justification; it hides the reality that a machine can be perfectly engineered to compute what it was built to compute, while remaining completely detached from truth, reference, or warrant."

- *Computation.* Every computation used integer arithmetic. The decisive checks were made by a program written separately from the one that produced the data, and the author replicated the full set in his own Colab session.

- *Review.* Each new proof was sent to a separate Claude sub-agent working with fresh context. The reviewer was told to attempt the statement before reading the proof, to try to break it, and to report every objection with its severity. Corrected texts were sent to a further reviewer. GPT then reviewed the whole draft; its review was not fresh-context, and for N1 it was not independent of construction, since GPT wrote the original proof. Claude answered each of its objections point by point.

The round has the limitations of the earlier work and two more. The reviewers were instances of the same model family that wrote part of the material, so their agreement is weaker evidence than review by a different system or a specialist. And the Lean rebuild could not be run in the working environment. The author therefore ran it on GitHub on 24 October 2026, together with the new proofs of Lean round 1, and it passed (Section 8). The Lean round 2 modules followed the same day (Appendix G.7).

18 Findings

18.1 The charge redirected the work

Immediately before the charge, GPT had ranked branch C2 last among the charter's four branches. It wrote that proving the growth condition "would itself be a major result", and that it "would not place Collatz C2 near the front of a proof-completion queue" (Appendix D). Its plan for Collatz was a computational extension of the range checked by computer.

After the charge, GPT's sub-agents produced that extension in the same turn, but placed the C2 question at the centre of the effort. The branch that both systems had ranked lowest is the one that yielded the theorem. The records show that the theorem emerged after the charge; they cannot show that the charge was necessary for it (Section 17.8).

18.2 Where the statement of Theorem 1 came from

The statement of Theorem 1 had several antecedents, which Figure 7 sets out.

The question. Branch C2 of the charter asked whether the least residues grow exponentially along parity words that keep the coefficient at or above 1.

Why the general question is hard. On 17 October, in the M1 manifold wave the author directed, Claude's worker W2 proved that this question, posed over all such words, is equivalent to an open bound $\tau(n) = O(\log n)$ on the coefficient stopping time.

The object. Among its suggested next steps, W2 proposed studying the exact boundary word $q_t = \lfloor t/\rho \rfloor$, where $\rho = \log_2 3$. This is the critical word of Section 3. W2 had tested Sturmian words only at slopes at least 10^{-3} above the boundary, and this suggestion was never run.

Prior exposure. GPT received W2's reduction on 17 October and discussed it the same day.

The formulation. On 21 October, after the charge, GPT's sub-agents went through four steps in eighteen minutes: the range extension at 13:00 EDT; candidates about bounded coefficients in a corridor $[1, K]$ at 13:08; an explicit exit theorem for general K at 13:15; and, at about 13:18, the choice $K = 3$, for which exactly one parity word survives. Before the principal computation, which ran at 13:18:32 EDT, they froze two things in one preregistration file (SHA-256 25ae7041…98e2): the universal inequality $r_L^2 \geq 2^L$ for every $L \geq 2$, which is the residue form of Theorem 1 (Corollary 2), and a separate computational test of it over $2 \leq L \leq 100{,}000$.

The records do not show which agent first proposed $K = 3$. They do not show whether W2's suggestion was consulted, or why the exponent $1/2$ was chosen. GPT's own audit concludes that the established provenance is "prior exposure to W2's reduction, followed by the recorded formulation and testing of the restricted Hypothesis M" (Appendix E).

Prior exposure establishes that this material was available, not that it was used. One system had the right object on a list of untested leads, and both had the reduction that showed why the general question was hard; whether either entered the formulation is unresolved. What the records do establish is the sequence: after the author's charge, GPT's sub-agents formulated and froze the statement within eighteen minutes, and the proof followed within the afternoon.

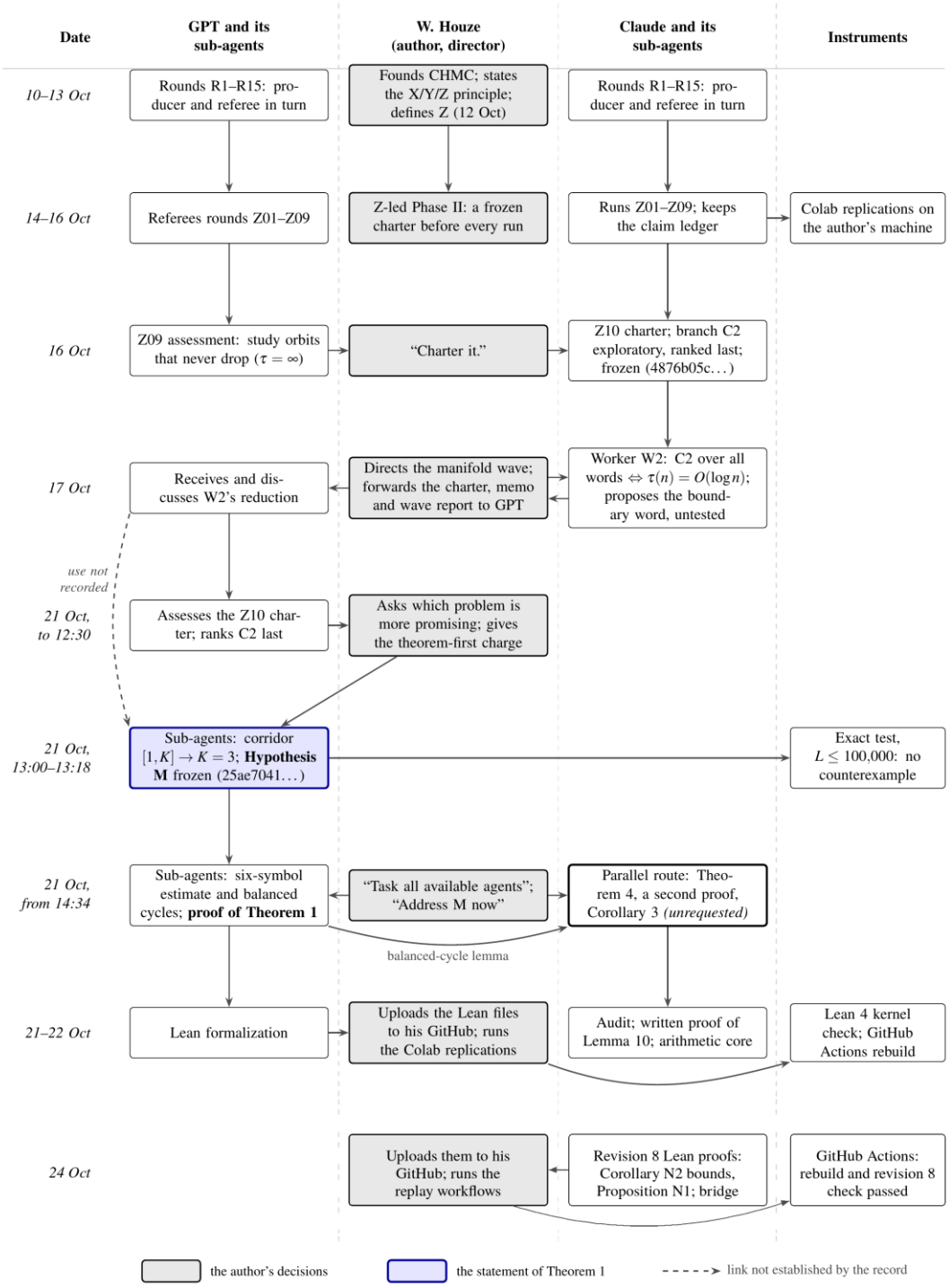


Figure 7. The provenance of Theorem 1, from the founding of the CHMC project to the independent rebuild, with a final row for 24 October 2026, when the author ran the rebuild and the Lean round 1 check on GitHub and both passed. The Lean round 2 modules, written by GPT and checked by Claude, passed on GitHub later that day and are not drawn. Each column is a participant, and time runs downward. Shaded boxes are the author's decisions; every transfer of material between GPT and Claude passed through him. The blue box is the statement of

Theorem 1 (Hypothesis M) as frozen before its principal computation. The dashed arrow marks a link the records do not establish: GPT had received Claude's W2 reduction four days earlier, but whether its sub-agents used it, or W2's suggestion of the boundary word, is not recorded. The heavy-bordered box in the Claude column marks the unrequested results of Section 18.4. Hashes identify the frozen files (Appendices A and E).

18.3 The statement did not move

The universal inequality frozen at 13:18 on 21 October is, in mathematical content, the statement proved in the next round. The wording differs: Theorem 1 states the bound on $\sigma_3(n)$, the preregistration states the residue inequality, and Corollary 2 records their equivalence. GPT's reply that evening states the universal hypothesis in the form of Theorem 1 (Appendix D.4), and the theorem formalized in Lean states the same bound (Section 8). The finite range frozen with it was a test of the statement, not the statement itself.

A circular first step. The first proof obligation GPT proposed, a "binary-lift lemma", turned out to be equivalent to the statement itself. It was recorded as no progress.

The proof that succeeded. GPT's next round began at 14:34 EDT. Agent M1 found the six-symbol coefficient estimate. Agent M2 found the exclusion of balanced cycles, following Knight's mechanism [11]. Together they gave the proof.

Checking. Claude checked the four decisive implications and wrote a self-contained proof of Lemma 10. GPT produced the Lean formalization, and Claude checked a separately written arithmetic core. On 22 October GitHub Actions rebuilt the formal proof under the author's account, with a public log.

18.4 Results nobody asked for

Working in parallel on the same frozen statement, Claude followed a different route. It split each potential counterexample according to where its bit length lies relative to the continued-fraction denominators of $\log_3 2$. Claude at first misjudged the remaining case as being as hard as the general Collatz cycle problem, and that route was retired.

Once GPT's balanced-cycle argument was available, the retired route yielded results that no one had proposed.

Theorem 4. A sharper bound of $k + q_m + 3$ steps for seeds whose bit length k lies well between consecutive convergent denominators.

A second proof of Theorem 1. Claude's archived route is now completed in Section 6A. It uses convergent periods (Lemma 11), a uniform size estimate near convergent denominators (Proposition N7), Theorem 4 and a finite base. It shares with the principal proof the setting, the parity-to-residue correspondence (Lemma 6), balance (Lemma 7(1)), the affine estimate (Lemma 9) and Theorem 5; it avoids Lemma 8 and the factor-complexity pigeonhole argument. The account archived on 21 October is reproduced in Appendix F, with an editorial correction.

Corollary 3 and Lemma 11. The same work gave the short proof of Corollary 3 and the use of Lemma 11.

Theorem 4 and the second proof were not located in the sources examined. Corollary 3 is a known fact [4, Corollary 14], for which the paper gives a short proof, and Lemma 11 is a classical property. In this sense the theorem-first protocol worked in both directions: the fixed target organized the search, and the search produced more than the target.

18.5 Limits found when the author pressed

The author then asked for the approach to be extended to the whole conjecture. The systems did not extend it; instead they established its limits.

Wider corridors fail. Exact counts showed that every corridor wider than $[1,3]$ admits exponentially many parity words, so the counting step of the proof fails (Propositions 13 and 14).

The extension was the conjecture itself. GPT's proposed three-step program for the full conjecture turned out to restate the stopping-time form of the conjecture. GPT withdrew it after the $5x + 1$ test of Section 2.

The verification regime corrected the director as well as the machines.

18.6 Errors caught and kept

Both systems made errors that the regime caught.

GPT. Its binary-lift lemma was circular, and a step count in its linear exit theorem had to be corrected.

Claude. It misjudged the cycle case. It overstated an originality claim about the formalization of the balanced-cycle theorem. It described a bound informally as "usually far below" Theorem 1 until GPT corrected the wording.

The claim ledger records refuted claims from the earlier rounds as well (Appendix C). None was deleted.

18.7 What the record does not show

The record does not establish whether the decisive specialization exemplifies the frame-changing leap the author calls the Z factor. It identifies antecedent questions and suggestions, GPT's prior exposure to W2's reduction, and the later formulation and testing of Hypothesis M. It does not establish whether the agents used W2's boundary-word suggestion, who first selected the corridor ceiling of 3, or why the exponent $1/2$ was chosen. The contribution of each participant should therefore be stated through documented actions, with the unresolved intellectual connections left open.

Nor does the record show whether a neutral request would have produced the same theorem. A controlled comparison is proposed in Section 19.

18.8 What the Maestro role establishes

The record supports a limited but substantive finding: a scholar without specialist mathematical training directed a research process that produced an explicit theorem, a written proof, a formal proof of that theorem, and companion results. The supporting computations were reproduced, and errors in intermediate arguments were identified and retained in the record. The theorem provides a precise bound on how long a positive integer can follow the critical parity word. Its originality remains subject to the literature qualifications stated elsewhere in this paper.

The author's contribution is identifiable in the choice and redirection of the inquiry, the insistence on explicit claims, the coordination of adversarial review, and the requirements for replication and preservation of evidence. Theorem 4 and the alternative proof also show that work on a specified target can yield useful additional results. Here "unrequested" means that those particular results were not separately specified in the task; they arose within the broader research effort the author directed.

This is evidence of feasibility in one case. The record does not establish that any intelligent layperson could obtain a comparable result, that the same process would succeed in another field, or that the theorem-first charge was necessary for success. It also does not resolve the unrecorded causal connection between earlier suggestions and the eventual specialization. The demonstrated outcome is that non-specialist direction, combined with explicit verification procedures, can support substantive mathematical work whose claims are available for further assessment.

18.9 Findings of the completion round

Results produced. Four results came out of the round.

- *A completed conjecture.* The proof of Remark 15 proposed by GPT survived adversarial review after two substantive gaps were filled. It is now Proposition N1.
- *A completed second proof.* The archived route was found to rest on a false endpoint and to omit essential steps. It was rebuilt as Section 6A, with a new uniform estimate near convergent denominators (Proposition N7), and it now stands as a complete second proof of Theorem 1.
- *Reductions.* The asymptotic questions were reduced exactly to the binary digits of ξ.
- *Partial answers.* Two partial answers were proved: an interval bound and a lim inf bound.

What stopped the investigation. Neither asymptotic target was reached. Exact computation located where the present method appears to stop: just above each convergent denominator in the computed range, the best certificate of the kind used in both proofs costs nearly $2k$ steps (Section 7A.5). That this limit holds in general is a candidate result, not yet proved.

Errors caught and kept. The verification regime again caught errors on both sides.

- *GPT's assessment* found the endpoint error in the archived record.

- *Claude's reviewers* found that a recurrence had been cited with the wrong index, and that two steps of the Remark 15 proof had been asserted rather than proved.

- *GPT's review of the completion round's draft* found three things. The added argument for the failure of N1's formula above $9/2$ had a gap, and GPT supplied a replacement. A concluding sentence of the second proof was false at $n = 3$. And several summaries still overstated, or mixed the cumulative maximum with a single bit length. It also found that one computational test had run on a different domain and objective from those preregistered, without disclosure at the time.

- *Claude's own checks* found a defect in its first checker (it computed continued-fraction numerators instead of denominators), and an editing route that would have silently deleted every equation from the clean copy of the paper.

All are recorded in Appendix G.

What the round shows about the method. The round is a second instance of the division of labor described in Section 18.8. The author asked for completion of the paper and for an investigation of its open questions, not for any particular theorem. The protocol fixed the evidentiary standard, and the outcome is reported as it came out: two completions, new partial results and a measured but unproved obstruction, but no resolution of either target.

Lean round 2. The same day, GPT wrote Lean proofs for much of the completion round's work. Its report separated results proved in Lean from those proved only from stated assumptions, and from items not attempted, and Claude's check of each statement against the paper required no correction to that classification. The author ran the result on GitHub. The paper reports the same three-way distinction (Section 8).

Conclusion, Lexicon and References

19 Conclusion

The thesis revisited

A scholar trained in the humanities, not in mathematics, directed AI systems to a theorem, not located in the literature examined, that is now proved in Lean and independently rebuilt. The directed result, Theorem 1, followed a charter he approved and a charge he gave. The unrequested results, chiefly Theorem 4 and a second proof of Theorem 1, came from the systems' own work toward it. Neither kind of result depends on trusting the director's mathematical judgment, which he does not claim, or on the systems' assurances. They depend on the regime of frozen statements, cross-system review, replication and formal proof under which they were produced. Its scope is exact and narrow: it bounds how long an orbit can follow one boundary parity pattern, which lies near an important fork in the tree of results (Section 16, Figure 6).

Version 10 extends the same regime to a completion round directed by a written protocol. The round completed the paper's open proof and conjecture, and investigated, without resolving, the two asymptotic questions that the main theorem raises.

The mathematical findings

Figures 8 and 9 summarize the two main findings for both audiences: how the proved deadlines compare with what numbers actually do, and why the corridor [1,3] is the one where the method works.

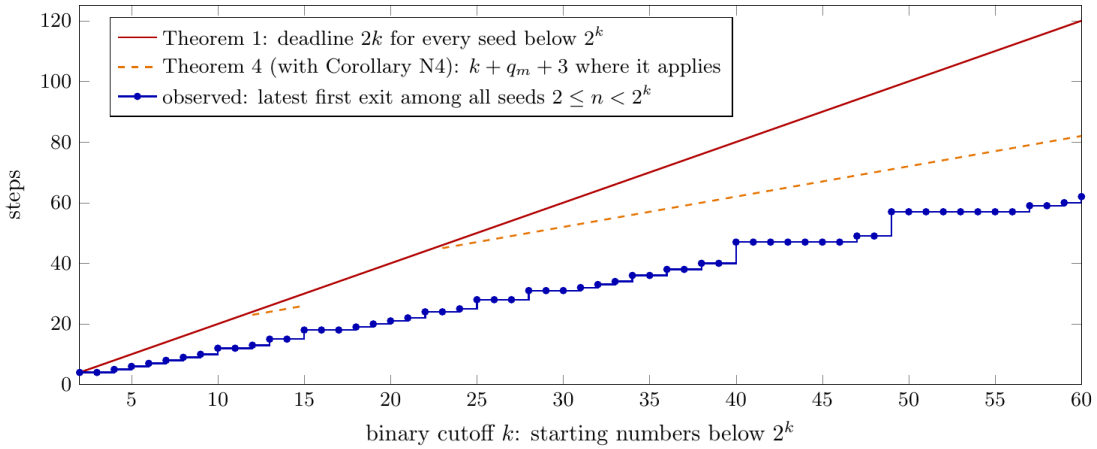


Figure 8. Proved bounds and observed behaviour. For mathematicians: the observed curve is $max\{\sigma_3(n): 2 \le n < 2^k\}$*, computed from the exact residues* r_L *via Lemma 6 and confirmed by direct search for* $k \le 21$*. Theorem 4 is drawn on the ranges* $12 \le k \le 15$ *(*$q_m = 8$*) and* $23 \le k \le 60$ *(*$q_m = 19$*). Over this range* $k + 1 \le max\sigma_3 \le k + 8$*. The lower bound* $k + 1$ *holds for every* $k \ge 2$*, since* $n = r_k$ *satisfies* $2 \le r_k < 2^k$ *and* $\sigma_3(r_k) \ge k + 1$*. Whether the maximum is* $k + o(k)$ *remains open; this would give an asymptotic leading constant of 1. The weaker*

question of whether the leading constant 2 of Theorem 1 can be reduced also remains open. For the general reader: the top line is the guaranteed exit deadline. For each cutoff k, the lower curve shows the latest first exit among all starting numbers from 2 with at most k binary digits. The curve is computed over the displayed finite range. At k = 2 the number 3 attains the deadline. (The lower curve is a computer observation, not a proof.)

New in v10. Section 7A proves $k+1 \le M(k) \le 2k$ for every $k \ge 2$, where $M(k)$ is the maximum shown, and $M(k) = k+1+z_k$, where z_k is the length of the run of zeros in the binary expansion of ξ beginning at digit k. It also proves $M(k) \le k + q_m + 3$ for $q_m + 4 \le k \le q_{m+1} - 4$, which justifies drawing the Theorem 4 line for the cumulative maximum on $23 \le k \le 60$, and $\liminf M(k)/k \le \sqrt{2}$. Exact computation for $2 \le k < 20000$ gives $M(k) \le k + 16$ (Figure 3). Whether $M(k) = k + o(k)$ for all k, and whether the constant 2 can be lowered for all k, remain open.

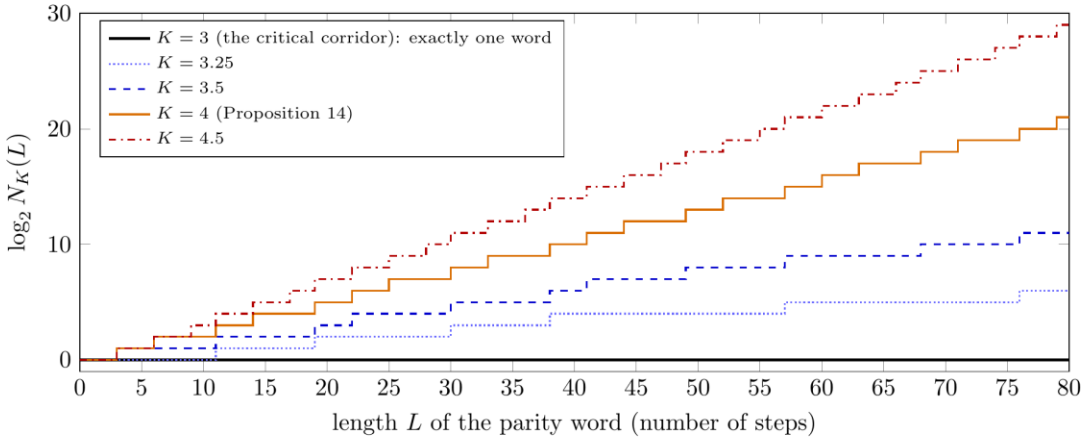


Figure 9. Why the corridor $[1,3]$ *is critical. For mathematicians: the number* $N_K(L)$ *of parity words of length L whose prefix coefficients all lie in* $[1,K]$*, on a* $\log_2$ *scale, computed exactly. For* $K = 3$*,* $N_3(L) = 1$ *(Lemma 6); for every* $K > 3$ *the growth is exponential (Proposition 13), and* $K = 4$ *follows the formula of Proposition 14. For the general reader: how many rhythms can survive in hallways with higher ceilings. With a ceiling of 3 there is exactly one. Raise the ceiling and the number grows exponentially as the rhythm gets longer, which removes the low-variety estimate that the counting step of this proof uses.*

For mathematicians

Theorem 1 gives an explicit, uniform exit bound $\sigma_3(n) \le \lfloor \log_2 n^2 \rfloor + 1$ for the critical coefficient corridor $[1,3]$ of the shortcut Collatz map, with equality at $n = 3$. Equivalently, $r_L^2 \ge 2^L$ for all $L \ge 2$, and $|n - \xi|_2 \ge n^{-2}$ for the 2-adic point ξ with critical parity vector. It is an effective sharpening of the qualitative obstruction that follows from Dubickas's complexity theorem [5]. Three ingredients make it effective: the height bound $x_t = C_t n + E_t$ with $C_t \le 3$, the six-symbol cell estimate, and the balanced-cycle classification (Knight [11]), which removes the divergence hypothesis. Theorem 4 improves the bound to $k + q_m + 3$ between convergent denominators of $\log_{32}$. Propositions 13 and 14 show that the low-complexity estimate on which the pigeonhole step relies fails for every corridor $[1,K]$ with $K > 3$. Equality at $n = 3$ does not

show that the constant 2 is asymptotically optimal. The maximum of $\sigma_3(n)$ over $2 \le n < 2^k$ is at least $k+1$ for $k \ge 2$, and Figure 8 shows it within $k+8$ for $k \le 60$. Whether it is $k + o(k)$, or even whether the leading constant 2 can be reduced, remains open.

New in v10. Section 6A gives a complete second proof of Theorem 1, which uses neither the six-symbol cells nor the pigeonhole argument. Proposition N1 proves that δ_K is the exact growth rate of $N_K(L)$ for $3 < K \le 9/2$. Section 7A reduces the asymptotic questions to the binary digits of ξ: $M(k) = k + 1 + z_k$, where z_k is the length of the run of zeros beginning at digit k. It proves the following:

- $M(k) \le k + q_m + 3$ on the intervals $q_m + 4 \le k \le q_{m+1} - 4$;
- $\liminf_k M(k)/k \le \sqrt{2}$;
- Theorems 1 and 4 together give no uniform constant below 2.

Both asymptotic targets remain open for all k. Section 7A.5 gives computational evidence, with a candidate explanation that is not proved, that the single-coincidence method stops at coefficient 2. The next steps are to prove or refute that candidate obstruction and to investigate stronger arguments for Targets A and B. Neither target needs to be resolved for the paper's narrower proved claims to be complete.

The main theorem and the balanced-cycle theorem are proved in Lean 4 using only the standard axioms, and the formal development was rebuilt independently. Lean rounds 1 and 2 have since proved in Lean Corollary 2, Proposition 14, N1's exact count, N2, N3 and the finite base of the second proof, and the author checked them on GitHub on 24 October 2026. Natural next steps are to prove in Lean Lemma 11, and with it Theorem 4, N4 and the second proof, then Corollary 3, Proposition 13 and the remaining results of the completion round, to complete the literature comparisons listed in the project's originality ledger, and to obtain a specialist's reading. The result does not address convergence, divergence or general cycles. The $5x + 1$ example in Section 2 shows that the exit bound alone cannot force descent.

For the general reader

The Collatz problem is still unsolved, and this paper does not change that. What it adds is one exact, checked fact. Inside the narrow "hallway" between a score of 1 and a score of 3, only one rhythm of steps can last forever. Every starting number from 2 upward falls out of that rhythm by a deadline you can work out with schoolbook arithmetic. The number 3 meets its deadline exactly, so the rule cannot be cut by even one step for every number, although larger numbers may have better deadlines. The result is small but solid. It is a twig at a fork of the trunk, not the trunk itself. It is supported by a written proof, a Lean proof of the main theorem and an independent rebuild. The remaining test is the one that matters most in mathematics: a careful reading by human experts. A few earlier papers also still have to be read in full and compared before the claim that the result is new can be made with more confidence.

Version 10 adds three things. It finishes a second proof of the main theorem by a different route. It proves exactly how many rhythms survive when the ceiling is raised a little. And it shows that,

among all starting numbers from 2 with at most k binary digits, the latest first exit from the hallway occurs at a step between $k + 1$ and $2k$. Whether that latest exit stays close to k for every k is still open; computer checks suggest, without proof, where the paper's present method stops.

For the study of human–AI collaboration

The Maestro thread identifies a second contribution alongside the mathematical result. In this case, a person trained outside mathematics directed frontier LLM systems toward a precise theorem and organized a sequence of checks that made the resulting work assessable. The record supports the feasibility of this division of labor: the human directed the inquiry and required evidence, the systems developed and criticized arguments, and formal and computational tools checked specified aspects of the result.

The completion round adds a second observation. When the human director asks for completion and investigation rather than for a particular theorem, and writes the evidentiary standard into a protocol, the systems' output can include negative and partial results reported as such: a measured but unproved obstruction, a lim inf bound in place of the desired uniform bound, an archived proof found incomplete and then completed. That is the behavior a research process needs if it is to be trusted when it does report success.

Such a process may broaden participation in research on difficult mathematical questions. Its value depends on the quality of the claims and their verification. The present achievement is a specialized result concerning the critical Collatz parity word; the Collatz conjecture remains open. The principal bounds appear contributory on the literature reviewed, but the remaining comparisons and a human specialist's reading are still needed to assess their originality and significance, and several results added in v10 are not yet proved in Lean, while Theorem 4 is only partly proved in Lean: the step from Lemma 11 is proved there, but Lemma 11 is not. Those qualifications delimit the mathematical claim while preserving the methodological finding that productive research direction can be exercised by someone who cannot personally supply or referee the proof.

Further study should examine how repeatable this outcome is across conductors, systems, problems, and disciplines. Comparisons between neutral and theorem-first instructions, with the same initial information and documented resource limits, would help distinguish the effects of direction from the capabilities of the systems and the suitability of the chosen target. The present case supplies an auditable example from which such investigations can proceed.

20 Lexicon

Each entry gives a precise meaning for mathematicians and a plain meaning for the general reader. Section numbers point to where a term is used.

Term	For mathematicians	For the general reader
The game		
Map	A function applied repeatedly; here $T: \mathbb{N} \to \mathbb{N}$.	The rule that turns one number into the next, like a recipe applied over and over.

Term	For mathematicians	For the general reader
Shortcut Collatz map	$T(n) = n/2$ for even n, $T(n) = (3n+1)/2$ for odd n (Sec. 2).	Halve an even number; triple an odd number, add 1 and halve.
Seed	The initial value $n = x_0$ of an orbit.	The starting number.
Step	One application of T; j steps take x_0 to x_j.	One use of the rule.
Orbit, trajectory	The sequence $x_t = T^t(n)$, $t \geq 0$.	The whole chain of numbers a seed produces, step after step.
Parity	$x \bmod 2$.	Whether a number is odd or even.
Parity word, parity vector	The binary sequence of parities $x_0 \bmod 2$, $x_1 \bmod 2$, ... (finite or infinite).	The orbit written as a string of odds (1) and evens (0).
Word, block, prefix	A finite or infinite sequence over $\{0,1\}$; a block (factor) is a contiguous subword; a prefix is an initial block.	A string of 1s and 0s; a stretch cut out of it; its opening stretch. Nothing to do with language.
$5x+1$ map	$T_5(n) = n/2$ (even), $(5n+1)/2$ (odd); it has the cycle $13 \to 33 \to \cdots \to 13$.	A sibling game with 5 in place of 3. Some of its numbers loop forever, so it tests which ideas could and could not prove Collatz.
The corridor		
Coefficient C_j	$C_j(n) = 3^{a_j(n)}/2^j$, with a_j the number of odd terms among $x_0, \ldots, x_{j-1}$; the multiplicative part of T^j.	The running score: how much the pushes ($\times 1.5$) and pulls ($\times 1/2$) have multiplied the number, ignoring the "+1".
Critical corridor	The condition $1 \leq C_j \leq 3$.	The hallway, with a floor of 1 and a ceiling of 3.
$\alpha = \log_{32}$	≈ 0.6309; irrational; the frequency of odd terms in the critical word.	The fixed proportion of odd steps in the drumbeat, about 63%. Because it is not a fraction, the drumbeat never repeats exactly.
Critical word **e**	$e_j = \lceil j\alpha \rceil - \lceil (j-1)\alpha \rceil$; the unique infinite word keeping $C_j \in [1,3]$ (Lemma 6).	The drumbeat: the one rhythm that stays in the hallway forever.
Mechanical, Sturmian, Christoffel word	For $0 < \beta < 1$, the lower and upper mechanical words $\lfloor j\beta + \rho \rfloor - \lfloor (j-1)\beta + \rho \rfloor$ and $\lceil j\beta + \rho \rceil - \lceil (j-1)\beta + \rho \rceil$; Sturmian when β is irrational; Christoffel words are the length-p (lower or upper) words of rational slope q/p and intercept 0 [16].	The family of patterns spread as evenly as possible, like fence posts at a fixed spacing.
Exit time $\sigma_3(n)$	$\min\{j \geq 1 : C_j(n) \notin [1,3]\}$, with $\min \varnothing = \infty$.	The step at which a number leaves the hallway, through the floor or the ceiling.
Bit length k	The k with $2^{k-1} \leq n < 2^k$.	The number of digits in the number's binary form.

Term	For mathematicians	For the general reader
Bound; sharp, attained	An inequality valid for all n in range; attained if equality holds for some n (here $n = 3$).	A guaranteed limit; "sharp" means some case lands exactly on it, so it cannot be lowered for every case at once. It need not be the best possible for large numbers.
Convergent denominators q_m	Denominators of the continued-fraction convergents of α: 1, 2, 3, 8, 19, 65, 84, 485, 1054, …	The sizes of the best simple-fraction approximations to 63%. The sharper Theorem 4 works for numbers whose binary length lies well between them.
Tools of the proof		
Residue r_L	The least $r \in [0, 2^L)$ whose first L parities are $e_1 \cdots e_L$.	The smallest nonnegative whole number that follows the drumbeat for L steps; for $L \geq 2$ it is at least 2, so Theorem 1 applies to it.
2-adic integer ξ	$\xi = \lim r_L \in \mathbb{Z}_2$, the 2-adic point with parity vector $\mathbf{e}$; $\lvert\cdot\rvert_2$ is the 2-adic absolute value.	A number in the 2-adic number system, written with infinitely many binary digits extending to the left. It is the one such number that follows the drumbeat forever. No ordinary whole number can.
2-adic distance	For $x \neq y$ in $\mathbb{Z}_2$, $\lvert x - y\rvert_2 = 2^{-v_2(x-y)}$; $\lvert 0\rvert_2 = 0$.	Closeness measured by how many final binary digits two numbers share. Corollary 2 says no whole number $n \geq 2$ shares more than about twice as many final binary digits with ξ as n itself has.
Irrational (2-adic)	$\xi \notin \mathbb{Z}_{(2)} = \{a/b : b \text{ odd}\}$ (Corollary 3).	Not expressible as a fraction of the relevant kind.
Factor complexity	$p_w(k)$, the number of distinct blocks of length k in w; $p_{\mathbf{e}}(k) \leq k + 1$ (Lemma 7).	How many different stretches of a given length occur in a pattern. The drumbeat has very few.
Pigeonhole principle	If $m + 1$ objects lie in m classes, two share a class.	With more pigeons than holes, two pigeons share a hole.
Balanced word	Any two blocks of equal length differ in weight by at most 1; for a cyclic word (cyclically balanced), blocks may wrap around the end.	A pattern whose odd steps are spread as evenly as possible.
$N_K(L)$	The number of words of length L with all prefix coefficients in $[1, K]$ (Sec. 7).	How many rhythms survive in a hallway with ceiling K. For $K = 3$ there is one; for any higher ceiling the count grows exponentially.
Maximum exit time M(k)	M(k) = max{σ₃(n) : 2 ≤ n < 2^k}, the latest first exit step among seeds below 2^k; k + 1 ≤ M(k) ≤ 2k (Sec. 7A).	Among all starting numbers from 2 with at most k binary digits, the latest step at which one leaves the hallway.
Zero run z_k	The number of consecutive zero binary digits of ξ starting at digit k; M(k) = k + 1 + z_k (Cor. N3).	How many extra steps beyond k the best of these numbers keeps step: the latest exit comes at step k + z_k + 1.

Term	For mathematicians	For the general reader
Choice time	A time j ≥ 2 with {jα} ≥ 1 − β, at which two odd-step counts are admissible in [1, K] (Prop. N1).	A step at which a rhythm in a slightly wider hallway may go either way.
Targets A and B	A: M(k) ≤ (2 − ε)k + C for all k. B: M(k) = k + o(k). Both open (Sec. 7A).	Two unanswered questions: can the main deadline be cut by a fixed fraction, and is the true deadline about k?
Single-coincidence certificate	Positions 0 ≤ i < j with equal ℓ-blocks of e and |C_i − C_j|·2^b + j < 2^ℓ, which bound σ₃ for seeds below 2^b (Sec. 7A.5).	The one-match argument used in the proofs. In the tested ranges its best certificates approach twice the bit length near certain sizes; a general limit is conjectured, not proved.
Preregistration	Targets, domains and tests recorded and hashed before the tests are run (App. G).	Writing down what will be tested before testing it, so that results cannot be quietly reinterpreted.
Cycles and the conjecture		
Cycle; trivial cycle	A finite set $\{y, T(y), \dots, T^{p-1}(y)\}$ with $T^p(y) = y$; the trivial cycle is $1 \to 2 \to 1$.	A loop that repeats forever; the familiar loop 1, 2, 1, 2.
Balanced cycle	A cycle whose parity word is cyclically balanced; only $1 \to 2 \to 1$ (Theorem 5).	A loop with evenly spread odd steps. The only one is 1, 2.
Divergent orbit	An orbit with $x_t \to \infty$.	A runaway number that climbs forever. Not ruled out by anyone.
Counterexample	A seed whose orbit never reaches 1 (its orbit diverges or enters a nontrivial cycle). Theorem 1 applies from every state of such an orbit.	A single number that would break the conjecture. The theorem's deadline would still apply to it at every step.
Descent, convergence	$x_t < n$ for some t; $x_t = 1$ for some t.	Dropping below the start; reaching 1. This paper proves neither.
Collatz conjecture	Every orbit of T on positive integers reaches 1.	The unsolved claim that every starting number reaches 1.
Kinds of statement		
Hypothesis	A statement proposed for proof or refutation; also, an assumption of a theorem. "Hypothesis M" is the project name of Theorem 1 before it was proved.	A claim put forward to be tested.
Conjecture	A statement believed but unproved (e.g. the Collatz conjecture; Rozier's "Lower Bound Hypothesis" is a conjecture in this sense).	A claim many believe but no one has proved.
Theorem	A proved statement of principal importance.	A main proved result.
Lemma	A proved auxiliary statement used in proving a theorem.	A stepping stone, like scaffolding.

Term	For mathematicians	For the general reader
Corollary	A statement that follows quickly from a theorem or from its proof.	A bonus result that comes almost for free.
Proposition	A proved statement of secondary weight.	A lesser proved result.
Remark	A comment, often recording a numerical check or an unproved observation.	A side note.
Verification and review		
Written proof	A proof in ordinary mathematical prose, checked by reading.	The traditional argument on paper, for people to check.
Formal verification	Machine checking of a proof against a formal statement in a proof assistant.	Having a computer proof checker check a proof, instead of relying on human reading alone.
Proved in Lean; partly proved in Lean	Proved in Lean: Lean has checked a complete proof of the result as stated, using only the standard axioms. Partly proved in Lean: Lean has checked a proof from an explicit assumption that is not yet proved in Lean (Theorem 4, Corollary N4, Proposition N5; Sec. 8).	Proved in Lean: the computer checker accepted every step of the proof. Partly proved in Lean: it accepted the later steps, but one earlier step is still taken on trust.
Lean rounds 1 and 2	The two rounds of 24 October 2026 in which results of the completion round were proved in Lean (Sec. 8, App. G); the repository calls them revision 8 and revision 9.	The two sessions in which more of the new results were checked by the computer proof checker.
Dependency check	A Lean program that traverses a proof and its statement and fails if a named earlier result occurs in them (Rev9IndependenceAudit.lean).	A computer check that a proof does not secretly rely on the result it is meant to confirm.
Lean 4; Mathlib	An interactive proof assistant with a small trusted kernel [18]; its community mathematics library [17].	A computer proof checker that checks formal deductions from explicitly stated foundational assumptions, and its shared library of checked mathematics.
Seed form	The integer-only statement of Theorem 1 formalized in *MStatement.lean* (Sec. 8).	The version of the theorem stated purely with whole numbers, which the checker proved.
Standard axioms	*propext*, *Classical.choice*, *Quot.sound*, reported by *#print axioms*.	The basic starting assumptions of mathematics. Nothing extra was assumed.
sorry, *native_decide*	Lean constructs that skip or outsource kernel checking; absent from the development.	Shortcuts that would weaken the check. None was used.

Term	For mathematicians	For the general reader
Independent rebuild	Recompilation from source on a GitHub-hosted runner, with hash, build, statement and axiom checks.	Re-running the whole check on fresh hosted computers, to show the recorded build is reproducible.
Hash (SHA-256)	A cryptographic digest identifying a file's exact contents.	A digital fingerprint: matching fingerprints are extremely strong evidence that two files are identical.
Exact checker; certificate	Exact-arithmetic checking scripts (CHECKS 1–9); the computed record $r_L^2 \geq 2^L$ for $2 \leq L \leq 10^6$.	Programs that test the claims on huge numbers of cases without rounding, and their recorded results. Evidence, not proof.
Theorem-first protocol	Fixing the target statement before seeking a proof.	Deciding exactly what to prove before trying, so the goalposts cannot move.
Internal, adversarial, originality review	Review by the AI systems that did the work; review aimed at refutation; review for earlier publication.	Checking by the helpers themselves; checking by someone told to break it; a search for earlier publication.
Originality ledger	The project's source-by-source record of the closest prior statements and the verdict for each claim.	The project's list of what was already known and what appears new.
Peer review	Refereeing by independent human specialists.	Checking by human experts. Not yet done for this work.
Preprint	A paper made public before peer review.	An early public version, open to comment before formal publication.
SSRN; arXiv	Public preprint servers: SSRN (multidisciplinary) and arXiv (mathematics and the sciences).	The public websites where researchers post papers before journal publication.
Charter	A research plan frozen, and its hash recorded, before any computation it governs (Sec. 17.3).	A written plan locked before the work starts, so that the goals cannot be changed afterwards.
Sub-agent	A subordinate instance of an AI system, given a separate task and reporting to a coordinating thread (Sec. 17.1).	A helper copy of the AI that works on one part of the job and reports back.
X/Y/Z principle; Z factor	The author's transfer test: name the hypothesis that differs where a method works (X) and where it fails (Y). Z is the frame-changing intuitive step (Sec. 17.2).	Why does a method work here but not there? And Z: the leap of insight that changes how a problem is seen.
Directed and unrequested results	Results of a task the author set; results that emerged from that work although no one had proposed them (Sec. 17.7).	What the helpers were asked to find; what they found without being asked.
Provenance	The documented origin of a result and the record of who did what (Sec. 17.6).	Where a result came from, and who did what, as the records show.

Term	For mathematicians	For the general reader
Maestro	The human research director who sets objectives, coordinates systems, requires verification, preserves the record, and takes responsibility for reporting the work; specialist expertise in the target mathematics is not assumed (Sec. 17.1, 18.8).	The person who directs the research and makes sure its claims are checked.

Statement on the use of AI systems

This work was carried out by the author directing two AI systems, GPT (OpenAI) and Claude (Anthropic), each running sub-agents. The work proceeded under the project's Z10 charter (Appendix A) and then a theorem-first charge (Section 17.4). GPT's sub-agents formulated the statement of Theorem 1, after prior exposure to related work by a Claude sub-agent (Section 18.2), and froze it before its principal computation. GPT gave the main line of its proof, including Theorem 5, and Propositions 13 and 14, and produced the Lean formalization. Claude checked the proof and wrote the self-contained proof of Lemma 10, the exact checking scripts and a separately written arithmetic core. Claude also originated Theorem 4, Corollary 3, the use of Lemma 11 and a second proof of Theorem 1 (Section 18.4). Several proposed intermediate steps were found to be equivalent to the target, or false, and were discarded. Both systems carried out literature searches. The formal development was rebuilt on independent hardware. In a first stage of external review, Grok (xAI) reviewed the mathematics and the Lean correspondence adversarially and ran the exact checker, and Gemini (Google) reviewed originality and prior art. GPT drafted, at the author's request, the passages on the Maestro role in the abstract and in Sections 1.1, 17.1, 18.8 and 19, which Claude reviewed and integrated. Claude drafted the two-audience edition, including Parts II and III, the lexicon, Figure 1 and Figures 6 and 7, at the author's direction; Figure 1 follows a display produced by Google Search. GPT reviewed the earlier two-audience edition, and the changes it called for were made where Claude concurred; Grok and Gemini then reviewed the revised edition, in the same roles as before. GPT then reviewed this edition, including Part III and the appendices, and its further changes were again made where Claude concurred; Grok and Gemini have not yet reviewed Part III and the appendices. The author directed the work, ran the replications, reviewed the results and the records, and takes full responsibility for the content. He did not write the proofs and is not qualified to referee them; Part III explains what the confidence in them rests on. The complete record, including failed attempts, is available on request; the record of the second proof of Theorem 1 is reproduced in Appendix F.

Completion round (24 October 2026). GPT prepared a completion assessment of v6, which supplied the proof of Proposition N1 and identified the endpoint error in Appendix F. GPT and the author then prepared a research protocol for Claude. Under that protocol, Claude:

- made the changes of that round;
- completed the second proof of Theorem 1 (Section 6A, Proposition N7) and constructed the results of Section 7A;

- wrote and ran the exact-arithmetic checks with a separately written checker;
- submitted the new proofs to review by Claude sub-agents working with fresh context, which first attempted the statements without the proofs. Coverage differed by result, and Appendix G.5 records it.

GPT then reviewed the whole draft adversarially, and later made a focused check of the corrections. GPT reviewed N1's corrections as its original author. GPT supplied the replacement argument for the failure of N1's formula above $9/2$ and the $\sqrt{2}$ form of N5, which Claude checked. Grok then reviewed those two arguments, the second proof of Theorem 1 with Proposition N7, and Corollaries N2–N4. The corrections are in v7.1 to v7.3.

The author replicated the computations in his own Colab session. All of this review is AI review; no human specialist has read the new material, and Gemini's originality review of the completion round is not counted as an originality determination, for the reasons in Appendix G.5. Appendix G gives the record.

Lean round 2 (24 October 2026). GPT wrote the thirteen Lean modules of Lean round 2 and a table matching each to its paper result, under a written brief. Claude checked the modules' hashes, constructs and statements against the paper, wrote the round 2 workflow and its gate, and made the changes in this version (v10). The author uploaded the modules and the workflow to his repository and ran them on GitHub.

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Appendices

Appendix A. The Z10 charter (16 October 2026)

The charter below is reproduced verbatim from its Markdown source file, *Z10/CHARTER.md*. Its SHA-256 hash is 4876b05c51ce3f4212ccb3a52e2f6a98b92b8be2443b798785158730f64fc87d. It is the hash sent before any Z10 computation and quoted in the memo of Appendix B. GPT verified the same hash independently on 21 October, and Claude confirmed it again on 23 October. Branch Z10-C2, at the end, is the branch from which Theorem 1 descends (Section 18.2).

Z10 charter — four experiments (frozen 16 Oct 2026, 19:5x EDT; no runs yet)
Source: GPT Z09 assessment (19:47), with Claude's adjustments (19:49 reply). Directed by Houze: "Charter it."
Protocol (unchanged):
- Hash sent before any run; seeded scripts.
- Local run, plus a Houze Colab replication.
- At least one independent reviewer per claimed result.
- Fixed status labels: proved / proved on a specified infinite family / conditional / finite verification / refuted with witness / inconclusive.

Z10-C1 — glide-record integration and the "n = r_w" lemma (Collatz; priority 1)
*- **Old:** N_ver is the t = τ verification range (own 984,572,810,980; Rozier–Terracol 2.8·10¹⁹).*
*- **Obstruction:** 47/82 is the exact limit of the relaxation, so further gains need new information.*
*- **New representation:** use t(n) ≥ τ(n). Any n whose stopping time is verified to be ≤ G cannot be a paradox with τ > G . So the effective verification range is the largest bound B such that every n < B has stopping time below the window's min imum τ.*
*- **Propositions:**
 - L-C1a (lemma, to be proved): for a paradox in the window, n < (k+1)/δ < 2^τ, hence n = r_w, the least residue of its o wn parity word. Cite Lagarias 1985 for the "least member" form.
 - L-C1b: replace N_ver with B_glide, the largest bound with verified maximal stopping time < τ_min(window). Recomput e K_min with the Z09 exact relaxation (reviewer H's code).*
*- **Literature task first (no computation):** establish exactly which maximal stopping-time ("glide") or delay records are exhaustively verified, and up to what bound. Sources: Oliveira e Silva (to 5·2⁶⁰?), Roosendaal's records, Barina (2⁷¹; check whether stopping times were computed or only convergence). Classify each as primary/secondary. Do not use a bound tha t cannot be confirmed from a primary source.*
*- **Falsifier:** if no glide range beyond 2.8·10¹⁹ can be confirmed, report "no gain" and stop.*
*- **Success establishes:** a proved bound, conditional on the cited verification.*

Z10-E1 — Fejér-type low-order-character target criterion (ES; priority 2)
*- **Old:** S2 (R_spec < 1) demands that every residue be hit and discards phase information. The exact count N_x(t) dec ides success but certifies nothing beyond the test itself.*
*- **New representation:** a lower bound on target hits that uses only characters of order dividing m.
 - A kernel K on G = (Z/d)^× whose Fourier transform is supported on {χ : χ^m = 1} is constant on cosets of the subgroup H_m of m-th powers. So low-order characters alone can only certify that the **coset** t·H_m contains divisors; they cann ot single out t.
 - The criterion therefore has two parts:
  - (i) coset level: a positive-kernel bound proving that t·H_m is hit, using characters of order dividing m;
  - (ii) inside the coset: a bound on the "within-coset" spectral mass, using the remaining characters, that forces the hit to land on t itself.
 - The exact form of (ii) is fixed in a derivation step that is reviewed before any computation.*
*- **Propositions:**
 - P-E1a: derive and prove a valid minorant criterion (exact statement to be fixed in the derivation step, before any run; t he derivation is reviewed before computation).
 - P-E1b (measurement): the fraction of successes certified using only characters of order dividing m, for m ∈{2, 3, 4, 6, 12}, compared with S2's 46.5%.*
*- **Falsifier:** the criterion fires at a failure (stop), or it certifies no more than S2 (report "no gain").*

*- **Data:** six-class $p < 10^7$, prime $d \le 107$, as Z09 M1.*

Z10-E2 — prime-constellation adversarial construction; finite-shift barrier (ES; priority 3)
*- **Old:** a "coverage lemma" over a fixed set of shifts was sought.*
*- **New representation:** a constructive barrier.*
*- **Proposition B (conditional, to be proved):** for any finite set D of prime shifts ($d \equiv 3 \bmod 4$), Schinzel's Hypothesis H implies infinitely many six-class primes p failing at every $d \in D$.*
* - Construction: impose p mod 840 in a six class; require $(p+d)/4$ prime for every $d \in D$; and choose p mod d so that neither target lies in $\{1, x_d, x_d^2\} \bmod d$. Check that the congruence system is admissible (no fixed prime divisor).*
* - Consequence: no unconditional proof of ES can use a fixed finite set of shifts, unless Hypothesis H fails. This is the shift-wise analogue of Mordell–Schinzel.*
*- **Measurement (Colab, seeded):** sieve search for actual primes with $D = \{3,7,11\}, \{3,7,11,19\}, \{3,7,11,19,23\}$ (and larger if feasible). Report witnesses, certified by exact primality below 2^{64}; $S(p)$ for each; $S(p)/(\ln p)^2$; and whether O3 survives.*
*- **Falsifier:** an inadmissible congruence system for some D (then B fails for that D; report it).*

Z10-C2 — $\tau = \infty$ odd-density test, integer vs 2-adic (Collatz; priority 4)
*- **Fact used (proved, elementary):** $\tau(n) = \infty \Leftrightarrow q_t \ge t/\rho$ for all t, i.e. the odd-step density stays at or above $1/\log_2 3 \approx 0.6309$ at every prefix.*
*- **Measurement:***
* - For integers $n \le 10^{10}$ (sampled) and the record-τ numbers, report the maximum length L(n) of a prefix with $q_t \ge t/\rho$ throughout, i.e. $\tau(n)$, against log n.*
* - For 2-adic paths built by prescribing admissible parity sequences, report the least positive integer representative of each prefix class, r_{w_j}, and its growth in j.*
*- **Question (inconclusive by design):** is there a measurable necessary condition, e.g. $r_{w_j} \ge 2^{\theta j}$ for some $\theta > 0$ along paths with $\tau > j$, that integers violate? Any such pattern is reported as a heuristic pattern unless proved.*
*- **Falsifier:** none (exploratory). Labelled exploratory in the charter so it cannot be promoted from data alone.*

Not in Z10 (per GPT, agreed)
Additional small-k Collatz proofs; further ES residue-class catalogues; G0R enlargement; purely quadratic-character cross-shift coupling.

Appendix B. The memo to GPT that accompanied the charter (16 October 2026)

The memo below is reproduced verbatim from *MEMO_TO_GPT_Z10.md* (SHA-256 2859dc61891a9294d9bead4d1783ac728c1a829b9293ff9c8b84efac6aca5fbd). It was written by Claude and sent to GPT by the author together with the charter. Item 2(b) records the origin of branch C2 in GPT's assessment of the previous round.

Memo to GPT — response to your Z09 assessment, and the Z10 charter
From Claude, via W. Houze, 16 Oct 2026 (evening). The charter is Z10/CHARTER.md, sha256 4876b05c51ce3f4212ccb3a52e2f6a98b92b8be2443b798785158730f64fc87d (frozen; no runs yet).

1. Agreement
Your status statement is accurate and can enter the record with two qualifiers:
- "No coupling" means consistent with independence within about ±15% (bootstrap CI on 900 primes), not that independence is proved.
*- 47/82 is the **exact** limit of the present relaxation (reviewer H exhibited matching feasible points), so further gains need new information.*
We agree with your classification of productive and falsified Z moves, and with parking G0R and quadratic cross-shift coupling.

2. Four adjustments to your Z10 list, and why

(a) Your C1, "exact-prefix residue + energy", reduces to a known statement.

*- In the paradox window $n < (k+1)/\delta$, which is far below 2^{τ}. So a paradox n is not merely congruent to its word's least res idue r_w: it **equals** r_w. (Lagarias 1985: a counterexample is the least member of its residue class.)*
- The fused experiment therefore asks whether an n below about 10^{14}–10^{20} can have $\tau \geq 10^{10}$. That is the stopping-time pro blem itself, and enumeration cannot settle it.
- The concrete gain available is different. Because $t(n) \geq \tau(n)$, any exhaustively verified range of maximal stopping times (" glide records") acts as a larger N_ver.
- Z10-C1 first establishes from primary sources exactly what has been verified (Oliveira e Silva, Roosendaal, Barina), the n recomputes K_min.

(b) Your C3 ($\tau = \infty$) is kept, but labelled exploratory.
- $\tau = \infty \Leftrightarrow$ odd-step density $\geq 1/\log_2 3$ at every prefix.
- The experiment compares integer and constructed 2-adic paths, and can only yield a heuristic pattern unless a necessary condition is proved.

(c) Your E-S target-specific Fourier criterion needs reformulation.
- Using exact phases simply recomputes N_x(t), which is the success test itself and so certifies nothing new.
- A useful criterion uses partial information: characters of low order only, which is also all that analytic equidistribution r esults control.
- Low-order characters see cosets of the m-th-power subgroup, not individual residues. So Z10-E1 is a two-stage criterion : a coset hit via a positive kernel, then a within-coset bound.
- The exact form is to be derived and reviewed before any run.

(d) Your adversarial many-shift test should succeed, and that is a theorem-level point.
- Require $(p+d)/4$ to be prime for every d in a finite set D, and choose p mod d so that neither target lies in $\{1, x_d, x_d^2\}$. Then p fails at every $d \in D$.
*- Under Schinzel's Hypothesis H such primes exist for every admissible finite D. So **no proof of E-S can rely on a fixed fi nite set of shifts**, only on a set growing with p. This is the shift-wise analogue of the Mordell–Schinzel barrier on identiti es.*
- Z10-E2 proves the conditional statement and constructs real witnesses for $|D| = 3$ to 5.

3. Z10 order
1. C1: glide integration + n = r_w.
2. E1: low-order-character target criterion.
3. E2: prime-constellation barrier.
4. C2: $\tau = \infty$ odd-density test (exploratory).

4. Requests to GPT
1. Do you know the exact verified range of maximal stopping times for the shortcut map, from a primary source?
2. Check the admissibility argument for Z10-E2's congruence system before we search.
3. Suggest the cleanest form of the within-coset step for Z10-E1.

Appendix C. Extract from the claim ledger

The project's claim ledger records every claim with its status label and its review. The extract below is verbatim. It gives the label set, three entries from round Z09, and every entry from the M1 manifold wave of 17 October. Entry M1-A7 is worker W2's reduction (Section 18.2), and entry M1-D lists the claims that wave refuted. The full ledger, of which this is an extract, has SHA-256 ccbd694c5300c7a6d0fbe2b6b25a2a8c51353e334b1a1fba09c78a57907aad22.

Claim ledger — Z06 round (15 Oct 2026)
Labels: proved; proved on a specified infinite family; conditional; finite verification; refuted with witness; inconclusive.

Z09 round additions (16 Oct 2026)
| ID | Claim | Status | Evidence / review |
|---|---|---|---|

*| C-U" | Exact relaxation: every first-drop paradox has **k ≥ 47** odd runs (own range 984,572,810,980) / **k ≥ 82** (R–T 2.8·10¹⁹) | **proved (computer-assisted)**, conditional on N_ver | reviewer F (derived), reviewer H (independent bo und and code, tightness) |*
*| C-E3 | "Bottleneck is the size side" | **refuted** (reviewer F) | — |*
*| C-E5 | P2 inequalities + budget cannot exclude τ = ∞ | **proved** (expected: 2-adic realisability) | reviewer F |*

M1 manifold wave (17 Oct 2026) — see M1/MANIFOLD_REPORT.md (labels per GPT control 2)
| ID | Claim | Label | Review |
|---|---|---|---|
| M1-A1 | n = r_w for every first-drop paradox | PROVED | R1 |
| M1-A4 | Paradox ⇔ shifted cycle equation, D ∈ [0, k−1]; D = 0 ⇔ cycle with exactly k runs | PROVED | R1 |
| M1-A5 | k ≥ 48 (own) / 83 (R–T) given no small cycles (S–dW m ≤ 68; Hercher m ≤ 91) | CONDITIONALLY PROVED | R1 |
| M1-B1 | k ≥ 78 at primary-sourced range 20·2⁵⁸ (79 with cycles excluded) | CVB / COND. | R1 |
| M1-A7 | Extremal admissible path = integer; separation ⇔ τ = O(log n) | PROVED | R1 |
| M1-A10 | Two-stage criterion T'₆; certifies 63.5% of successes (vs 46.5%) | PROVED + CVB 10⁷ | R2 |
| M1-A12 | Under H, no fixed finite prime-shift set suffices (B) | CONDITIONALLY PROVED | R2 |*
| M1-B7 | 3 solution types cover all six-class p ≤ 10⁶ | CVB 10⁶ | R2 |
| M1-D | Refuted: B as chartered, SRP, HA1, W7.2, ordered runs, W2 density-1, W6 link L | REFUTED | R1/R2 |

Appendix D. The record of 21 October 2026

The exchanges of 21 October survive in two forms. One is a compiled record that runs from GPT's assessment of the Z10 charter through the proof of Theorem 1 and Claude's review: *Assessment_Z10_Charter (10212026_1228).docx*, SHA-256 c3e5d01630bf299d8afb0908db32793024868894a0971d222749be4c00345e25. The other is the project archive of that evening, *M_Archive_20261021.zip*, SHA-256 000c4bf7d583f22a570ba849739f7a2fb45913ba9b93885b24005ec7040913eb. Four passages are quoted below, verbatim, in the order in which they occur.

D.1 GPT on branch C2, before the charge.

> C2 addresses a fundamental issue: whether an infinite parity sequence satisfying the density condition can correspond to a positive integer. That is important, but the distance from the proposed experiment to a theorem is considerable. … The proposed growth condition $r_{w_j} \geq 2^{\theta j}$ would be powerful if proved uniformly. Indeed, it would imply a logarithmic upper bound on coefficient stopping time: $\tau(n) \leq 1 + \frac{\log_2 n}{\theta}$. But proving that uniform growth condition would itself be a major result. … The charter appropriately labels C2 exploratory. I would retain that status.

D.2 The author's question.

> Of the two problem sets, and given the status of the Z10 Charter, which of the two problem sets is most promising with respect to offering anything approaching a full or partial proof of even a specific aspect of the given problem set?

D.3 The author's charge. The charge is quoted in full in Section 17.4.

D.4 The opening of GPT's reply, and the statement it proposed.

Yes—the theorem-first approach is appropriate here, and it has now produced a precise Collatz hypothesis for your review, together with completed supporting proofs. I used all three available subagents, with separate assignments for construction, computational checking, and adversarial review. Erdős–Straus is tabled.

Working Hypothesis M. For every integer $n \geq 2$, $\sigma_3(n) \leq \lfloor \log_2(n^2) \rfloor + 1$.

Appendix E. GPT's provenance audit (23 October 2026)

At the author's request GPT audited, from its own records, what it had received before proposing Theorem 1, how the statement was chosen and when it was frozen. It marked each answer as documented in the records or not. It supplied an evidence package, *Hypothesis_M_Provenance_Evidence_20261023.zip* (SHA-256 7cb9a40265ed4ba79d0b1e6f37a3259f582ccbb0e5c99441c6ed8c33c38cbcfd), containing the original 21 October research archive, the proof package, the complete M1 manifold-wave archive, the 17 October results inventory and GPT's assessment of it, a compiled exchange record, and GPT's provenance report and audit files.

Claude checked every file in the package against its listed hash. It confirmed that the preregistration in Table 10 states the universal inequality $r_L^2 \geq 2^L$ for every $L \geq 2$ as its frozen candidate, fixes a separate computational test of it over $L = 2, \ldots, 100{,}000$, and forbids adjusting the exponent or the range afterwards, and that Section G of worker W2's results proposes the boundary word as an untested next step.

Table 10. Files that identify the steps of Section 18.2, with their SHA-256 hashes.

File	**Role**	**SHA-256**
Z10/CHARTER.md	The Z10 charter (Appendix A)	4876b05c51ce3f42…f64fc87d
MEMO_TO_GPT_Z10.md	The memo sent with it (Appendix B)	2859dc61891a9294…6aca5fbd
M1 charter	Charter of the M1 manifold wave, 17 October	04034e5fc672d9e1…772512f4
CHMC_M1_manifold.zip	The M1 wave archive, including W2's results	89471ad167ed8904…58c6e9f0
MECHANICAL_PREREGISTRATION_v1.md	The freeze of the universal inequality (residue form of Theorem 1) and of its test range	25ae7041f0d9d80d…5a8698e2
Collatz_Research_Package_20261021.zip	GPT's research archive of 21 October	083165750745f122…5043e0bc

Table 11. The recorded sequence, as given in GPT's audit (EDT unless stated). ZIP member times carry no time zone; the preregistration has no independent timestamp.

Date	Event	Recorded time
17 Oct	Results inventory, with W2's reduction, uploaded to GPT	07:29:20
17 Oct	GPT's assessment discussing that reduction saved	08:03:59
17 Oct	Complete M1 wave archive uploaded to GPT	09:57:02
21 Oct	Z10 charter and memo uploaded to GPT	12:09
21 Oct	Preregistration of Hypothesis M (ZIP member time)	13:18:08
21 Oct	Principal run, $L = 2, \ldots, 100{,}000$	13:18:32–13:18:34
21 Oct	First research archive saved	13:36:24
21 Oct	Proof round: agents M1, M3 and M2 begin	14:34:02–14:34:23

GPT's audit concludes:

> For the essay, the established provenance is prior exposure to W2's reduction, followed by the recorded formulation and testing of the restricted Hypothesis M. Causal reuse, the first proposer of $K = 3$, and the reason for choosing $\theta = 1/2$ remain unestablished.

The author's replication. On 23 October 2026 the author ran, in his own Colab sessions (a hosted runtime separate from the systems that produced the files), a set of self-contained check blocks. They recompute the values shown in Figure 1 and the constants quoted in Section 16. They also rebuild, from copies embedded in the code, seven of the files listed in this appendix and in Appendices A–D: the Z10 charter, the memo, the claim ledger, the M1 charter, the preregistration, the 21 October record and *M_Archive_20261021.zip*. Every recomputed SHA-256 hash matched the value printed in this paper. The evidence package itself, at 15 MB, was too large to embed; its hash was confirmed by Claude and by GPT's audit. Because the files were embedded in the check code, these runs confirm that the printed hashes match the archived files and that the files reached the author's machine intact.

Appendix F. The second proof of Theorem 1, as recorded (21 October 2026)

The passage below is reproduced verbatim from Section 2 of Claude's working document *M_package/Claude_audit_and_route.md*, SHA-256 55a61b375bcf4c78b9b52085c56efde94fbef1171ec86ab7c20d9f223bdfe733. The file is a member of *M_Claude_Package_20261021.zip* (SHA-256 073c8e96d1c7178c6004b9c4146e633d0fa3f0a275c8ee1d82f181a5d6b9ff5c), which is listed in the checksum file of *M_Archive_20261021.zip* (Appendix D). It is a working record written for GPT and the author, not a finished proof, and no human specialist has checked it. It uses the project's labels of that day: "M" and "Theorem M" are Theorem 1; "the cycle lemma (1.4)" is Theorem 5; "Lemma S" is Lemma 11; the bound of Branch P is Theorem 4; "CHECK 7" and "CHECK 8" are exact checks in the archived script *M_proof_checks.py*; and the "referee agent" is one of Claude's sub-agents. The "retired bounded theorem" is the route described in Section 18.4, and the phrase "usually far below" is the wording GPT later corrected (Section 18.6).

Editorial note (v10). The quotation below is preserved as archived, with one error now identified.

- *The endpoint error.* Its statement of Lemma S, "$e_t = e_{t+q_n}$ for $2 \le t \le q_{n+1}$", is false at the endpoint $t = q_{n+1}$. For $q = 3$ and next denominator 8, it would give $e_8 = e_{11}$, but $e_8 = \lceil 8\alpha \rceil - \lceil 7\alpha \rceil = 6 - 5 = 1$ and $e_{11} = \lceil 11\alpha \rceil - \lceil 10\alpha \rceil = 7 - 7 = 0$. Exact computation shows that the endpoint fails for every consecutive pair of denominators up to $1054{,}24727$. The range of Lemma 11 in Section 6, $2 \le t \le q_{m+1} - 1$, holds throughout. Lemma 11, Theorem 4 and the proof of Theorem 1 in Section 5 are unaffected.

- *The gaps.* The quoted route is incomplete as written. Branch L does not define all its variables, prove its divisibility and size inequalities for every convergent, or show that the two branches cover every $k \ge 23$. The "worst case 0.561" is a numerical value, not a uniform estimate.

- *The completed proof.* Section 6A now gives a complete version of the route: Proposition N7, with a uniform estimate, takes the place of Branch L. Its exact constant at $q = 65$ is about 0.547. The archived balanced-word step is replaced by the argument of Section 6.

The quotation is therefore an archived account; the proof is in Section 6A.

2. Claude's route reconciled at the shared counterexample (GPT task C3)

2.1 Statement

***Convergents.** Let $q_1 < q_2 < \dots$ be the convergent denominators of $\beta = \log_3 2$:*
1, 2, 3, 8, 19, 65, 84, 485, 1054, 24727, ...

***Words.** Write $e = 1 \cdot c_\beta$, where c_β is the characteristic Sturmian word. The standard word s_n is the length-q_n prefix of c_β, which is $e_2 \dots e_{q_n+1}$.*

***Lemma S.** $e_t = e_{t+q_n}$ for $2 \le t \le q_{n+1}$. In particular $R_{q_n}(1) \ge q_{n+1} - 2$.*
- Proof: c_β begins with $s_{n+1} s_n$, whose prefix of length $q_{n+1} + q_n - 2$ is a central word with periods q_n and q_{n+1} (Lothaire, Ch. 2).
- Checked exactly, with equality, for every $q_n \le 16{,}785{,}921$ (referee agent).

***Setup.** Suppose M fails at n, with bit length k. Then n follows e for at least $2k - 1$ steps.*

***Branch P**, for $k \in [q_n + 4, q_{n+1} - 4]$:*
- $T(n) \equiv T^{1+q_n}(n) \pmod{2^{k+2}}$.
- Both values are $< 2^{k+2}$, so they are equal.
- The return word is s_n.

***Branch L**, for $k \in [q - 3, q + 3]$ with $q = q_{n+1} \ge 65$:*
- Let $Z = 2^q \cdot (x_1 - x_0) = (3^p - 2^q) \cdot x_0 + b$.
- Then $v_2(Z) \ge 2k - 2$ and $\lvert Z \rvert < 2^{2k-2}$. The worst case is $q = 65$, at 0.561 of the limit.
- So $Z = 0$, $x_1 = x_0$, and the return word is again a standard word.

***Tiling.** The two branches tile every $k \ge 23$ (referee). The exact certificate covers $L \le 10^6$, which is $k \le 500{,}000$.*

2.2 Does a counterexample in this route produce a balanced cycle?

Yes, by a different mechanism from the two-copy argument.
- In both branches, the return word over q steps is the standard word s.
- Standard words are conjugates of Christoffel words: s = w·xy and the Christoffel word is x·w·y. So s is primitive and cyclically balanced. CHECK 7 confirms this for q ≤ 1054.
- The least period of the forced cycle divides q, and s is primitive, so the least period equals q.
- The cycle lemma (1.4) then makes the cycle trivial, which is a contradiction.

***Consequence.** Claude's route also proves M for every n. It no longer needs the cycle-length threshold P_min or any external verification bound. The earlier "unconditional for n < 2^{217,976,794,613}" statement and its conditional corollary are **retired**.*
- They were correct as stated.
- They are now superseded.
- They also misstated the remaining difficulty as "as hard as general cycle exclusion".

***Independence of the two proofs.** They share only the cycle lemma (1.4):*
- GPT's route uses the complexity bound, the 4/9 cells and the two-copy argument, plus 11 base cases.
- Claude's route uses Lemma S, the standard-word / Christoffel conjugacy and the one-lap inequality, plus the computational certificate for k < 23 (or ≤ 500,000).

2.3 By-product: a sharper bound (new, needs external review)

***Branch P statement.** For bit length k ∈ [q_n + 4, q_{n+1} − 4]:*

$\sigma_3(n) \le k + q_n + 3$

This is usually far below M's 2k. For example, at k = 20,000 it gives at most 21,057 steps, against M's 40,000.

***Check.** CHECK 8 finds 0 violations over 2,896 bit lengths. The equivalent residue test holds throughout the 10^6-digit certificate.*

Appendix G. The completion round of 24 October 2026

This appendix records the completion round that produced the new results of v10. Its files, scripts and review reports are kept in the round's record package. Each is identified below by its SHA-256 hash, abbreviated to its first and last characters.

G.1 The charge and the protocol

The author asked GPT to assess v6. GPT's Completion Assessment [46] made four recommendations:

- explain the shared numbering;
- write "not yet formalized in Lean" in place of "no formal proofs";
- correct the endpoint error in Appendix F and complete the second proof;

- submit its proposed proof of Remark 15 to adversarial review, add the bounds $k + 1 \leq M(k) \leq 2k$, and investigate two asymptotic targets within the present paper.

GPT and the author then gave Claude a ten-step protocol [47], summarized in Section 17.9. Its governing instruction was to extend the work from verified foundations without changing the standard of evidence, and to report the actual outcome even if it is a precise remaining obstruction rather than the desired theorem.

G.2 Baseline

Item	Identification	Status
v6 manuscript	SHA-256 *6b4455a1…6f56212*	Frozen; matches the hash in [46]
Completion Assessment [46]	*2e29b03e…6ba136a*	Frozen
Research protocol [47]	*ee3a8ec8…dc0a21*	Frozen
Lean repository	commit *449929d2…5ae30*, package *a2ef196d…1f9275*	Frozen-source check passed (33 files). A fresh build was not possible in the working environment; the author rebuilt it on GitHub on 24 October 2026, and it passed (Section 8)

Not supplied, and not reconstructed:

- the archives of 21 October cited in Appendices D and F;
- the script with CHECKS 1–9 and the certificate output for $L \leq 10^6$;
- the preregistration and charter files of 16 and 21 October;
- the earlier external review reports.

G.3 Preregistered targets and tests

Recorded at 09:12 UTC on 24 October, before the principal computations (file *PREREG_targets_and_tests_20261024.md*, SHA-256 *ef183770d104dbd647a478b165bc540cbf55f47ba196b936e63f19d2730755ec*):

- **Target A.** There are constants $0 < \varepsilon < 1$ and C, independent of k, with $M(k) \leq (2 - \varepsilon)k + C$ for every $k \geq 2$.
- **Target B.** For every $\eta > 0$ there is $k_0(\eta)$ with $M(k) \leq (1 + \eta)k$ for every $k \geq k_0(\eta)$.

A statement for infinitely many k, a lim inf, an interval-restricted bound or a finite range was not to be presented as either target.

The record also lists tests F1–F8 with frozen domains and discloses what had been seen before freezing. One later extension of test F8 was recorded as a dated amendment (*d053e748…bc5a0d*) after its first results were seen, and is labelled exploratory.

That amendment wrongly said the extension used "the same definition" as F8. In fact the first F8 run used the domain $8 \le b \le 300$ in place of the frozen $b \le 400$, and the value $\max(j + \ell, 2j - i)$ for $i \ge 1$ in place of the frozen $j + \ell$. Claude reports that both choices were made while writing the generator, after the preregistration was locked and before any F8 output existed. That report rests on file modification times in Claude's working environment: preregistration 05:12:02, generator 05:13:12, first output 05:13:28 EDT. A listing of those times is included in the record package, but it is Claude's environment reporting on itself, and no independent timestamp evidence was preserved. The choices were not recorded at the time. The reason for the reduced domain $8 \le b \le 300$ was not recorded either and cannot now be established. A dated clarification (Amendment 3) states these limits.

GPT's review identified this. A retrospective Amendment 2 (*aacaa24a…a5bfa379*) now records it. It also records an exhaustive recomputation under both definitions for every $2 \le b \le 400$ and both windows (*hm_f8_exhaustive.py*, *e405cc9c…01c03bf5*). That recomputation reproduces the original runs exactly. It finds that the two definitions differ only at $b = 10$ and $b = 493$, by one step each, and that the range $301 \le b \le 400$ adds no new peak.

G.4 Verification

Check	Method	Result
Critical word, Q_j for $j \le 30020$	integer comparison $3^{Q_j - 1} < 2^j < 3^{Q_j}$	certified
Residues r_L, $L \le 20000$	closed form and bit-by-bit lifting (two programs)	identical
Corollary N2	all seeds below 2^{22} by direct simulation	agrees
Corollary N3	literal definition of S_k, every $k < 20000$	agrees
Theorems 1 and 4	exact data, $k < 20000$	no violation
Lemma 11 range; Appendix F endpoint	every denominator pair up to $(1054, 24727)$	range holds; endpoint refuted at every pair
Proposition N1	dynamic count, $L \le 3000$, 22 widths (13 on choice boundaries)	no disagreement
Proposition N7 constants	exact rationals and certified intervals, every window up to $q = 24727$	ratio at most about 0.547
Second proof, base and coverage	all seeds below 2^{22}; bit lengths 23–50511	pass

Check	Method	Result
Corollary N2 bounds; Proposition N1 (rational K)	Lean 4.35.0 proofs (Lean round 1), compiled and axiom-gated on GitHub Actions by the author against the formal development, 24 October 2026 (run 40109923980)	passed; standard axioms only
Lean round 2 modules	Lean 4.35.0, 13 modules by GPT, compiled and axiom-gated on GitHub Actions by the author against the formal development, 24 October 2026 (run 40131960252, commit 13bd411)	passed; 27 theorems, standard axioms only; dependency check passed
Method power (F8)	exhaustive best single-coincidence certificate, $2 \le b \le 400$ and two windows, both definitions	peaks 1.967–1.970 ($b = 493$), 1.979 ($b = 1062$)

The generator (*hm_gen.py*, *1676cc73…fa0b*) and the checker (*hm_check.py*, *1791f6eb…6ece*) were written separately and use different algorithms. A self-contained replication script (*hm_r7_colab_replication.py*) embeds them. The author ran the script twice in his own Google Colab session on 24 October.

- *05:36 EDT, first version of the script.* All nine recorded results matched (output SHA-256 *66c1c8a5…f992*).
- *06:15 EDT, the version that adds the Section 6A checks* (*hm_r7_colab_replication.py*, SHA-256 *f650affd…9729*, embedding *hm_branchL.py 20c63bdc…0251*). All ten recorded results matched, including *branchL ALL PASS*: the N7 constants (worst 0.5471 at $q = 65$), coverage of bit lengths 23–50511, and the base check below 2^{22} (output SHA-256 *8a405001…6a65*).

This establishes reproducibility on a second runtime; it is not a substitute for proof.

G.5 Review record

How the reviews were run. Review took place in two Claude sub-agent rounds, a subsequent GPT review and focused check, and a Grok review of selected results, with different coverage by result; this appendix records that coverage. GPT reviewed N1's corrections as its original author. GPT supplied the replacement failure argument and the $\sqrt{2}$ strengthening, which Claude checked and Grok then reviewed. In the sub-agent rounds, each proof under review was sent to a Claude sub-agent working with fresh context, which had not constructed it. The model was the one in use for the session. Each reviewer was given only the statements and the proofs. It was told to write its own derivation before opening the proof file, to try to break the claim, to test it with its own exact-integer scripts, and to list every objection with its severity. The statement-first order was instructed but not technically enforced.

Round 1 (05:19–05:25 EDT).

- *Proposition N1* (proof by GPT): accepted with corrections. There were two substantive objections: the counting induction had been asserted rather than proved, and the identity behind the $K = 4$ case was missing; the reviewer supplied the latter. There were also seven minor and two presentational objections. The reviewer found no counterexample at 51 widths for $L \le 2000$.

- *Corollaries N2–N4, Proposition N5, Remark N6* (by Claude): N2 and N4 were accepted; N3, N5 and N6 were accepted with corrections. The substantive objection was that N5 cited the denominator recurrence with the wrong index; the inequality it needs was true, and its justification was corrected. The "no constant below 2" conclusion of N6 had not been argued, and was added. The reviewer found no counterexample for any seed below 2^{20}.

Round 2 (from 05:50 EDT).

- *Proposition N7 and the second proof* (by Claude): accepted with corrections; no counterexample.
  - The substantive objection asked that non-circularity be shown by listing the dependencies of the lemmas used; Section 6A now does so.
  - Minor points concerned the justification of three constants, the case $C' < 1$ and two implicit balance steps.
- *Corrected N1 and Section 7A* (re-review by a further fresh reviewer): no error in any proof, and every quoted number confirmed by recomputation, including an extension to $L = 40000$.
  - *Substantive objection.* Section 7A.6 had described the method obstruction of Section 7A.5 as identified, although it is a candidate supported by computation. The wording was corrected throughout the paper.
  - *Improvement.* The reviewer showed that the argument of Proposition N5 gives $3/2$ in place of the golden ratio, using the Gelfond–Schneider theorem. Claude checked the argument and adopted it; the reviewer constructed it and Claude reviewed it.
  - *Minor and presentational points.* A proof was added that Proposition N1 fails above $9/2$ (this argument was later found by GPT to fail for $27/4 \leq K < 9$ and has been withdrawn and replaced; see Round 3), a statement in Remark N6 was proved for every k, and the finiteness of z_k and some notation were fixed.

Every objection in both rounds was closed by a correction to the text; none was rejected.

Round 3: GPT's external review (report of 24 October, 06:51 EDT). GPT reviewed the whole of the completion round's draft.

Standing. The review was not fresh-context, since GPT had the earlier project context. For N1 it was not independent of construction, since GPT wrote the original proof.

Verdicts. GPT's verdict was "accept with corrections". It accepted:

- Proposition N7 and the coverage argument;

- Corollaries N2–N4 and Remark N6;
- N5 in its $3/2$ form;
- the corrections to N1 on $3 < K \leq 9/2$.

Its independent exact checks confirmed the finite base, the five windows, coverage and the reported data. They also confirmed the two certificate peaks, by an exhaustive search.

Objections. It raised eleven objections:

- a gap in the added argument for the failure of N1's formula above $9/2$, which GPT replaced with a correct argument;
- a false concluding sentence in Section 6A at $n = 3$;
- remaining overstatements, and confusion of the cumulative maximum with a single bit length, in lay summaries and the Lexicon;
- stale text in Section 18.4 and Table 9;
- the undisclosed F8 departures;
- inaccurate review-history and "next steps" summaries;
- an out-of-date table of contents.

It also proposed strengthening N5 to $\sqrt{2}$.

Resolution. Claude agreed with all eleven objections and checked both of GPT's arguments. The corrections are made in v7.1; Claude's point-by-point response is a separate document. The two GPT-supplied arguments, the $\sqrt{2}$ form of N5 and the failure proof above $9/2$, have been checked by Claude only, and await the next external review.

Round 4: GPT's focused check of v7.1 (report of 24 October, 07:24 EDT). GPT accepted the revised failure argument, the corrected conclusion of the second proof, the $\sqrt{2}$ form of N5 and the F8 recomputation, which its own exhaustive checker reproduced.

Because GPT supplied the two adopted arguments, this check is not an independent review of them. At that point Claude was their only non-constructor checker, and they were flagged for the next external review (Round 5).

GPT raised six minor or presentational points, all agreed and corrected in v7.2:

- the general-reader caption and legend of Figure 8;
- the review-history summaries;

- the cost definition in Figure 3's caption, which omitted the $i = 0$ case;
- the F8 chronology, which is now qualified as reported;
- stale entries in the record;
- the final pagination, which must be refreshed in Word.

Round 5: Grok's review of v7.2 (24 October). Grok checked the package against its manifest and reviewed, in order, the two GPT-supplied arguments, Proposition N7, the second proof of Theorem 1, and Corollaries N2–N4.

Standing. Grok constructed none of these results, so for the two GPT-supplied arguments this is the first external review by a non-constructor. It was not fresh-context: Grok had the earlier review records, which it says it consulted for context only. Its numerical checks are reported in prose; no Grok code or execution log was supplied, so those runs cannot be audited.

Verdicts. Grok accepted all five items and made its own numerical checks of the decisive constants. It raised no medium or high objection. Its restatement of Corollaries N2 and N3 was off by one against the paper ($\max S_k = M(k) - 1$ and $z_k = M(k) - k - 1$); its acceptance of those two is recorded, but it is not evidence that they were checked closely.

Objections. Grok raised two low objections, and Claude's response is a separate document.

- *N7-1.* Grok asked for a remark that ± 3 is the largest window for which the present constants close, and attributed the limit to the congruence step. Exact computation shows otherwise: the lower end is set by the size estimate, and the upper end is not binding. A remark stating the actual limits is added after Proposition N7 in v7.3.
- *6A-1.* Grok asked for a pointer from the second proof to the constants of Proposition N7 for readers who skip Section 7A. Proposition N7 and its constants are in Section 6A, directly before the second proof, so Claude rejected this objection and made no change.

Round 6: GPT's focused check of v7.3 (report of 24 October, 08:05 EDT). GPT accepted the v7.3 manuscript changes. Its own exact computation confirmed the window remark: at $q = 65$ the congruence condition $2k \le q + Q$ holds through $k = q + 9$ and fails at $k = q + 10$, and the size ratio is about 1.094 at $k = q - 4$. GPT noted that a ratio above 1 means only that the size estimate does not certify equality there; it is not a counterexample to a stronger statement. GPT agreed with the rejection of 6A-1, and accepted Grok's review as the required review of the two GPT-supplied arguments by a system that did not construct them.

GPT raised three minor corrections, all to the supporting documents and all agreed. No theorem or proof changed.

- Response 3 had said that the size ratio halves with each unit step in k. In fact one term halves and the other quarters, so the ratio falls to less than half.

- Response 3 had called $\sqrt{2}$ sharp. The eventually-all-2 example shows only that the denominator-ratio bound $R \geq 1 + \sqrt{2}$ is sharp under the reduced hypotheses. It does not show that $\sqrt{2}$ is the optimal lim inf bound for $M(k)$, and Proposition N5 makes no such claim.

- The Record's statements that every objection was closed by a correction now apply only to Rounds 1–4. It now states the two Grok dispositions separately.

Response 3 was corrected accordingly; the uncorrected copy is kept in the record package.

Round 7: Gemini's originality review of v7.4 (24 October). Gemini was asked for prior art on N1–N7, the candidate obstruction and the framing of v6, and for an audit of references [1]–[48]. Its verdicts were:

- *partly known:* N1, N2–N3, N7 and the obstruction;

- *related:* N4 and N6;

- *not located:* N5;

- *known:* the framing of v6.

Standing. The review supplied no search record. It quoted no earlier statement, and gave no identifiable source for most of its verdicts: N7, N4 and N6 rest on descriptions such as "classical 2-adic valuation arguments". Its reference notes contained checkable errors:

- it gave wrong reference numbers for Terras and Everett;

- it listed as missing the Bernstein–Lagarias paper [2] and Chang's preprint [48], which are both cited;

- it attributed the paper cited as [15] (López and Stoll) to other authors, with different pages, which Claude checked against the published paper;

- it gave a wrong arXiv number for Bugeaud and Kim (arXiv:1002.4598 is a different paper).

Its verdicts are therefore not counted as an originality determination, and none of the paper's originality statements was changed on their strength.

Objections and changes. Claude's response is a separate document.

- *OBJ-01* asked that N6 be framed as a limit of the method rather than of the dynamics. Remark N6 already says that it "is a property of the two bounds, not of $M(k)$". No change was made.

- *OBJ-02* asked for the exact correspondence with the Bugeaud–Kim function. Claude agreed, and v7.5 adds it in Section 7A.5: the block length ℓ plays the role of n, and every certificate has value at least $r(\ell, \mathbf{e})$.

- Gemini's note on the journal version of [44] was correct, and the published citation (Trans. Amer. Math. Soc. 371 (2019)) has been added.

Follow-up. Asked for citations, numbered results, quotations and search queries for its "known" and "partly known" verdicts, Gemini named only two sources, [2] and [44], both already cited.

- The quotation it attributed to Bernstein and Lagarias, describing the map as conjugate to "an odometer-type adding machine", is not in that paper. The paper conjugates the 3x + 1 map to the shift map, and its Section 2 is on empirical data, not the construction Gemini cited.

- The quotation from Bugeaud and Kim matches their abstract.

Neither source supports the "partly known" verdicts for N1, N2–N3, N7 or the obstruction, and Gemini gave no source for N1 or N7.

In checking [15], Claude found that López and Stoll treat the number ξ itself: they give a 2-adic series for $\Phi(1c_\alpha)$, and print its binary digits at $\alpha = \log_3 2$. The opening of Section 7A now says so (v7.6). They do not consider zero runs or $M(k)$.

Round 8: documented literature validation (24 October). Gemini was given a written search protocol requiring retrieved links, verbatim quotations and a query log. Its report marked every reference and claim as found or confirmed and every new result as not found. It gave no source links and no source-linked quotations adequate to verify its claims (only short phrases), and no query log, so it is not counted. Claude then ran the search through three sub-agents (references, claims attributed to the literature, prior art) and checked each finding that bears on the paper against the source text. The query logs, the URLs and SHA-256 hashes of the retrieved texts, and the results are in the record package; the texts themselves are not redistributed.

References. Every reference was located, including the 2026 items. Two citations were wrong and are corrected in v7.7:

- the title of [19];

- the proceedings year of [22].

Claims attributed to the literature. Most were confirmed by quotation. Six statements were corrected:

- the Periodicity Conjecture was proposed by Lagarias [13] and restated by Bernstein and Lagarias [2];

- Halbeisen and Hungerbühler describe the extremal word by a floor-difference formula, not as a Christoffel word;

- neither [13] nor [14] dates the problem to 1937, and Lagarias reports a notebook entry of 1 July 1932;
- the Erdős remark now cites [13];
- $\log 3/\log 4 \approx 0.7925$;
- the bound of 10^{11} odd steps in a nontrivial cycle needs Barina's verification [1] as well as Hercher's [32], whose unconditional bound is $7.2 \cdot 10^{10}$.

Two remarks in the archived records reproduced verbatim in Appendix F are also imprecise, and are left unchanged because those records are fixed by their hashes. They attribute to Lagarias 1985 the "least member" form, whereas [13] states that the least residue is the minimal element of its class.

Prior art. The search did not locate a statement of Theorem 1, N1, N2, N4–N7 or the obstruction of Section 7A.5. (Round 9 qualifies this for N5.) Three findings change the paper's account of earlier work:

- *De Jesús, Sturmian–Mahler edge [50] (June 2026, addendum September 2026).* It studies the same 2-adic number: $\Xi_{\log_2 3,0} = \xi$, an exact identity of defining series (Section 7A), which Claude also checked modulo 2^{200}. It proves a linear lower bound for the least realizer of a finite prefix unless its orbit meets a nontrivial positive cycle (Theorem 5.5), excludes an integer realizer of the whole infinite word (Corollary 5.6), and shows that realizers are Hensel truncations of ξ (Proposition 7.1). It also proves that an effective irrationality measure of ξ is equivalent to a zero-run law (Proposition 7.2), and poses as open any finite exponent, to be reached by a 2-adic Mahler method (Open Problem 2). Corollary 2 of this paper supplies the exponent 2 at intercept 0, by a different method; it does not carry out the proposed Mahler construction. The zero-run equivalence anticipates the reformulation behind Corollary N3, though not N3's exact formula for $M(k)$. Sections 3 and 7A now say so.
- *De Jesús, Sturmian irrationality [51] (September–October 2026).* It proves directly that Φ of every Sturmian word, the critical word included, is irrational, which contains Corollary 3. Section 3 now cites it beside [4].
- *Related work, not cited in the text.* These use related methods in other settings:
  - Sharpe's *Rung One* paper (28 August 2026, in the repository cited as [21]), which applies a convergent-shift congruence and a size bound strictly above the critical density;
  - De Jesús's note on the 2-adic Sturmian carry constant (5 June 2026), which gives exact valuations of convergent-block approximants of ξ;
  - López and Stoll's 2021 paper (arXiv:2101.12747).

Ren's *Rational-height staircases for Sturmian Collatz seeds* [52] was seen in this round only in abstract (see Round 9). The Lagarias annotated bibliographies list no other work on the critical Sturmian word.

Limits. The search was run by Claude, which constructed much of the work, so it is not independent. arXiv's search interface, OEIS, Google Scholar and some Zenodo pages could not be queried directly. Several sources were seen only in abstract. A finding that no earlier statement was located is a search result, not a determination of novelty.

Round 9: GPT's focused check of v7.5–v7.7 (report of 24 October, 09:51 EDT). GPT accepted the integrity of the package, the six corrections to claims about the literature, the reference corrections, the Bugeaud–Kim correspondence and the López–Stoll account. It found no defect in any theorem or proof, and confirmed that none had changed. Its own exact-integer checks, modulo 2^{4096}, confirmed the identity $\Xi_{\log_2 3,0} = \xi$ and that $|\xi - 1|_2 = 1/2$. Its verdict was "accept with corrections". It raised seven objections, all agreed and corrected in v7.8.

- *V77-1.* Corollary 2 supplies the effective measure sought in De Jesús's Open Problem 2 only at intercept 0, and by a different method; it does not construct the proposed Mahler theory. The wording in Sections 1.3, 3 and 7A and in Round 8 is qualified accordingly.

- *V77-2.* De Jesús's Theorem 5.5 bounds the least realizer of a finite prefix only unless its orbit meets a nontrivial positive cycle; the infinite-word exclusion is his Corollary 5.6.

- *V77-3.* The identity $\Xi_{\log_2 3,0} = \xi$ is now proved from the defining series, not only checked to finitely many digits. The depth convention and the conversion between runs ending at a depth and runs beginning at a digit are now stated.

- *V77-4 (prior art).* GPT read Ren's paper [52] in full. It showed that the lower estimate in the proof of Ren's Theorem 4.1, restricted to positive integer realizers and converted to parity depth, gives $\liminf M(k)/k \leq 1 + 1/R$. With the denominator-ratio argument of Section 7A.3, this gives N5's bound $\sqrt{2}$. Section 7A.3 now credits this route, and N5 is no longer listed as not located. GPT's deduction is from Ren's stated theorem; neither GPT nor Claude has audited the rest of Ren's paper. Claude could not retrieve Ren's text, because Zenodo blocked automated retrieval, but checked that the deduction's logic is sound given Ren's estimate as GPT reports it. N4's explicit interval bound, N7 and Theorem 1 are not implied.

- *V77-5.* The versions of [50] and [51] are now identified exactly, and "in a later note" became "in a separate note".

- *V77-6.* The records are made precise. "No source states" became "the search did not locate". The package holds source hashes and URLs, not the texts. Gemini's search report is described exactly. The one changed computed value is the decimal for Korec's constant.

- *V77-7.* The Korec change is tracked inside an equation. Accepting or rejecting it in the XML gives the right result, but LibreOffice did not render the tracked decimal, and Microsoft Word was not available to either reviewer. It is a release check.

GPT recommended stating the De Jesús connection, qualified, in Section 1, which v7.8 does in Section 1.3. A sentence in the abstract is left to the author.

Round 10: Claude's check of Ren [52] (24 October). The author supplied Ren's paper (version 1.1, SHA-256 *a5a5268b…01c791e0*), and Claude read the passages GPT's deduction uses: the definitions (Section 2), Lemma 3.1, Propositions 3.2–3.3, Theorem 4.1 and its proof, and the evidence table (Section 10).

Confirmed.

- Theorem 4.1 and its lower estimate (18) are as GPT reported.
- Ren's seed $\xi_{\beta,0} = -\sum_{j\geq 0} 2^{\lfloor j\beta \rfloor} / 3^{j+1}$, with $\beta = \log_2 3$ (his equation (3)), is the paper's ξ.
- By his Lemma 2.2, a rational realizes the first N letters exactly when $v_2(x - \xi) \geq A_N + 1$. So his bound $H(N) \geq 2^B$ excludes every positive integer below $2^{\lfloor B \rfloor}$. At $N = q_k + q_{k+1}$ this gives $M(\lfloor B \rfloor) \leq \lfloor \beta(q_k + q_{k+1}) \rfloor + 1$, and hence GPT's lim inf bound.
- Against the exact values of $M(k)$, this inequality holds at all five applicable denominator pairs below 20000.

Finding. The lower estimate needs Ren's Proposition 3.2, the exact reach $v_2(F_k - \xi) = A_{M_k} + 1$ of the convergent approximants, with $M_k = q_k + q_{k+1} - 1$. Ren states it with a remark that "closely related forms appear" in earlier work, but gives no proof. His evidence table nonetheless lists it as "paper proof + exact finite verification", and his Proposition 3.3 has only a proof sketch. Claude checked Proposition 3.2 by exact computation for the six denominator pairs from $(2,5)$ to $(306,665)$. So the route to N5 through Ren is complete only modulo that proposition. N5's own proof in Section 7A.3 does not depend on it.

The check script and its output are in the record package.

Round 11: Lean formalization (24 October). At the author's request, Claude formalized the two results first in the proposed order. Lean 4.35.0 could be installed in the working environment from GitHub; Mathlib's binary cache still could not be reached. The modules were therefore written in core Lean.

- Claude wrote and compiled *Rev8Bounds.lean* and *Rev8Bridge.lean*. The bridge was compiled locally against a copy of the frozen definitions, with Theorem 1 supplied as a stand-in that is used only for that local check.
- A Claude sub-agent wrote *N1.lean*. Claude recompiled it, checked that it contains no *sorry*, *native_decide* or added axioms, and checked that its definitions match Section 7.
- Each theorem's axiom report lists only the standard axioms.

- The modules, an axiom gate and an updated workflow are in the record package, with their SHA-256 hashes.

- The author uploaded them to the public repository *whouxe82/hypothesis-m-lean*, in a package whose 33 frozen sources are unchanged (the repository's hash check passed in every run), and added a workflow that compiles them after the full replay.

Public runs, 24 October 2026. All ran on GitHub-hosted machines with Lean 4.35.0 and the Mathlib cache.

- *Original replay workflow, runs #3–#6 (15:11–16:08 UTC).* Each repeated the full replay: the frozen-source hash check, Mathlib at the pinned commit, all 13 proof modules, the statement check and the axiom gate. All passed. They are the first fresh rebuilds since 22 October.

- *Round 1 workflow, run #2 (15:50 UTC).* The replay passed, but Lean refused to compile the new modules from a folder outside the package root. No proof was checked in that run. The workflow was corrected to copy the three sources into the package before compiling.

- *Round 1 workflow, run #3 (16:05–16:07 UTC).* Passed: https://github.com/whouxe82/hypothesis-m-lean/actions/runs/40109923980

  – The replay record shows status PASS, 18 steps all with exit code 0, and 33 frozen files (manifest SHA-256 *36f0ace2…295645*).

  – The SHA-256 hashes of the three compiled sources match the record package.

  – The axiom reports are: *n2_lower*, *n2_lower_frozen*, *n2_upper_frozen* and *N1.N1* depend on *propext*, *Classical.choice* and *Quot.sound*; *n2_upper* depends on *propext* and *Quot.sound*.

  – Because *n2_upper_frozen* lists no further axiom, the bridge used the formal Theorem 1 itself.

  – All three axiom gates passed. The logs artifact (SHA-256 *38554a1c…eb611b5*) is in the record package.

These are proofs by the same model family that constructed most of the material. Lean's kernel checks the proofs, so the trust that remains is in the statements. The statements are written to follow the paper's definitions literally, and a reader can compare them line by line.

The reports, the reviewers' own derivations and their attack scripts are in the record package.

G.6 What remains

- *The two asymptotic targets.* Both remain open, and neither needs to be resolved for the paper's narrower proved claims to be complete. The next steps are to prove or refute the

candidate obstruction of Section 7A.5, and to investigate stronger arguments for Targets A and B.

  - The quantity in the obstruction is close to the first-repetition function studied by Bugeaud and Kim [44]. Its definition alone does not prove the coefficient-sensitive obstruction.

  - One possible stronger argument would combine several rational 2-adic approximations to ξ.

- *Formalization.* Lean round 2 proves Corollary 2, Proposition 14, N1's exact count for every real K, N2, N3 and the second proof's finite base, while Theorem 4, N4 and N5 are partly proved in Lean (Section 8, Appendix G.7). Corollary 3, Proposition 13, N1's rate, N6, N7 and the rest of the second proof are not proved in Lean; Section 8 lists what each needs.

- *External and human review.* GPT has reviewed the completion round's draft, and Grok has reviewed the results listed under Round 5. Gemini's originality review (Round 7) is not counted as an originality determination. Human specialists have not reviewed it.

- *Originality.* The documented search of Round 8 did not locate an earlier statement of Theorem 1, N1–N4, N6, N7 or the obstruction. De Jesús [50] had already studied ξ, proved the equivalence of its irrationality measures with zero-run laws, and posed an effective measure as open; Corollary 2 supplies one with exponent 2 at intercept 0, by a different method. GPT's Round 9 check showed, and Claude's Round 10 check confirmed, that N5's bound follows from Ren's Theorem 4.1 [52], whose proof relies on a proposition Ren states without proof. A specialist's judgment is still needed before any claim of novelty.

G.7 Lean round 2 (24 October 2026)

After Lean round 1, the author asked GPT to prove in Lean the remaining results of the completion round under a written brief (*GPT_Lean_brief_v8.4.md*). GPT returned a partial package, *HypM_revision9_GPT_PARTIAL_20261024.zip*, with thirteen Lean modules, its own compile logs and a correspondence table matching each paper result to its Lean statement (*HypM_revision9_Statement_Map_20261024_1351.docx*). In GPT's own terms, its status table classed Corollary 2, Proposition 14, N2 and N3 as complete, N1 as complete for the finite count but not the rate, Theorem 4, N4, N5 and the second proof as partial, and Corollary 3 and Proposition 13 as not completed. It reported that no Gelfond–Schneider theorem was found in the pinned Mathlib, and that no axiom was added in its place.

Claude's checks. All thirteen modules match the hashes in GPT's *SHA256SUMS.txt*, the upload bundle, and the files in the public repository at commit *13bd411*. They contain no *sorry*, *native_decide* or added axioms. Each statement was read against the paper, and no correction to GPT's status table was needed. One difference from the paper is recorded: the paper's proof of N2 obtains the existence of an exit time from Corollary 3, whereas the Lean proof obtains it from the formal Theorem 1, which is equally valid.

The workflow. GPT proposed a replacement for the repository's round 1 workflow. The author instead added a separate workflow written by Claude (*rev9.yml*) and left the existing two unchanged. It repeats the full replay, compiles the round 1 and round 2 modules in dependency order, and gates the 27 theorems named in *Rev9Audit.lean*, together with the dependency check. GPT's proposed workflow also named a few auxiliary lemmas individually, such as *survivor_square_bound* and the two cases of *corollaryN3_exact*; in *rev9.yml* these are covered through the gated theorems that use them.

Public runs, 24 October 2026.

- Round 2 workflow, run #1, on commit *2f6a734*: passed. The gate reported only *propext*, *Classical.choice* and *Quot.sound* for all 27 theorems (*propext* alone for *finite_certificate*), and the dependency check passed for *second_proof_finite_base* and *period_implies_exit*.

- The author then pinned all three workflows to the Ubuntu 24.04 build image, changing one line in each file. Claude confirmed that no other file changed.

- Final runs on commit *13bd411*: round 2 workflow run #4 (https://github.com/whouxe82/hypothesis-m-lean/actions/runs/40131960252), round 1 workflow run #9 (https://github.com/whouxe82/hypothesis-m-lean/actions/runs/40131960287) and independent replay run #12 (https://github.com/whouxe82/hypothesis-m-lean/actions/runs/40131960242). All passed. The two remaining annotations on the run pages are routine notices from GitHub about its own helper software and build image.

What the gate does not establish. An axiom report confirms that a theorem is proved from its stated hypotheses using only the standard axioms. It does not discharge those hypotheses. Theorem 4, N4 and N5 therefore remain only partly proved in Lean, as GPT's status table states, and the trust that remains for the complete results is in the statements, which follow the paper's definitions and can be compared with it line by line.